
2023 최신 반도체 집적회로 설계기술 워크샵

Low-Dropout (LDO) Voltage Regulators

– From Basics to Recent Design Trends

(presented in A-SSCC 2022)

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November 22, 2023

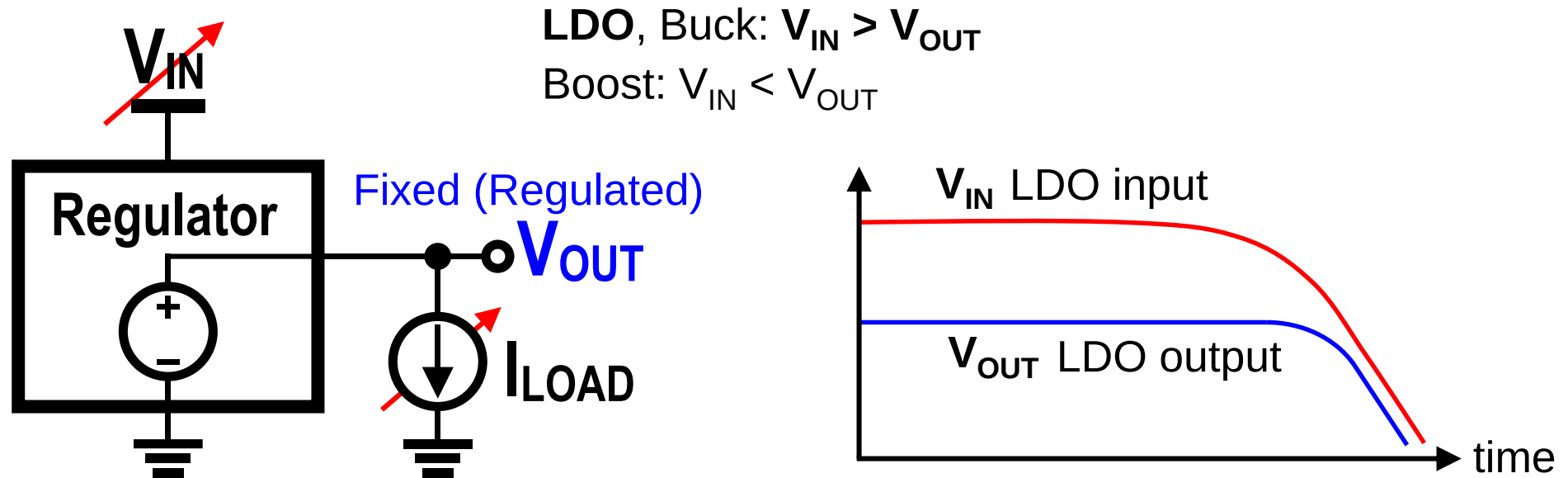
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- Basics of LDO Regulator**
- Key Performance Metrics & LDO Configurations**
- Frequency Response & Stability**
- Fast Transient-Response LDO Designs**
- Power Supply Rejection (PSR)**
- Energy-Efficient Ultra-Low-Dropout (Triode) LDO**
- Conclusion**

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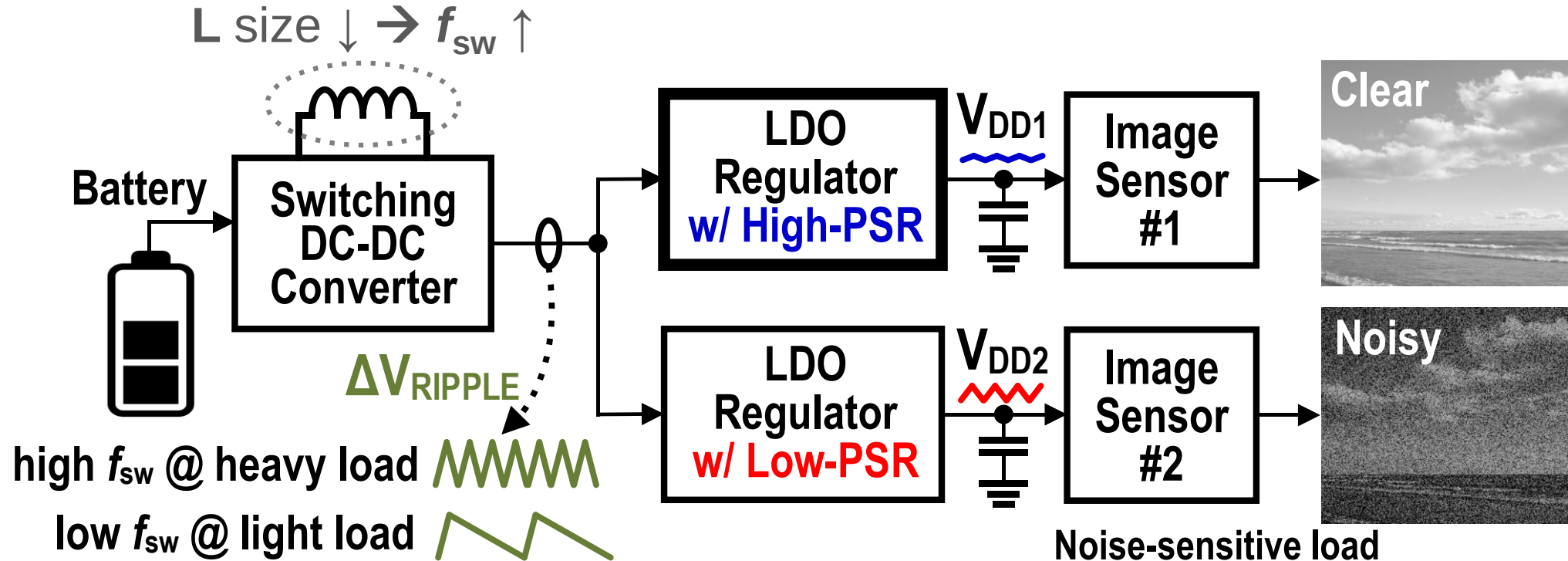
- ❑ **Basics of LDO Regulator**
- ❑ Key Performance Metrics & LDO Configurations
- ❑ Frequency Response & Stability
- ❑ Fast Transient-Response LDO Designs
- ❑ Power Supply Rejection (PSR)
- ❑ Energy-Efficient Ultra-Low-Dropout (Triode) LDO
- ❑ Conclusion

What is the Voltage Regulator?



- ❑ Ideal Regulator?: an voltage source (V_{OUT}) used to be V_{DD} of a load system
- ❑ Keep V_{OUT} fixed voltage under any situations
 - V_{IN} varies, but V_{OUT} is fixed $\rightarrow$ Regulator's Line Regulation
 - I_{LOAD} varies, but V_{OUT} is fixed $\rightarrow$ Regulator's Load Regulation
- ❑ Always " $V_{IN} > V_{OUT}$ " due to the LDO's resistive voltage generation

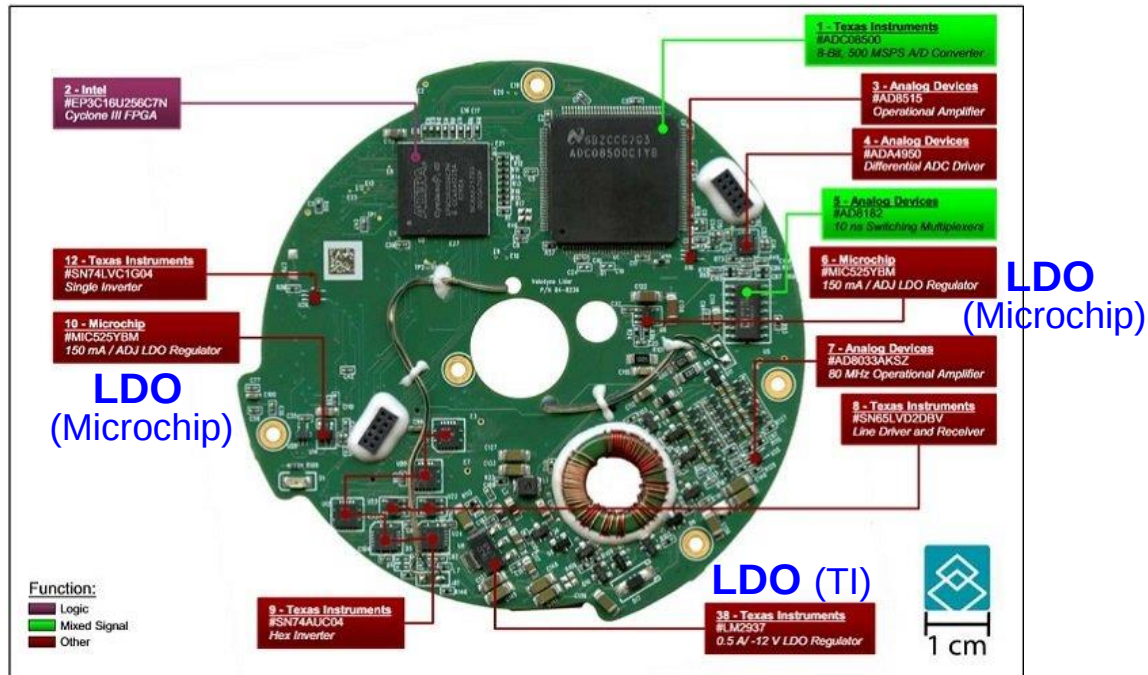
Key Features of LDO Regulator



- ❑ Ripple Suppression (Power-Supply Rejection, **PSR**) → Used as a Post-Regulator
- ❑ Impedance Isolation (Low Z_{OUT} like as a buffer)
- ❑ Low Output Noise (No switching operation) → Useful for noise-sensitive loads

Usages of LDO Regulator

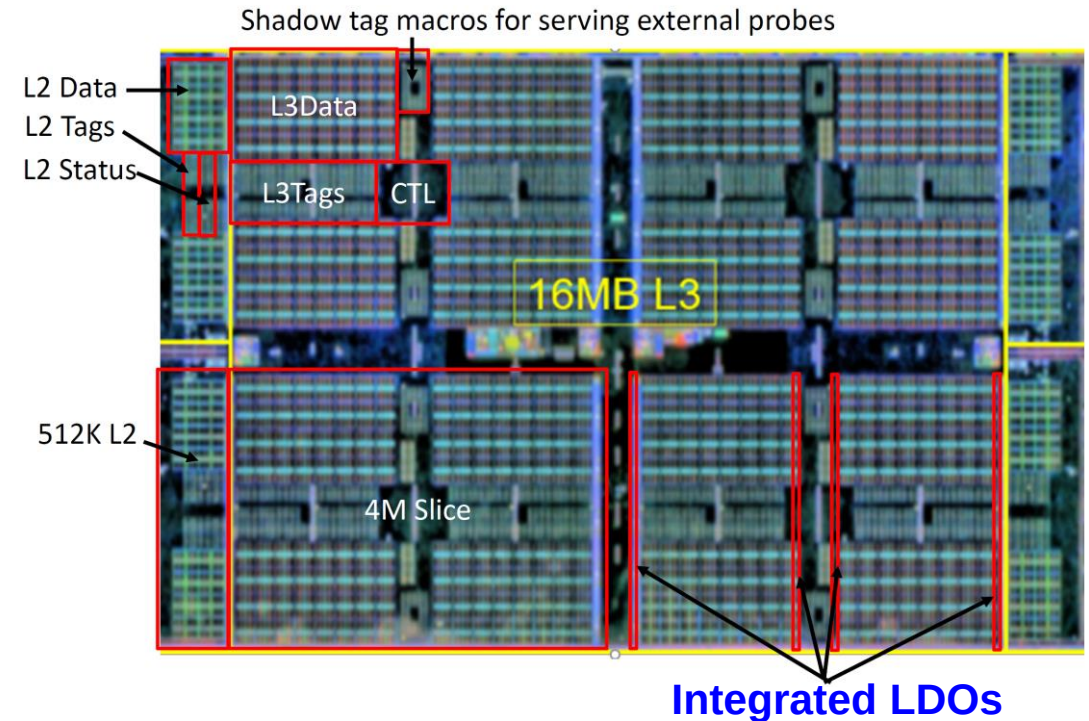
Discrete LDO chips



Velodyne LIDAR Puck VLP-16 Sensor
(source: Tech Insights)

- **Clean voltage-supply** (low-noise of LDO)
- **Step-down DC-DC** conversion

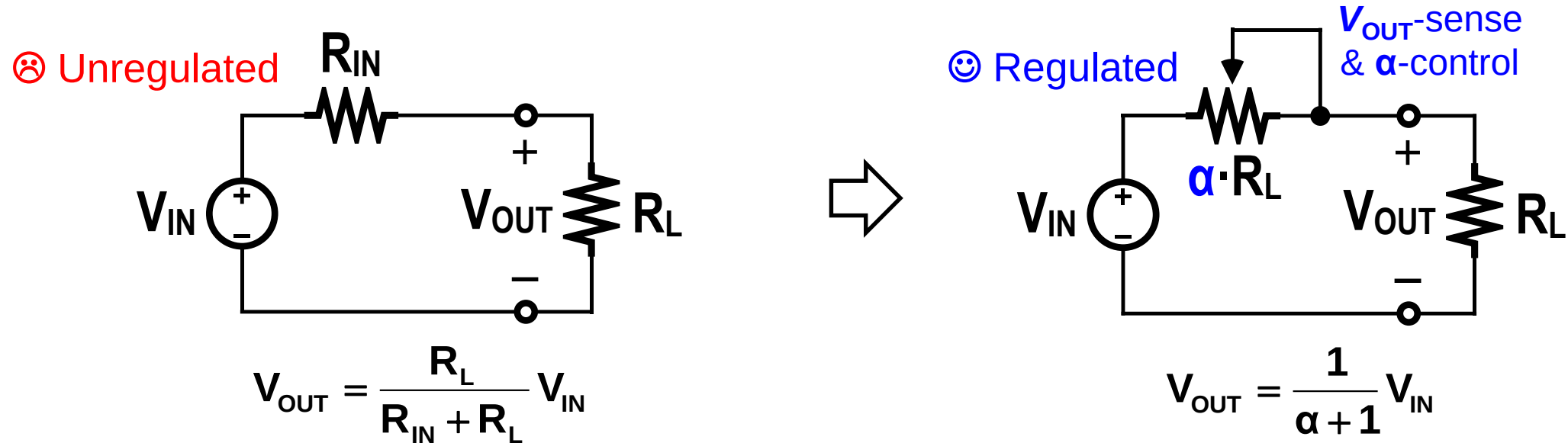
SoC-Integrated LDO IPs



AMD 7-nm Zen 2
(ISSCC'20)

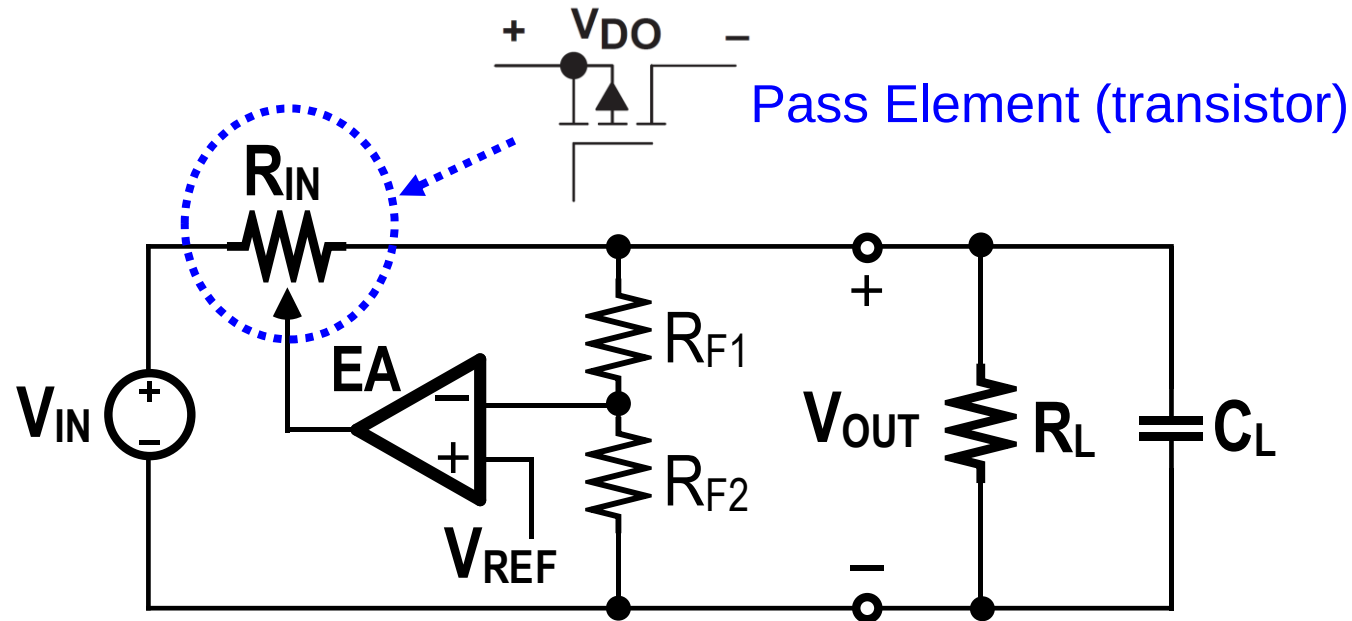
- **Per-Core voltage-scaling** (compactness)

Conceptual Realization of LDO Regulator (1/2)



- ☐ V_{OUT} generated using a **resistive** divider
 - Fixed divider ratio → sensitive to load current (R_L) changes
- ☐ Feedback loop **regulates** R_{IN} ($= \alpha \cdot R_L$) such that it is always a desired fraction of load current
 - V_{OUT} is independent to load current (R_L)

Conceptual Realization of LDO Regulator (2/2)

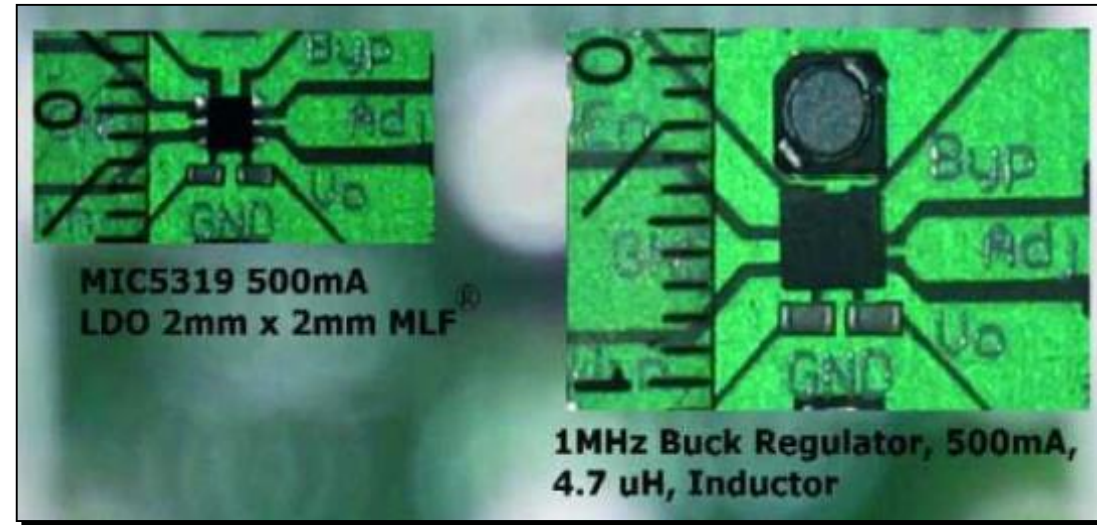


- Feedback loop adjusts R_{IN} such that $V_{OUT} = \left(\frac{R_{F1} + R_{F2}}{R_{F2}} \right) V_{REF}$
- V_{OUT} is ideally independent to V_{IN}
 - Feedback resistors $(R_{F1} + R_{F2}) \gg R_L$

Comparison of Power Converter Topologies (1/2)

	Linear Regulator (LDO)	Switched Inductor DC-DC	Switched Capacitor (e.g., Charge-Pump)
Transfer Media	Resistive	Inductive	Capacitive
Efficiency	Low 😞	High 😊	Medium
Noise	Very Low 😊 (no switching)	High 😞	Medium
Output Current	Low to Medium	Low to High	Low
Step-Up	X	O	O
Step-Down	O	O	O
Footprint	Small 😊	Large 😞	Medium

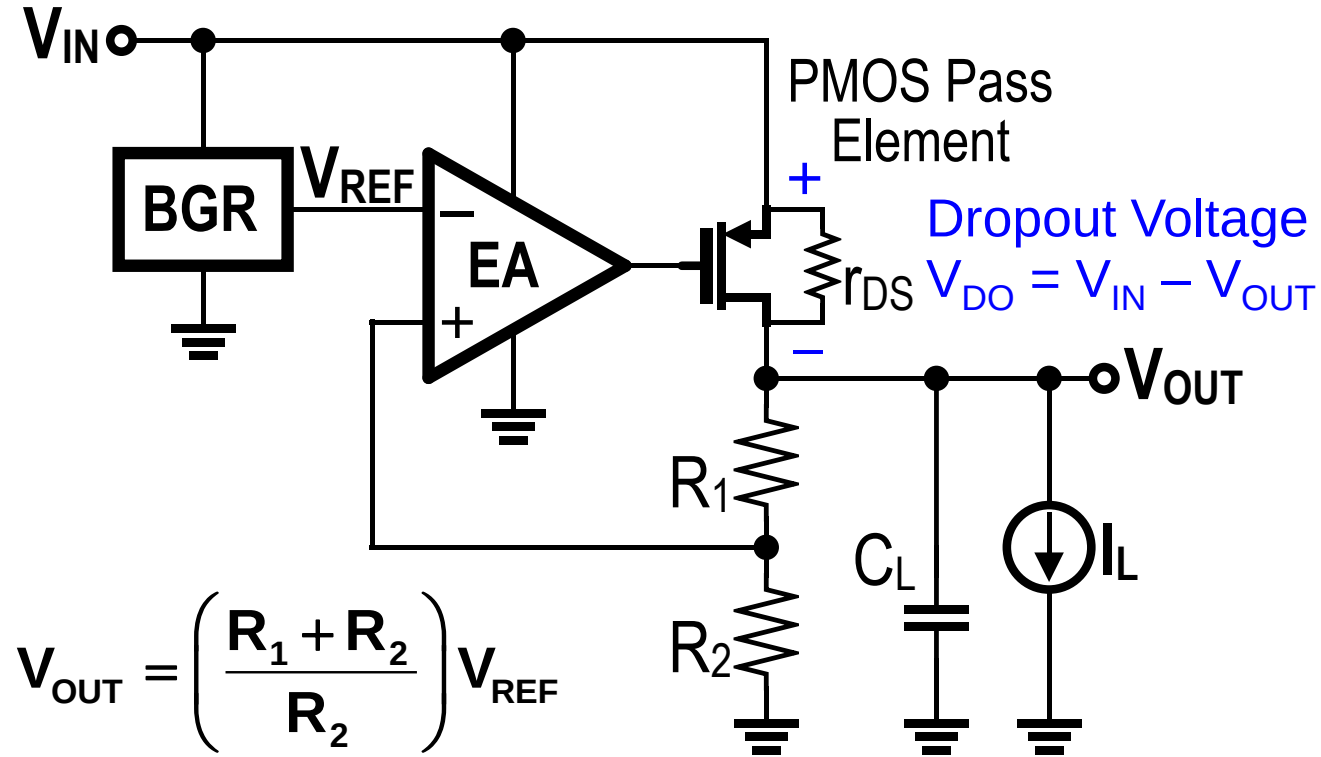
Comparison of Power Converter Topologies (2/2)



	CPU @ 24MHz	CPU @ 48MHz
LDO Regulator	165 μ A/MHz	158 μ A/MHz
Switched Inductor DC-DC	92 μ A/MHz	96 μ A/MHz

❑ **LDO regulator:** small footprint & low noise, but low efficiency

Basic Architecture of LDO Regulator



- **Bandgap (BGR):** fixed reference voltage V_{REF}
- **Error-Amplifier (EA):** comparing V_{REF} vs. V_{OUT} , and providing feedback loop-gain (T)
- C_L : used to “filter” ripple and noise

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Key Performance Metrics

- Regulation (in DC-domain)

- Line/load regulation

- Voltage (V_{OUT}) accuracy

- Dropout voltage (V_{DO})

- Power efficiency

- Quiescent current (I_Q)

- Current efficiency

- Input/output range

- Input (V_{IN}) range
- Output (V_{OUT}) range
- Max. output current ($I_{L,MAX}$)

- Transient response

- Voltage droop & Recovery time
- Loop-bandwidth (GBW)
- Slew rate

- Stability

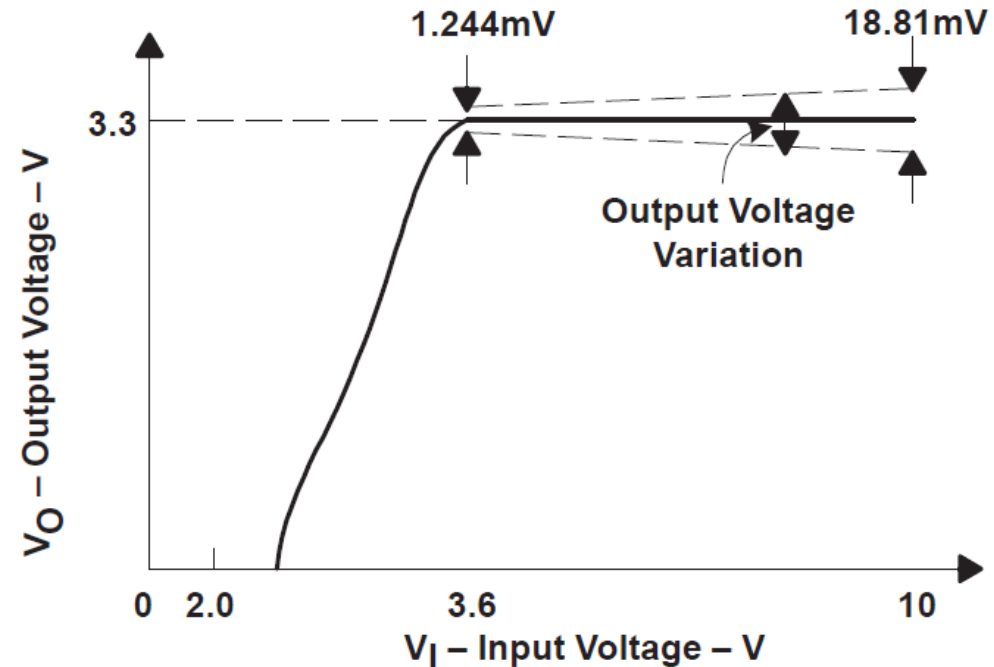
- Load (I_L) dynamic range
- Output capacitor (C_L)

- Power supply rejection (PSR)

- Output noise

- RMS noise voltage within BW

Performance Metrics: Line Regulation (1/2)



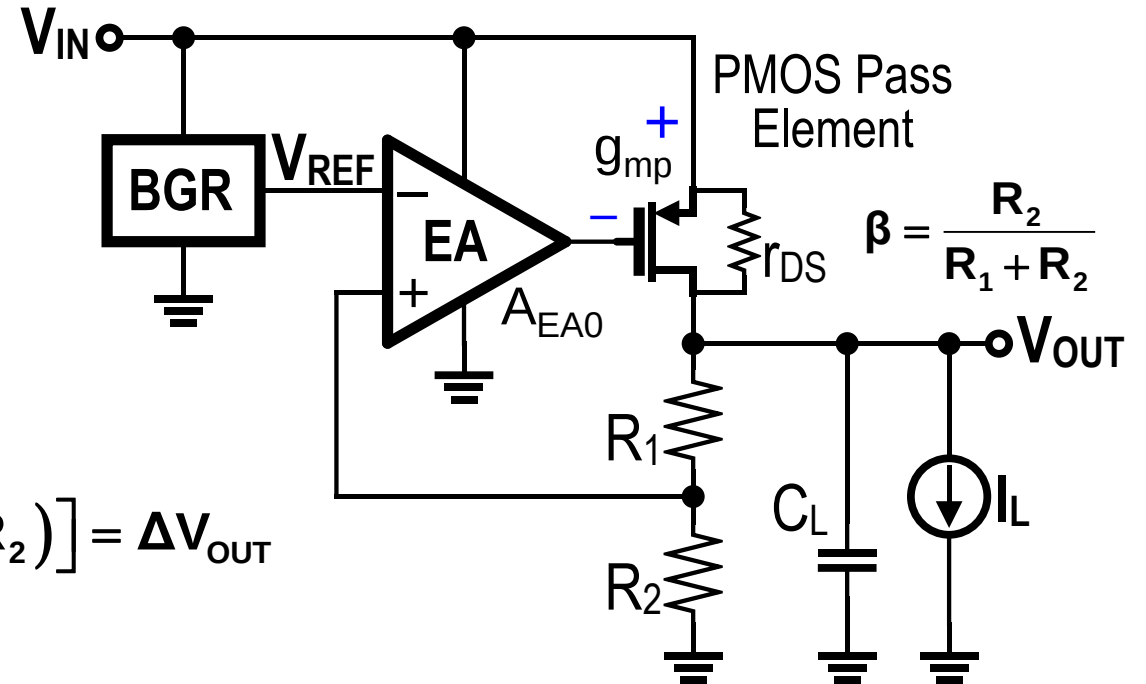
(source: TI tech. review)

□ Ability to maintain desired V_{OUT} with varying V_{IN}

■ Line Regulation = $\Delta V_{OUT} / \Delta V_{IN}$

■ Unit: [mV/V] or [%/V]

Performance Metrics: Line Regulation (2/2)

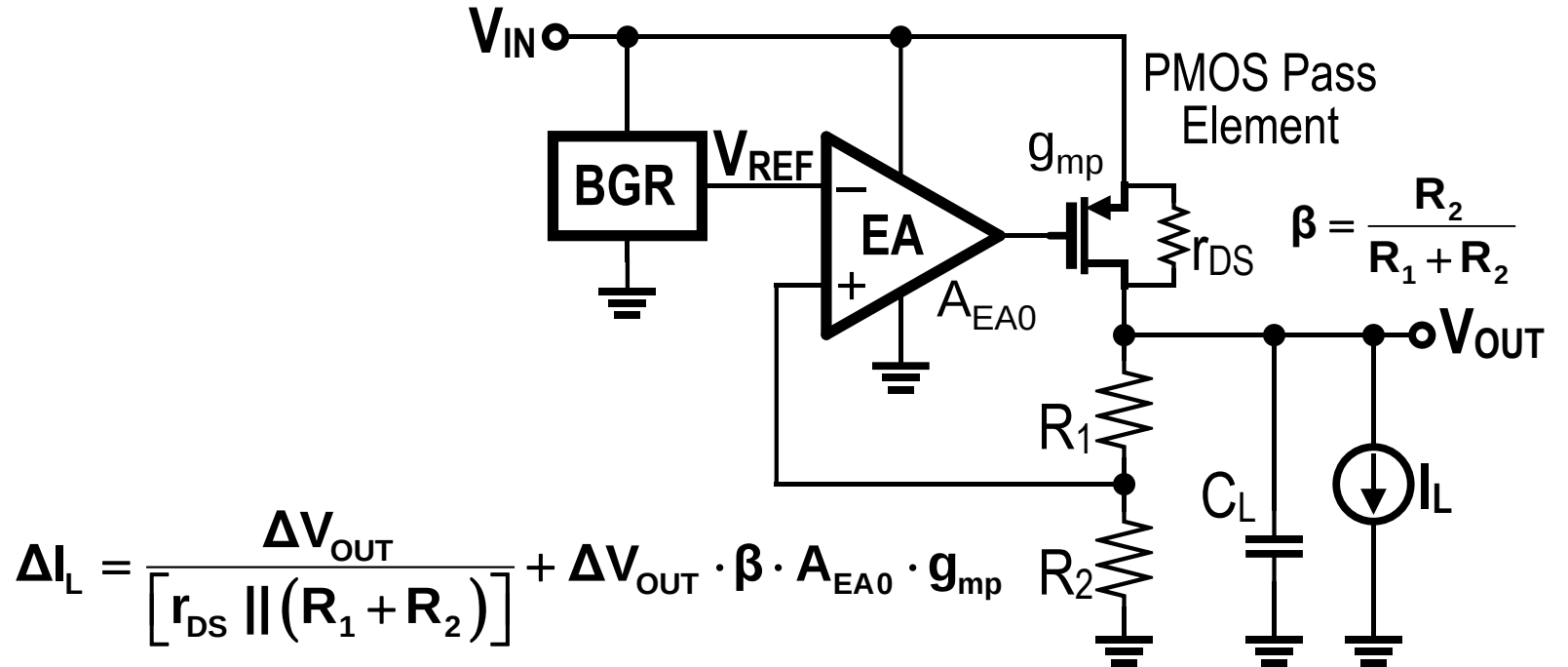
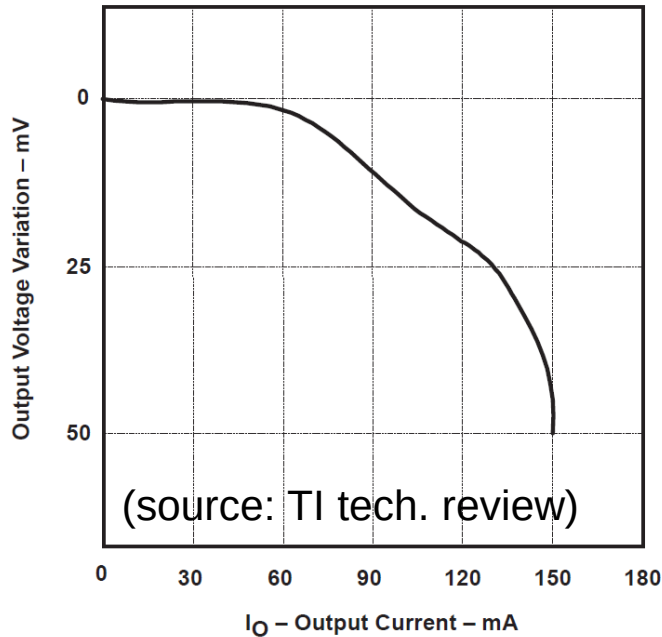


$$(\Delta V_{IN} - \Delta V_{OUT} \cdot \beta A_{EA0}) g_{mp} [R_L \parallel r_{DS} \parallel (R_1 + R_2)] = \Delta V_{OUT}$$

$$\square \text{ Line Regulation} = \frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{IN}}} = \frac{g_{\text{mp}} [R_L \parallel r_{\text{DS}} \parallel (R_1 + R_2)]}{1 + \beta A_{\text{EA0}} g_{\text{mp}} [R_L \parallel r_{\text{DS}} \parallel (R_1 + R_2)]} \approx \frac{1}{\beta A_{\text{EA0}}}$$

- Changes in V_{IN} suppressed by error-amp (EA)'s gain (A_{EA0})

Performance Metrics: Load Regulation



$$\Delta I_L = \frac{\Delta V_{OUT}}{r_{DS} \parallel (R_1 + R_2)} + \Delta V_{OUT} \cdot \beta \cdot A_{EA0} \cdot g_{mp}$$

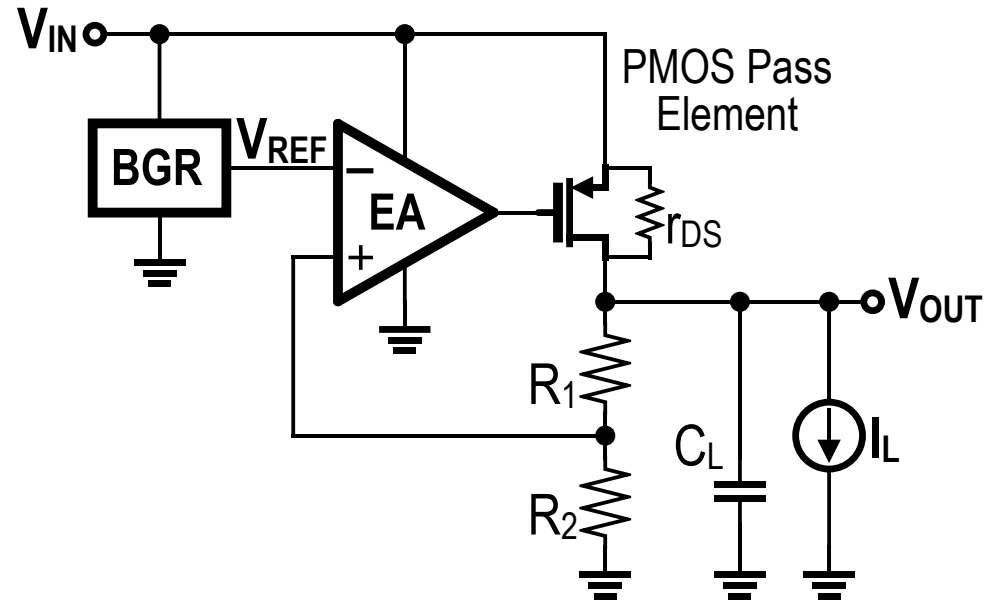
□ Ability to maintain desired V_{OUT} with varying I_L

$$\square \text{ Load Regulation} = \frac{\Delta V_{OUT}}{\Delta I_L} = \frac{r_{DS} \parallel (R_1 + R_2)}{1 + \beta A_{EA0} g_{mp} [r_{DS} \parallel (R_1 + R_2)]} \approx \frac{r_{DS}}{1 + T_0}$$

□ Dependent of feedback loop-gain (T_0)

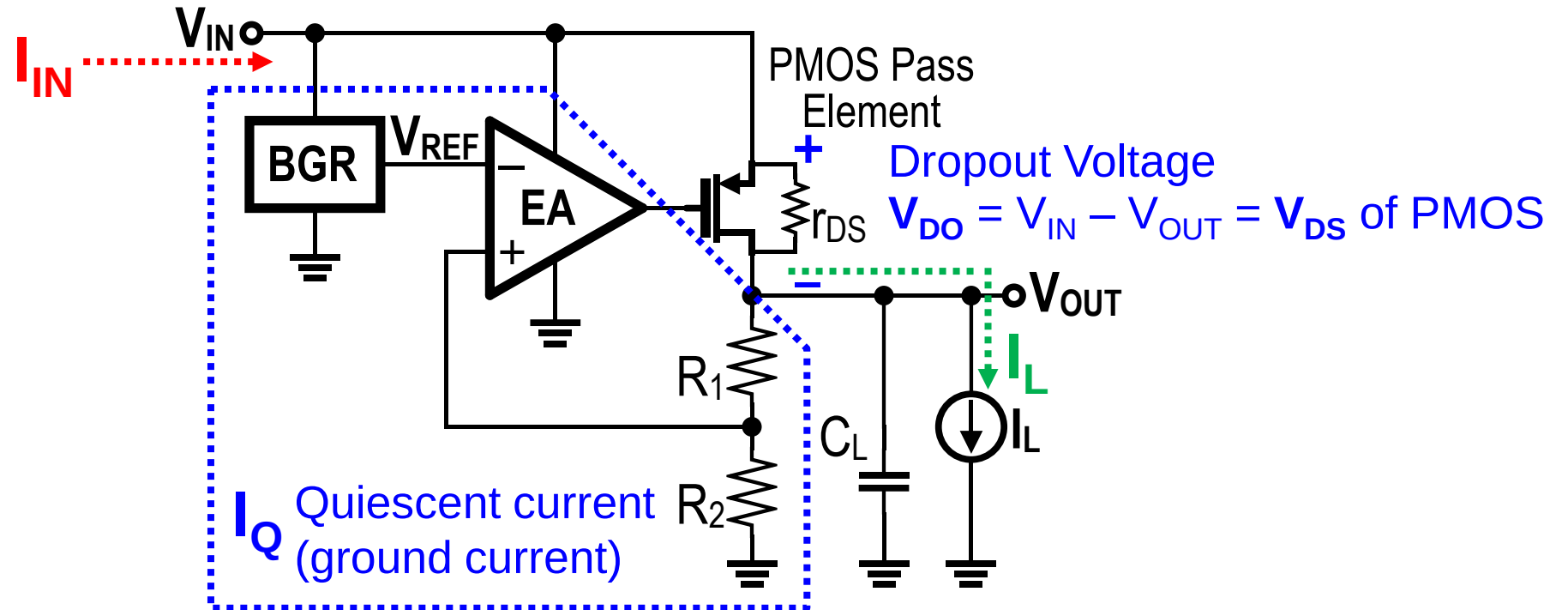
Performance Metrics: Voltage Accuracy

- Line/Load regulation: ΔV_{LR} , ΔV_{LDR}
- Reference voltage drift: $\Delta V_{O,REF}$
- Error-Amp offset drift: $\Delta V_{O,EA}$
- Feedback resistor tolerance: $\Delta V_{O,R}$



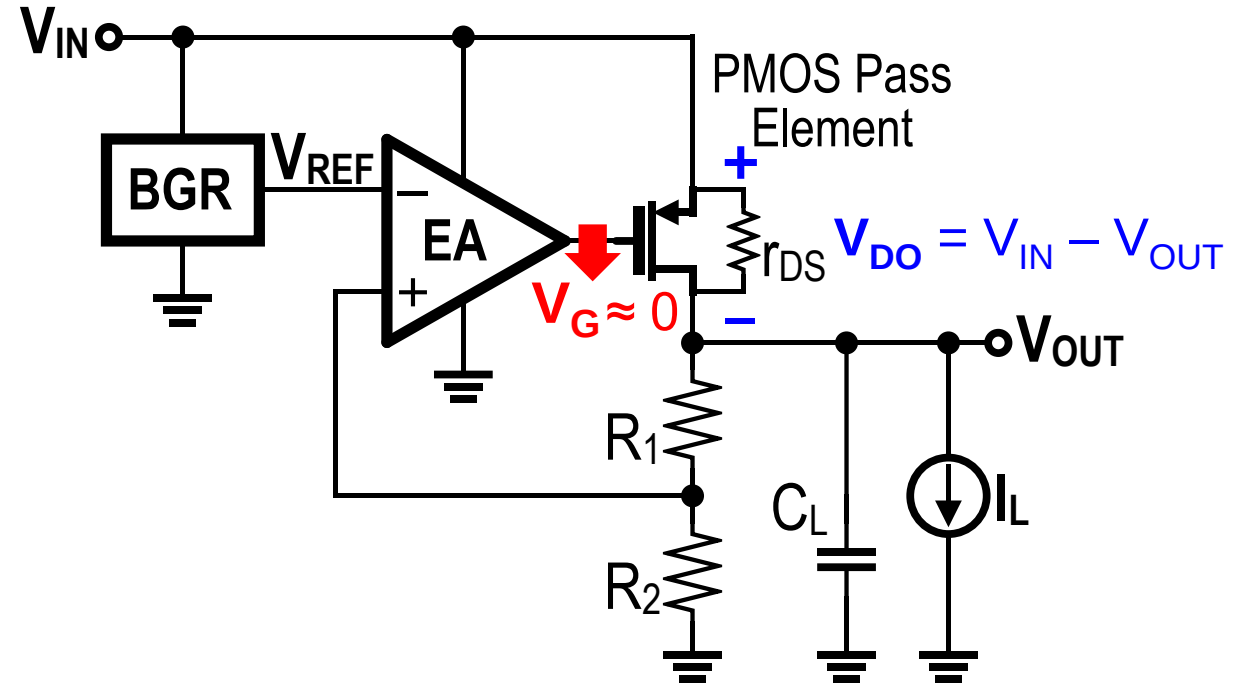
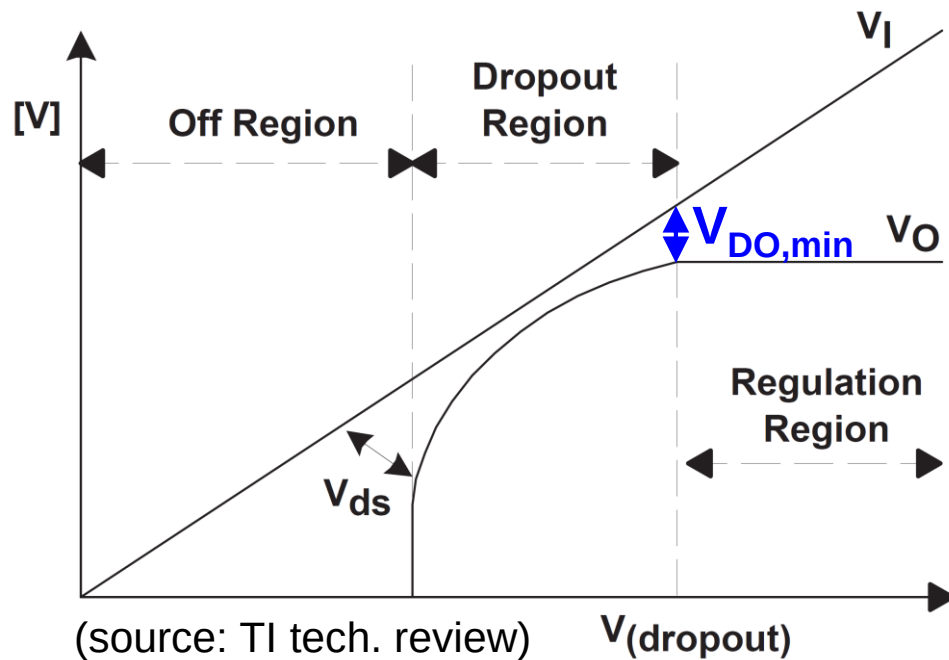
$$\square \text{ LDO's accuracy} = \frac{|\Delta V_{LR}| + |\Delta V_{LDR}| + \sqrt{\Delta V_{O,REF}^2 + \Delta V_{O,EA}^2 + \Delta V_{O,R}^2}}{V_{OUT}} [\%]$$

Performance Metrics: Power Efficiency (η)



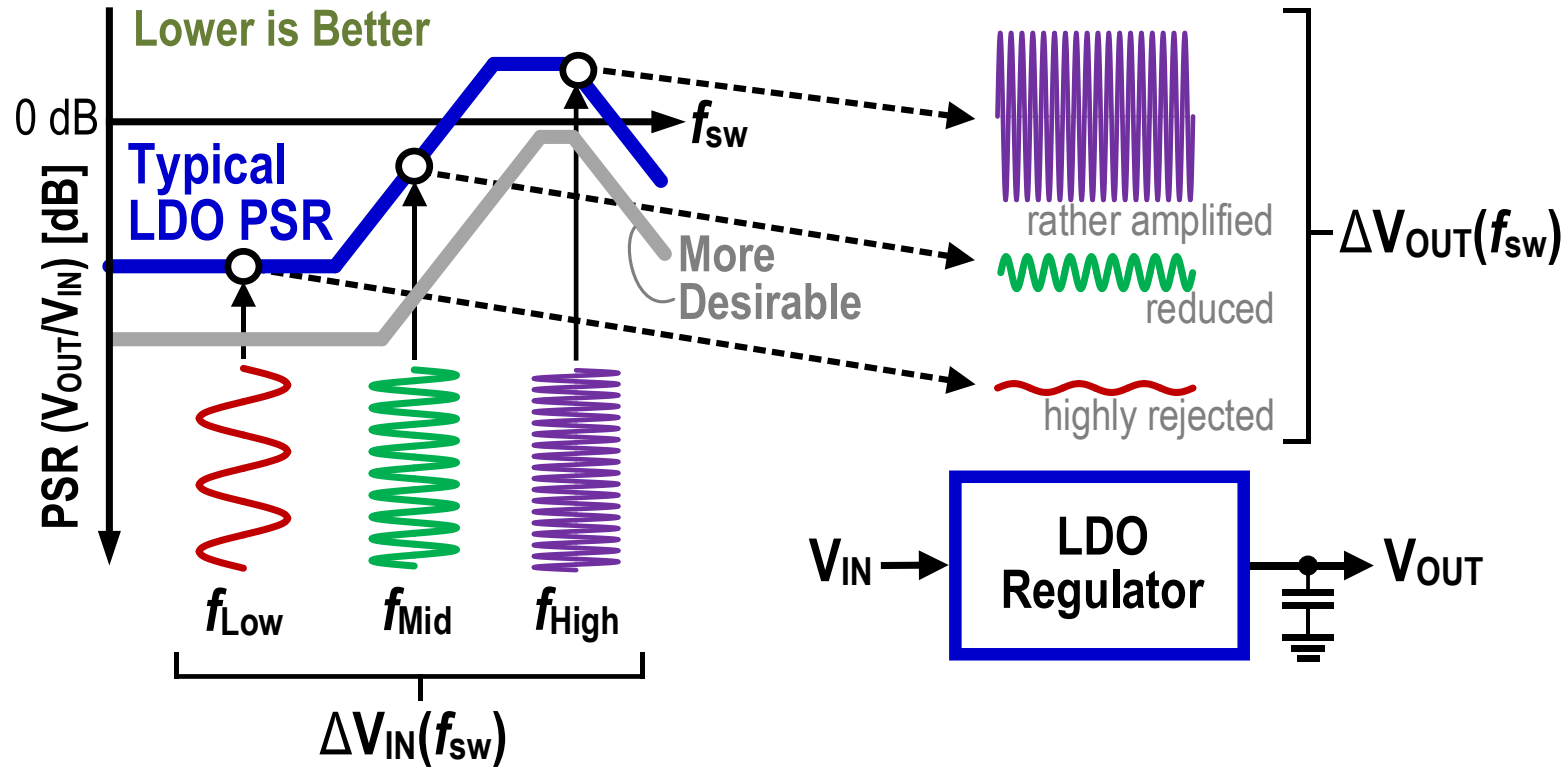
$$\square \text{ Efficiency } (\eta) = \frac{V_{OUT} \cdot I_L}{V_{IN} \cdot I_{IN}} = \frac{V_{OUT} \cdot I_L}{V_{IN} \cdot (I_L + I_Q)} = \frac{(V_{IN} - V_{DO}) \cdot I_L}{V_{IN} \cdot (I_L + I_Q)} \quad [\%]$$

Performance Metrics: Min. Dropout ($V_{DO,min}$)



- Min. dropout voltage ($V_{DO,min}$): minimum *regulating* point @ full-load ($I_{L,max}$)
 - At $V_{DO,min}$ & $I_{L,max}$, the EA's output (V_G) reaches the minimum limit ($\approx 0V$)
 - Conventional LDOs have $V_{DO,min}$ of 50mV ~ 200mV
 - High aspect ratio (W/L) of pass element $\rightarrow$ large parasitic capacitance (C_G) ☹

Performance Metrics: Power-Supply Rejection

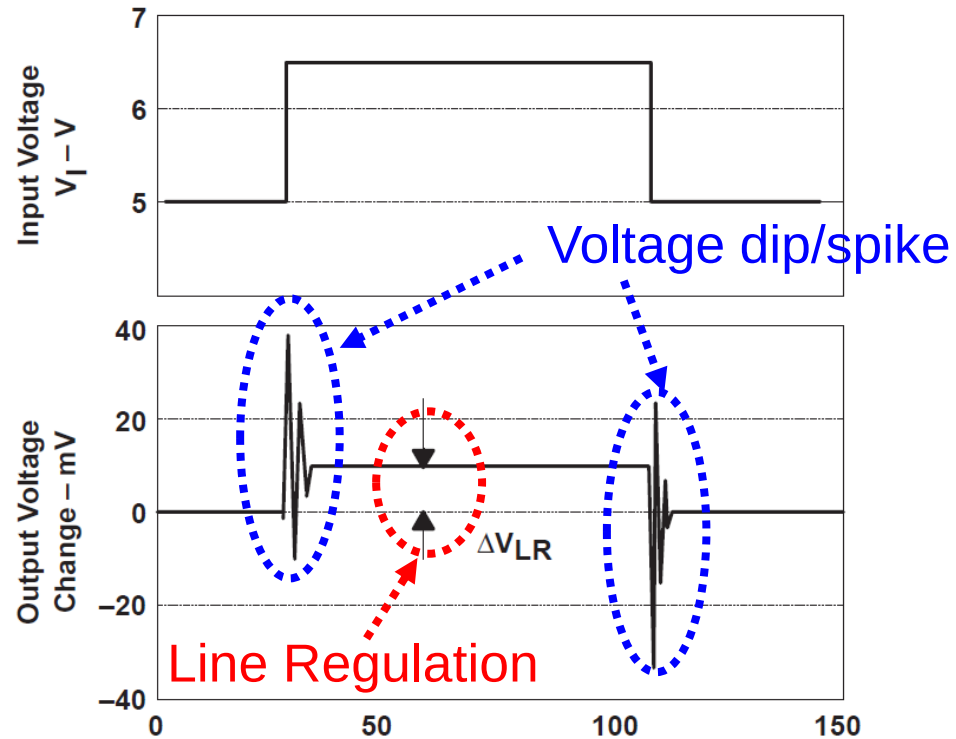


□ Power-supply rejection = $\text{PSR}(f) = \frac{\Delta V_{OUT}(f)}{\Delta V_{IN}(f)}$

- Similar to line regulation, but measured ΔV_{OUT} for *small-signal* ΔV_{IN} variations
- **PSR** = ripple rejection capability

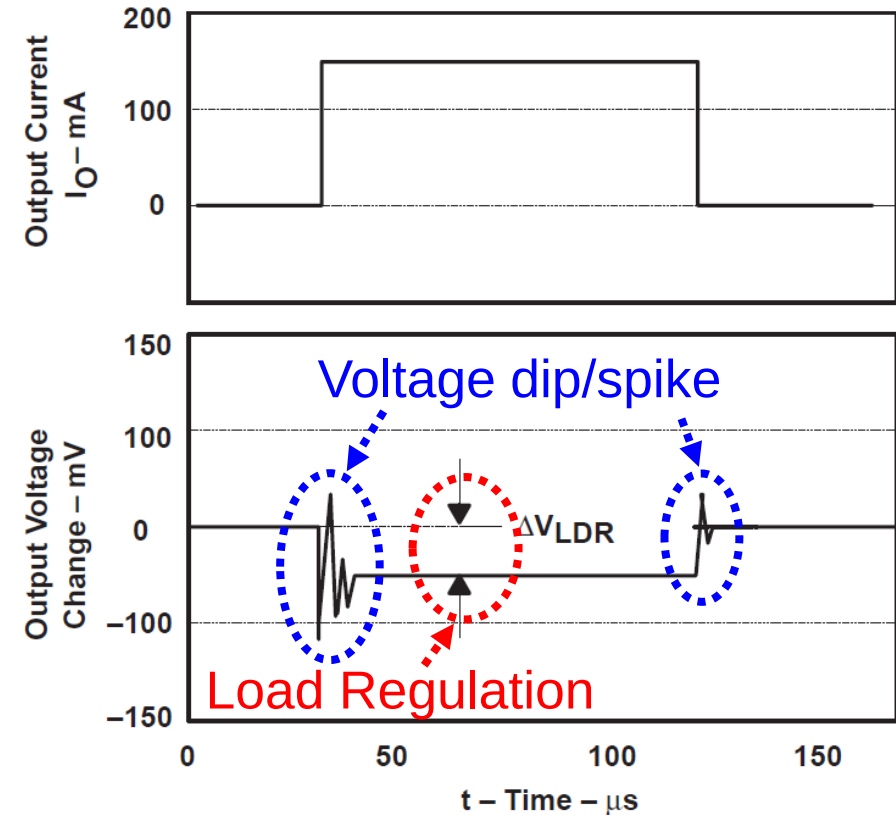
Performance Metrics: Transient Responses

□ Line (ΔV_{IN}) transient response



(source: TI tech. review) t - Time - μs

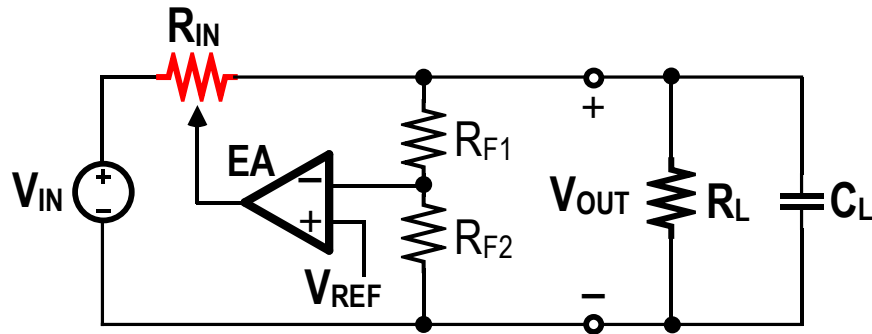
□ Load (ΔI_L) transient response



■ ΔV_{OUT} is dependent of load-step(ΔI_L), C_L , loop bandwidth (**GBW**), and slew rate (**SR**)

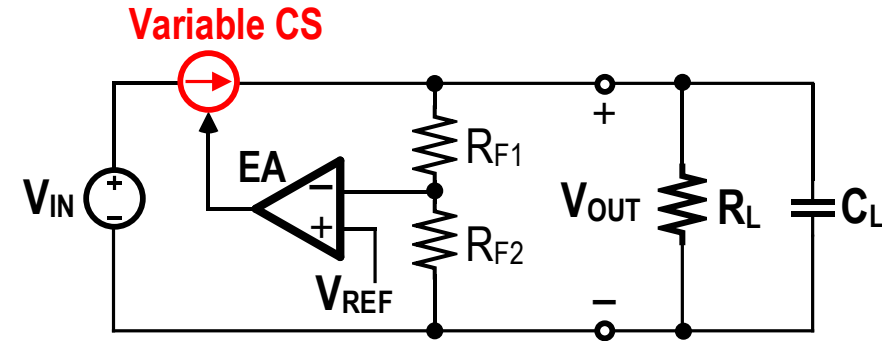
Various Types of LDO (1/4)

□ Triode LDO



- Pass element works in triode region
- Small V_{DO} (high η) 😊
- Low regulation due to low loop-gain (T) 😞
- Bad PSR due to small R_{DS} 😞
- Normally not preferred

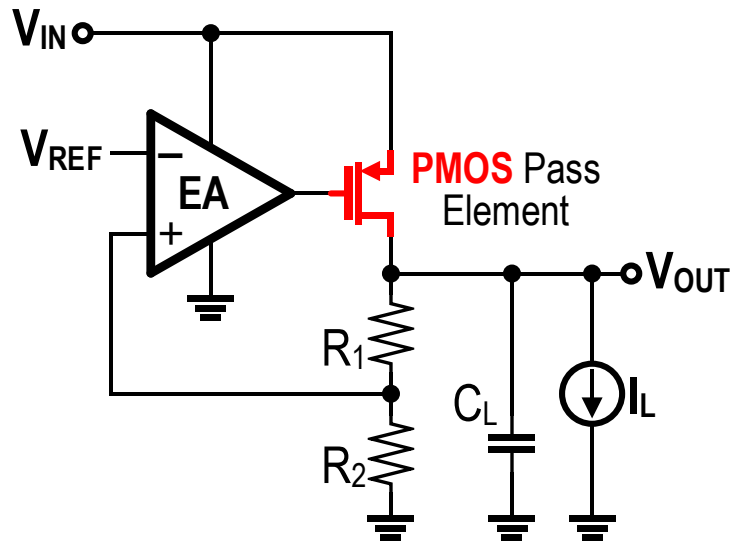
□ Saturation LDO



- Pass element works in saturation region
- High V_{DO} (lower η) 😞
- Good regulation performance due to high loop-gain (T) 😊
- Good PSR owing to large R_{DS} 😊
- Adopted in most LDOs

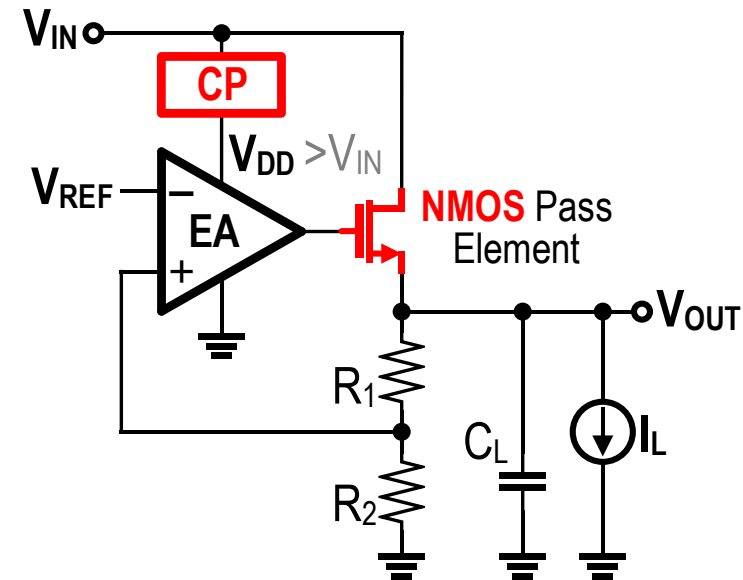
Various Types of LDO (2/4)

□ P-type LDO



- Pass element = **PMOS**
- Better regulation performance (higher loop-gain) 😊
- Stability design issue ☹️

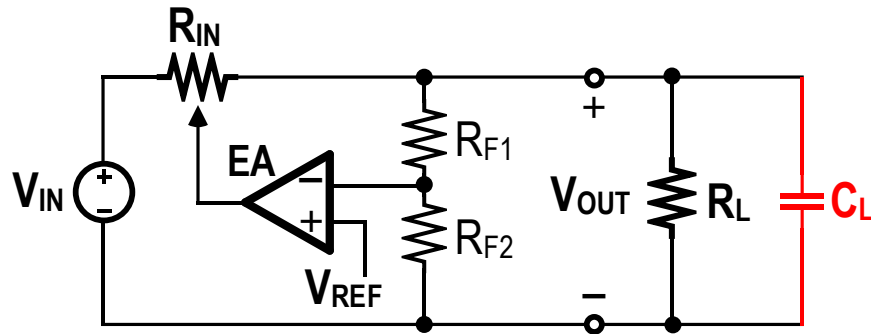
□ N-type LDO



- Pass element = **NMOS**
- Lower regulation performance ☹️
- Easier stability design 😊
- Additional charge-pump (V_{DD}) ☹️

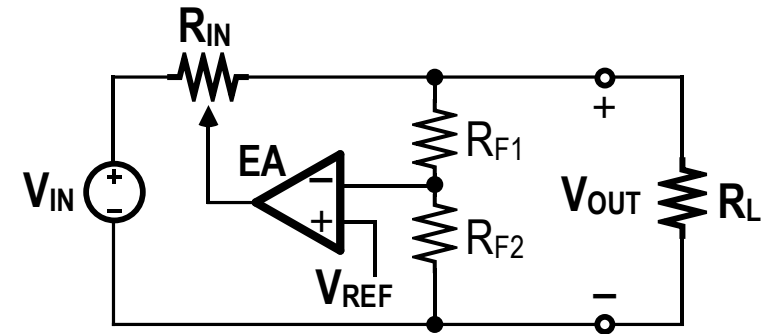
Various Types of LDO (3/4)

□ Cap LDO



- External capacitor (C_L) is required
- Larger PCB footprint ☹️
- Low voltage droop @ ΔI_L 😊
- Good PSR @ high freq. 😊
- Narrow bandwidth ☹️
- Stability issue over wide I_L ☹️

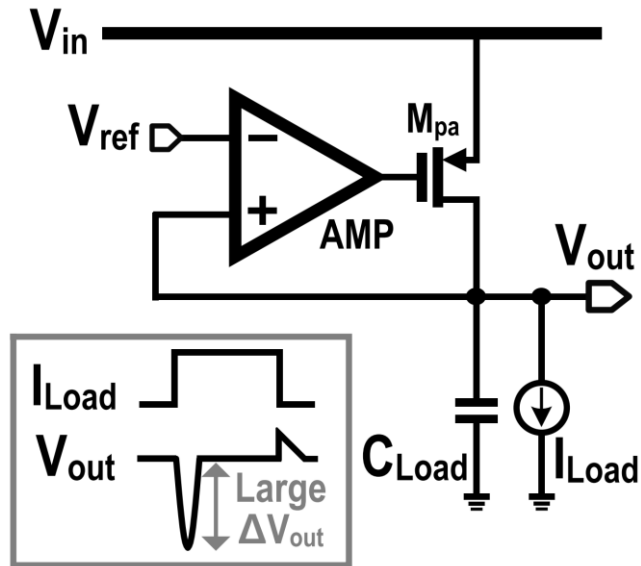
□ Capless LDO



- No "external" component (C_L)
- Compact or fully-integrated 😊
- High voltage droop @ ΔI_L ☹️
- Bad PSR @ high freq. ☹️
- Wider (faster) bandwidth ☹️
- Easier stability compensation 😊

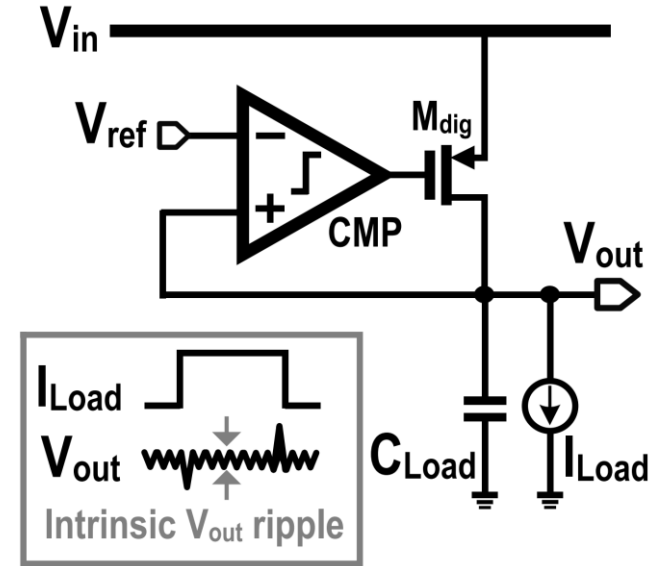
Various Types of LDO (4/4)

□ Analog LDO



- Analog control & single pass-TR.
- Excellent PSR 😊
- No ripple & clean V_{OUT} 😊
- Re-design @ different tech. nodes 😞
- Hard to work in low V_{IN} 😞

□ Digital LDO

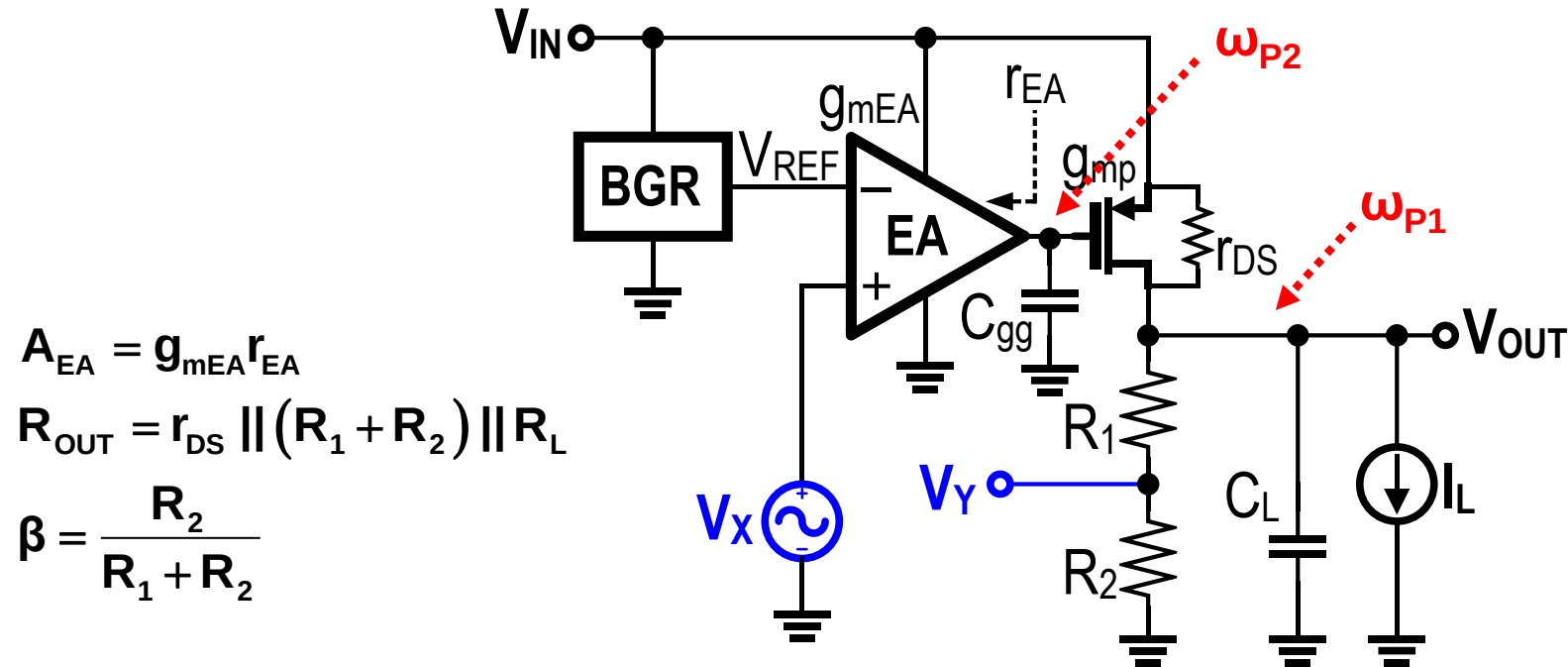


- Digital control & segmented pass-TR.
- Scalable design @ various CMOS tech. 😊
- Digitally synthesizable 😊
- Limit cycle oscillation (LCO) ripple 😞
- Friendly to low V_{IN} 😊

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Feedback Loop-Gain's Transfer Function: $T(s)$



$$A_{EA} = g_{mEA} r_{EA}$$

$$R_{OUT} = r_{DS} \parallel (R_1 + R_2) \parallel R_L$$

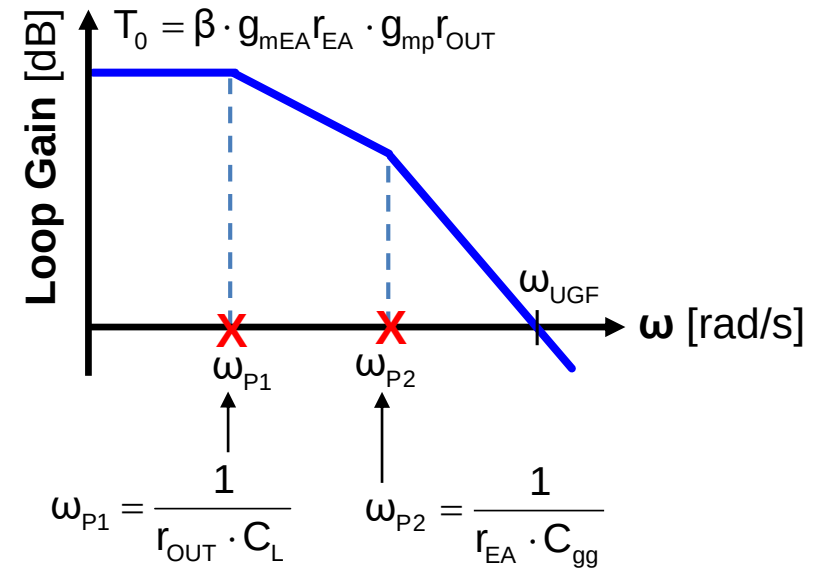
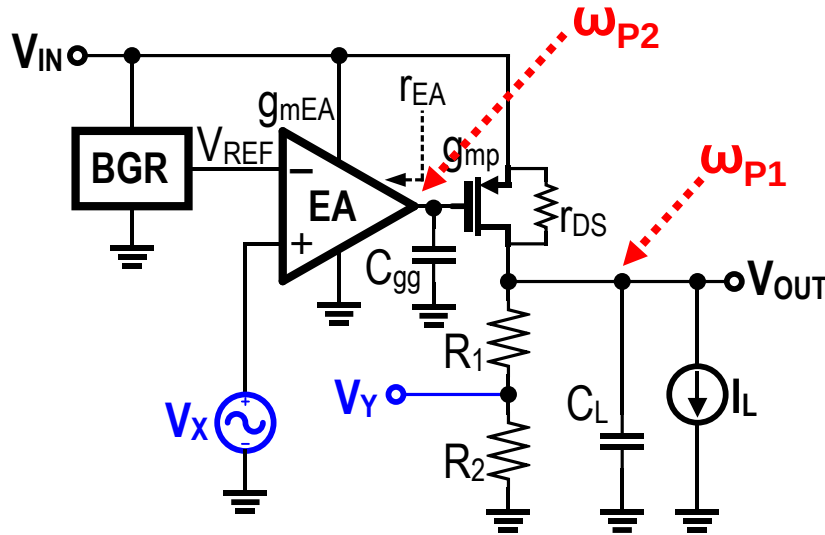
$$\beta = \frac{R_2}{R_1 + R_2}$$

□ Transfer function of feedback loop-gain:

$$T(s) = -\frac{V_Y}{V_X} = A_{EA} \cdot g_{mp} R_{OUT} \cdot \beta \frac{1}{\left(1 + \frac{s}{\omega_{P1}}\right) \left(1 + \frac{s}{\omega_{P2}}\right)}, \text{ where } \omega_{P1} = \frac{1}{R_{OUT} \cdot C_L} \quad \omega_{P2} = \frac{1}{r_{EA} \cdot C_{gg}}$$

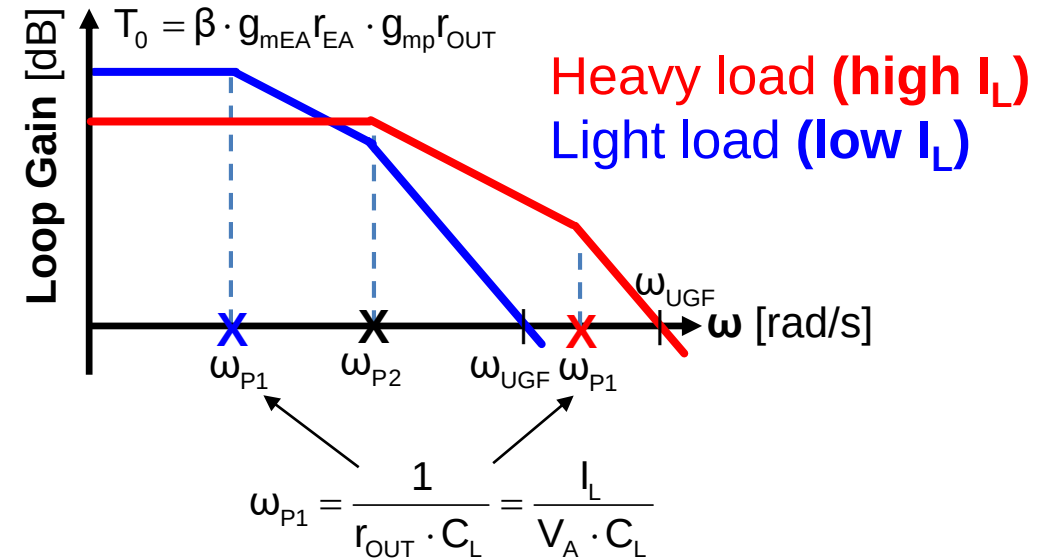
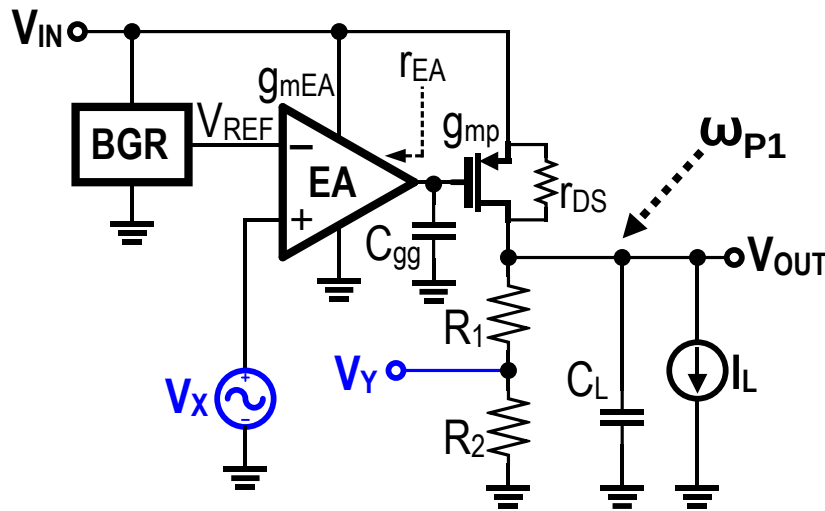
*Note: C_{gg} becomes higher for low V_{DO} & high I_L

Basics of Frequency Response



- Unity-gain frequency (ω_{UGF} , GBW) determines the LDO's transient response
 - Wide bandwidth (high ω_{UGF}) = fast transient speed
- Closely-spaced poles compromise stability
 - " $\omega_{UGF} \leq \text{non-dominant pole}$ " is desirable: just 1-pole under ω_{UGF}
 - Frequency (stability) compensation is necessary

Load Current (I_L) Dependency of Output Pole



□ Total R_{OUT} (open-loop) = $R_{OUT} = r_{DS} \parallel (R_1 + R_2) \parallel R_L$

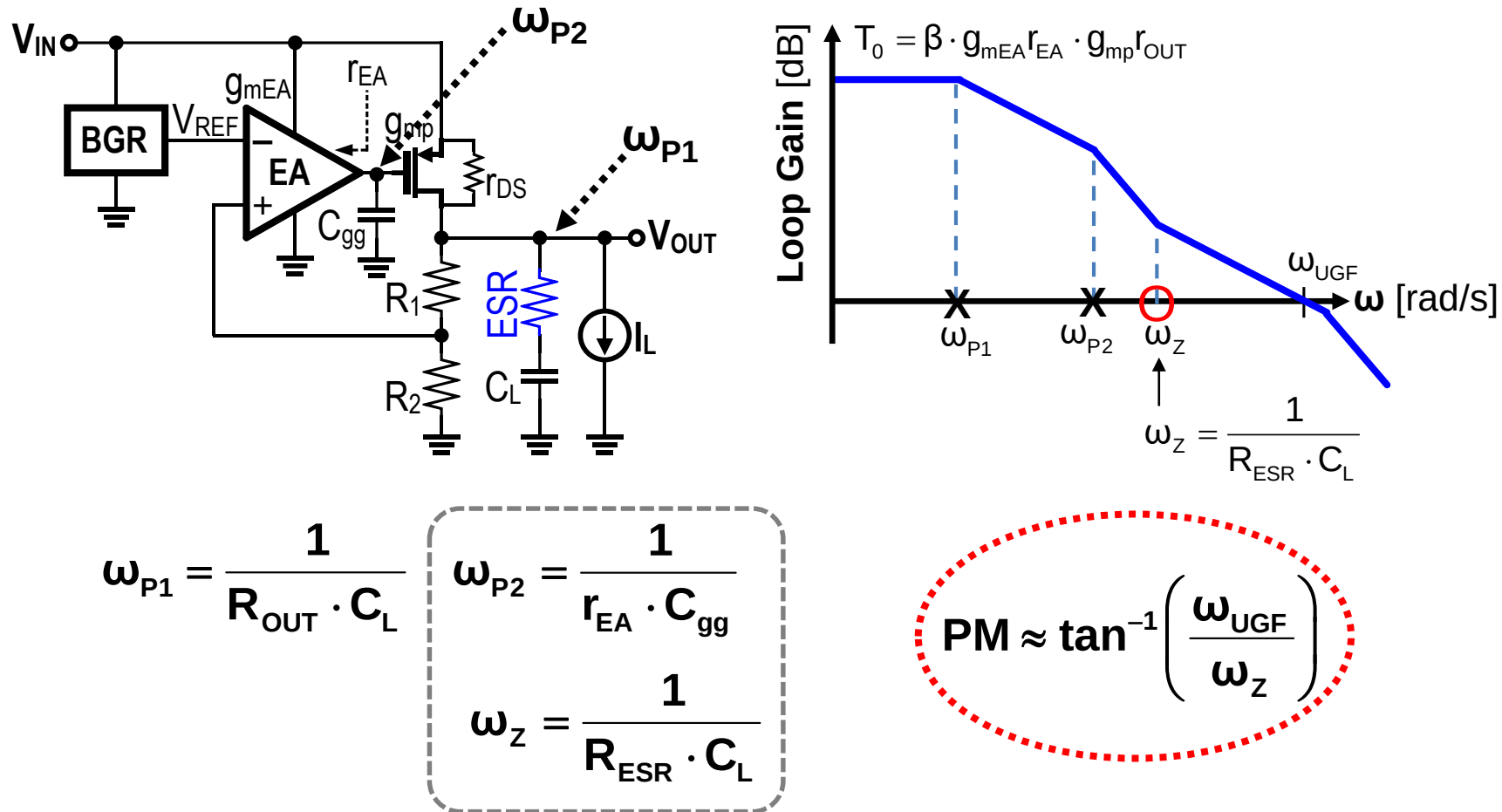
□ Why cap-LDO (w/ C_L) is difficult to be stable?

■ Load current (I_L) variation significantly changes the pole of ω_{P1}

$$\omega_{P1} = \frac{1}{R_{OUT} \cdot C_L} \approx \frac{1}{r_{DS} \cdot C_L} = \frac{I_L}{V_A \cdot C_L}$$

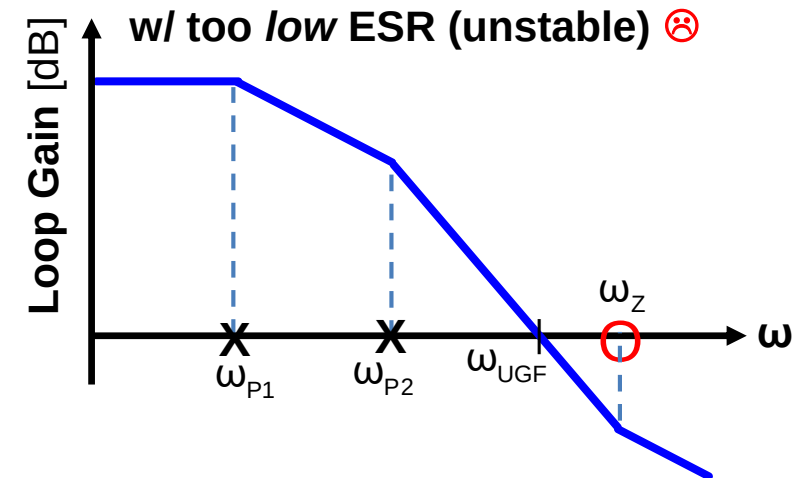
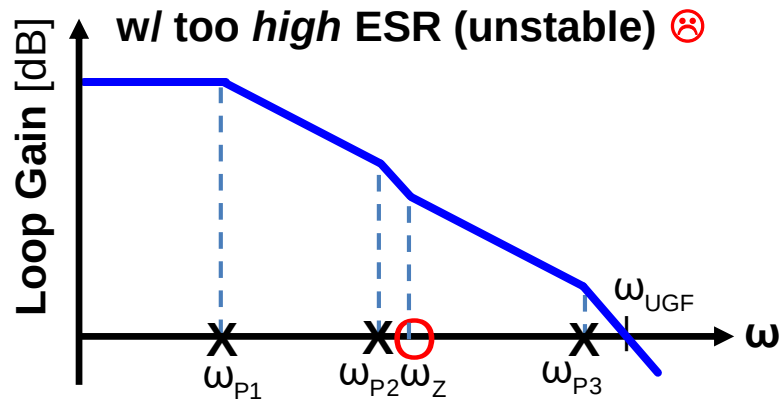
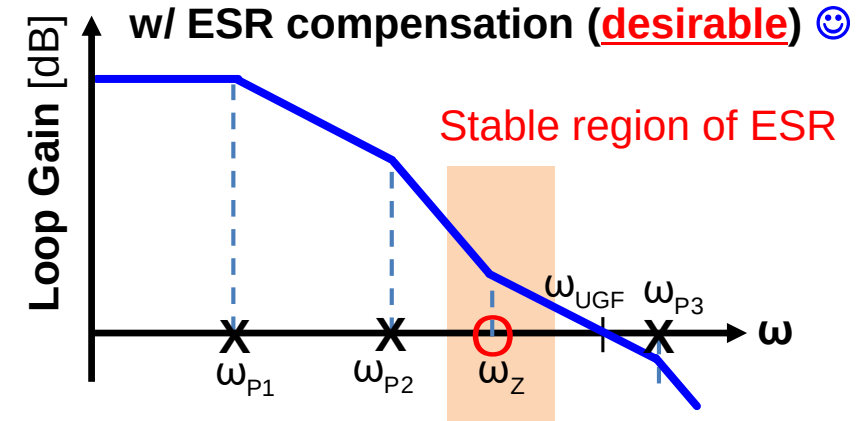
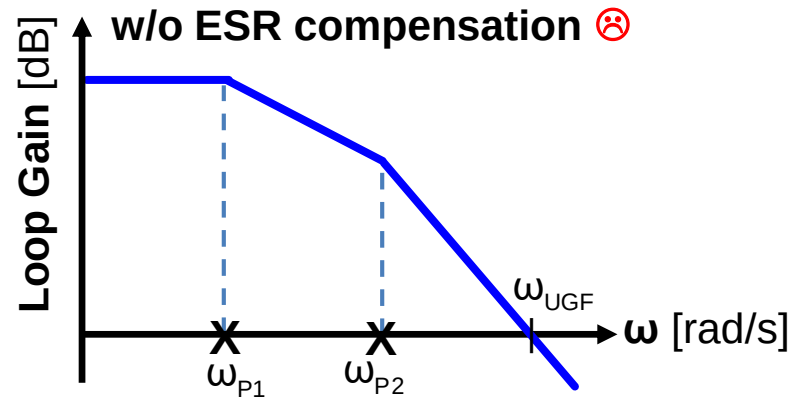
Stability Compensation I: Zero-Insertion (1/3)

- Zero (ω_z) insertion by adding **ESR** (equivalent series resistance) in C_L



Stability Compensation I: Zero-Insertion (2/3)

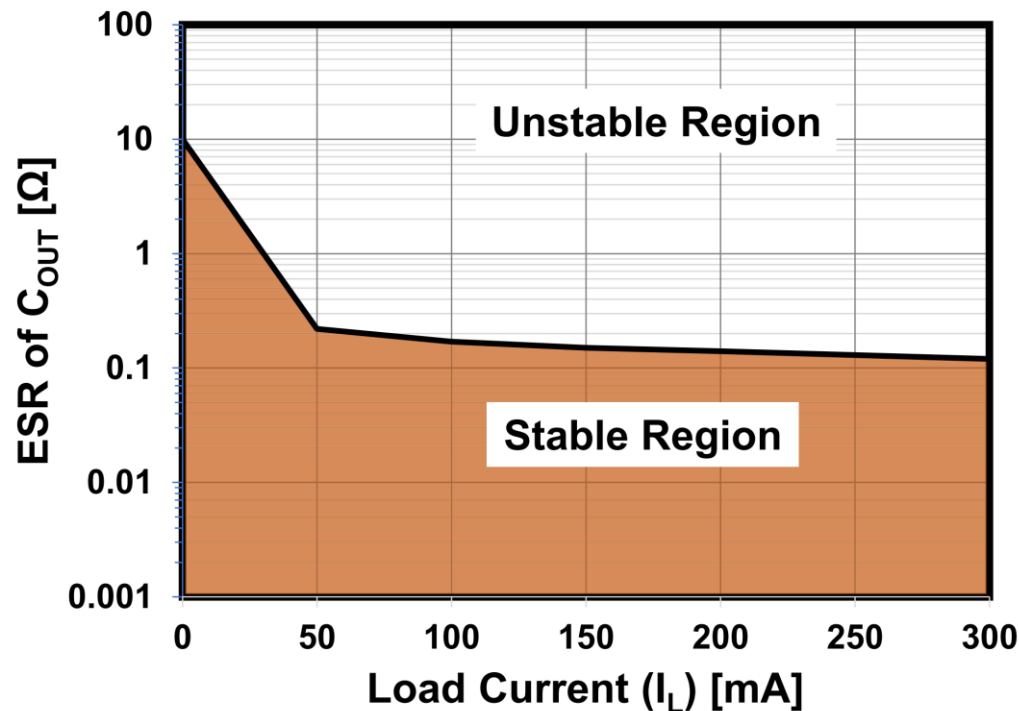
- Choose the right ESR to secure the phase margin



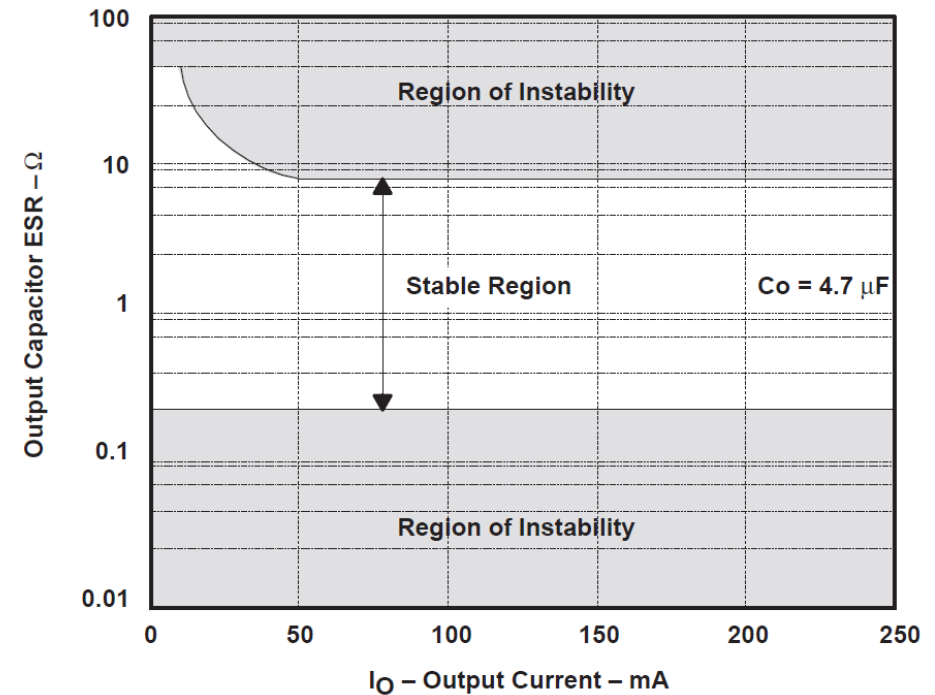
Stability Compensation I: Zero-Insertion (3/3)

□ **Tunnel-of-Death** graph: desirable range of stable ESR

■ ESR on C_L may cause a large spike/dip at load transient response ☹️



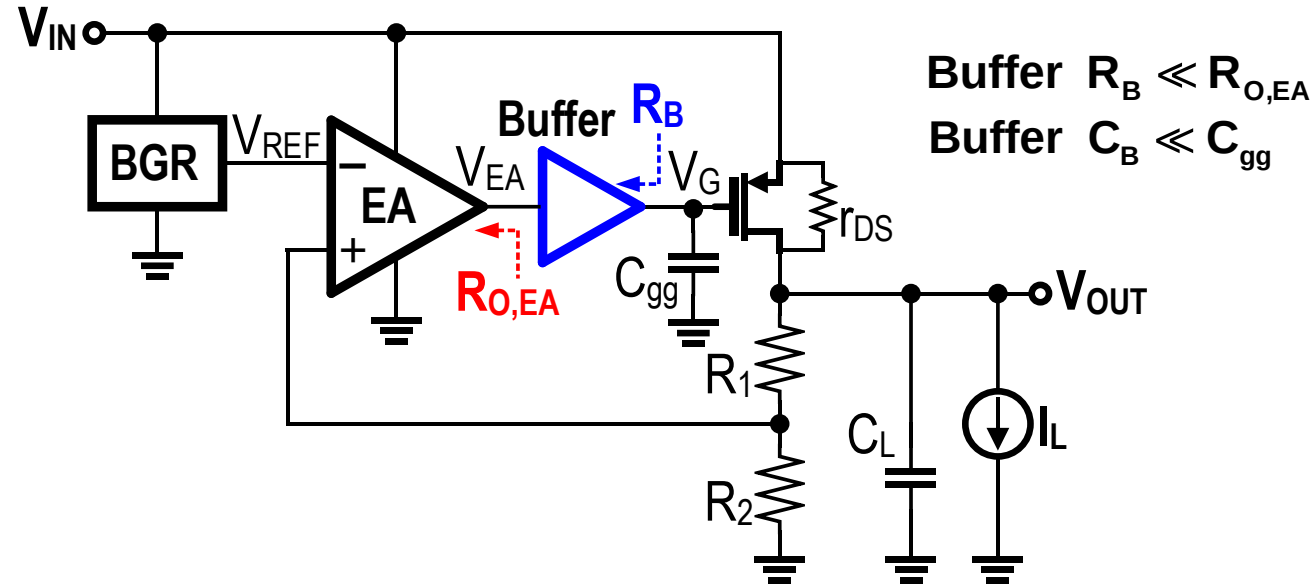
Kim & Kim, *IEEE TPEL* 2020



Texas Instrument, LDO

Stability Compensation II: Gate Buffering

- Shield C_{gg} from loading the error-amplifier output using a **Buffer**

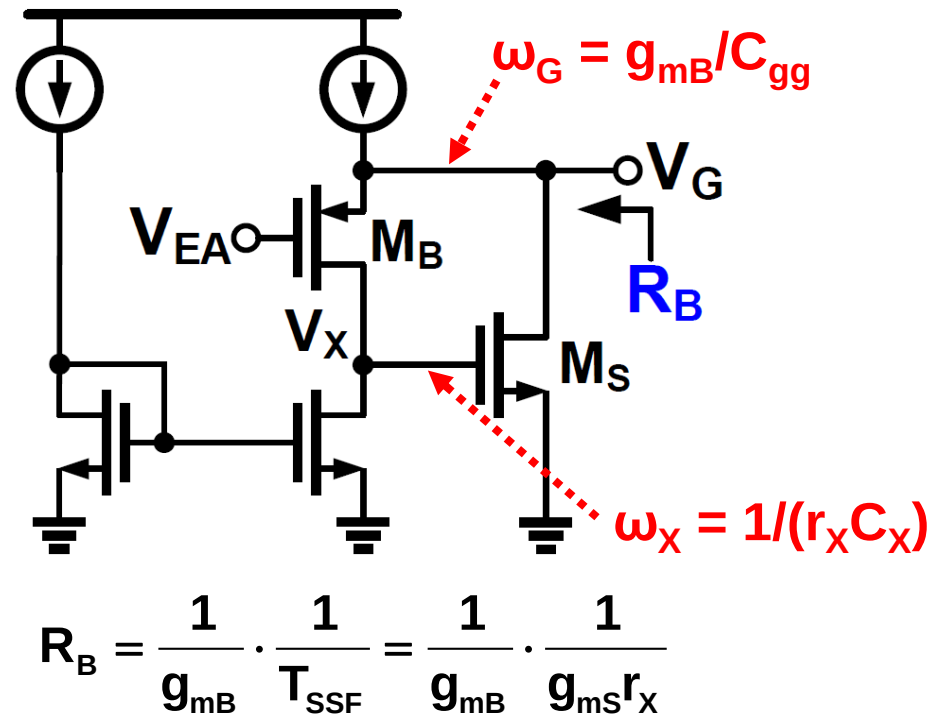


■ w/o gate buffering: $\omega_{P2} = \frac{1}{C_{gg} R_{O,EA}}$ (low-frequency one pole)

■ w/ gate buffering: $\omega_{P2,A} = \frac{1}{C_B R_{O,EA}}$ $\omega_{P2,B} = \frac{1}{C_{gg} R_B}$ (high-frequency two poles)

SSF-based Gate Buffer Design (1/3)

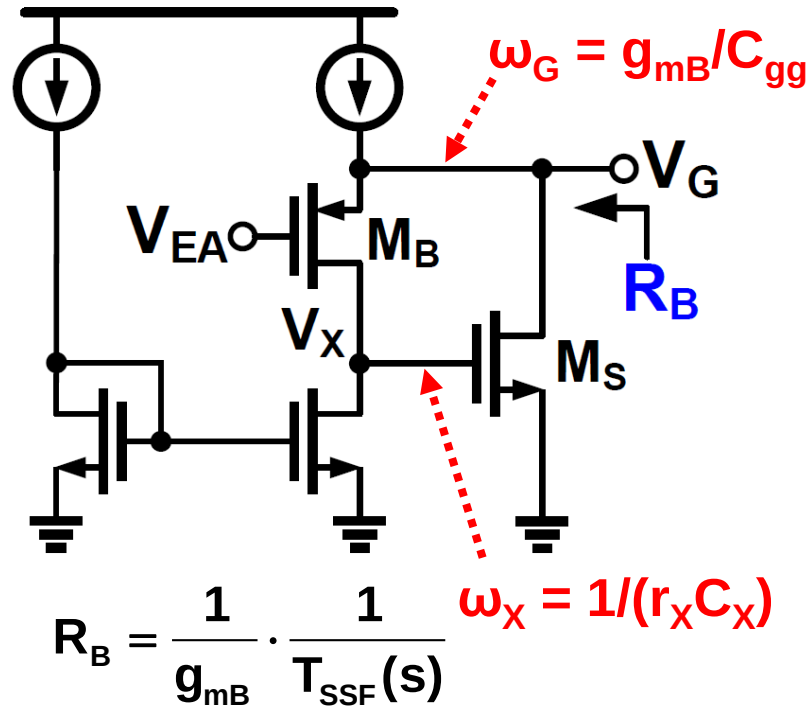
- Gate buffer design with Super Source Follower (SSF)



- Local feedback loop further reduces R_B by loop-gain (T_{SSF})
- SSF loop-gain (T_{SSF}) has closely-spaced two poles: ω_G & ω_X
 - Local stability should be also considered in the design of SSF-based gate buffer ☹

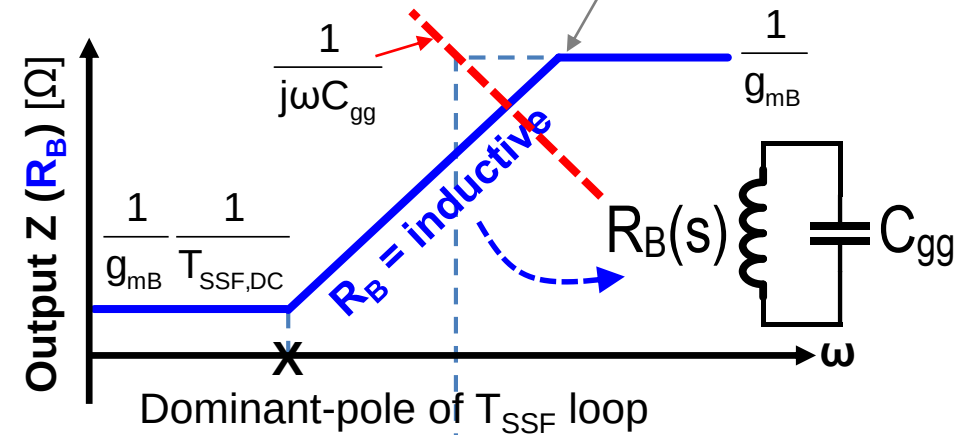
SSF-based Gate Buffer Design (2/3)

□ AC-peaking at SSF output

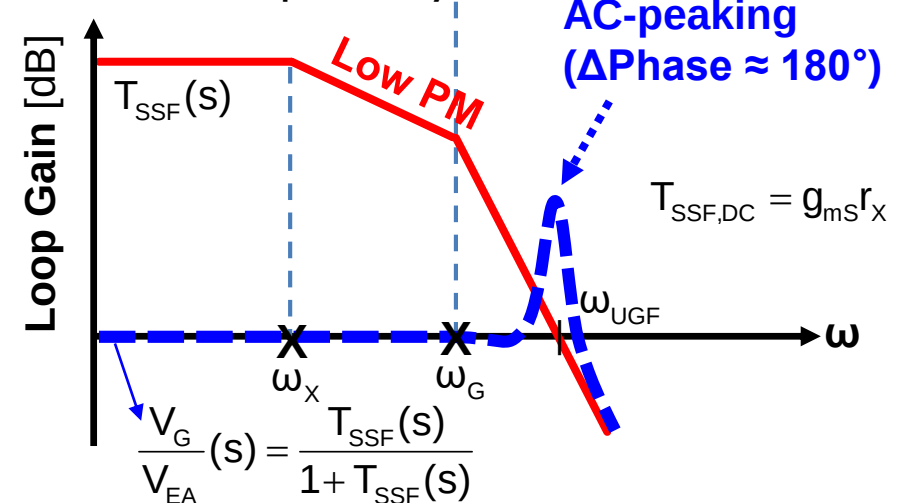


AC-peaking →
Complex pole-pair effect at $V_G/V_{EA}(s)$ ☹️

■ Impedance analysis $|T_{SSF}(j\omega_{UGF})| \leq 1$

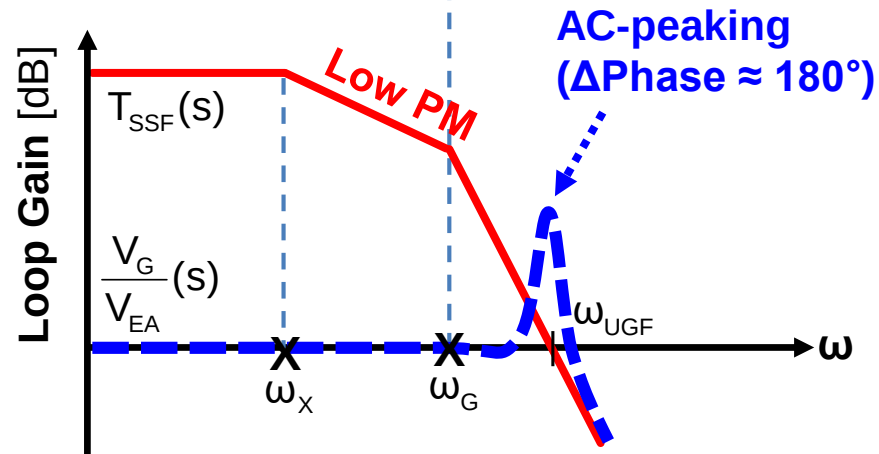
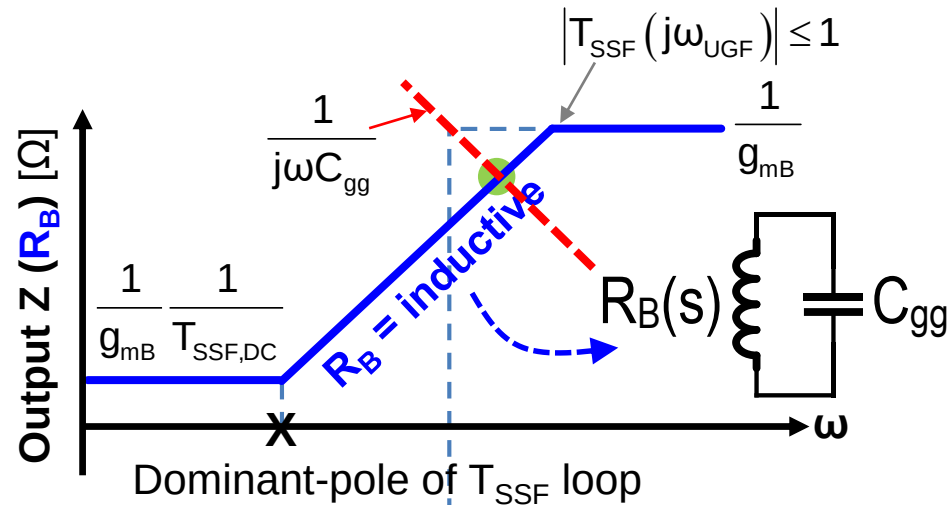


■ Closed-loop analysis

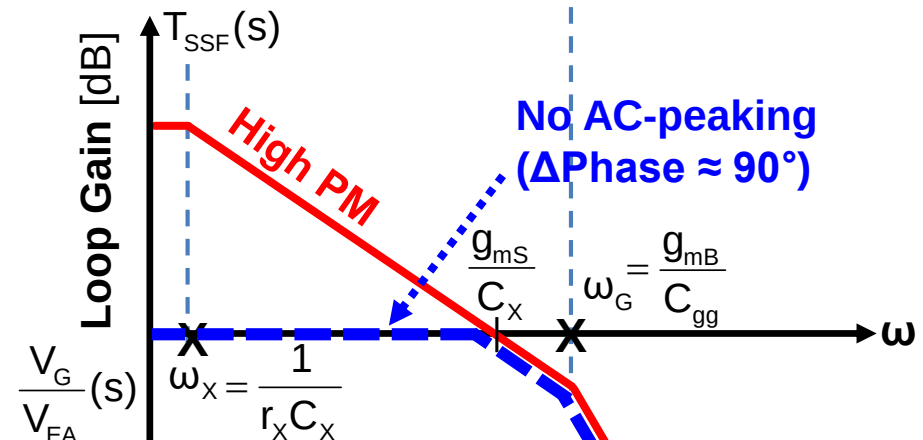
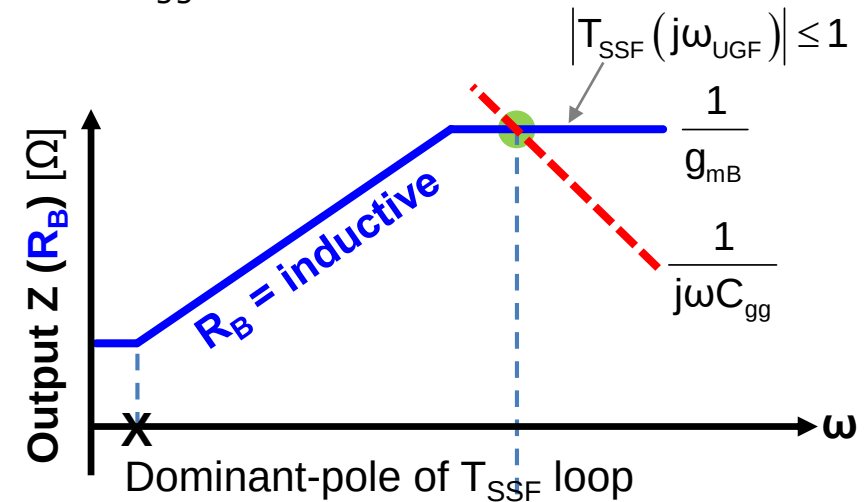


SSF-based Gate Buffer Design (3/3)

□ **AC-peaking** solution: Capacitive impedance ($1/sC_{gg}$) does not cross the *inductive* R_B



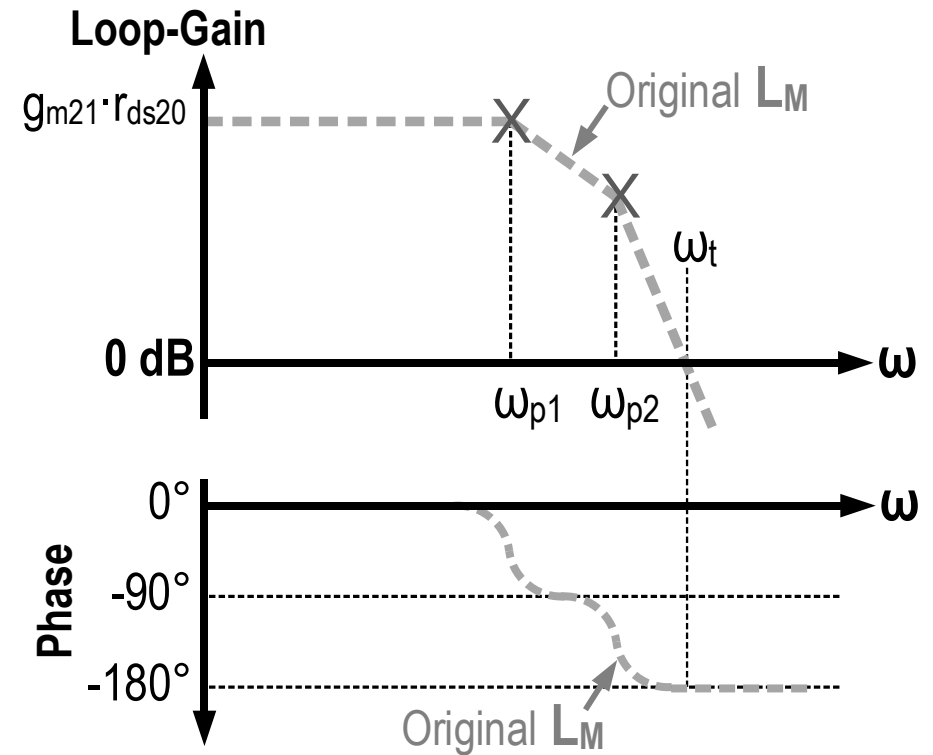
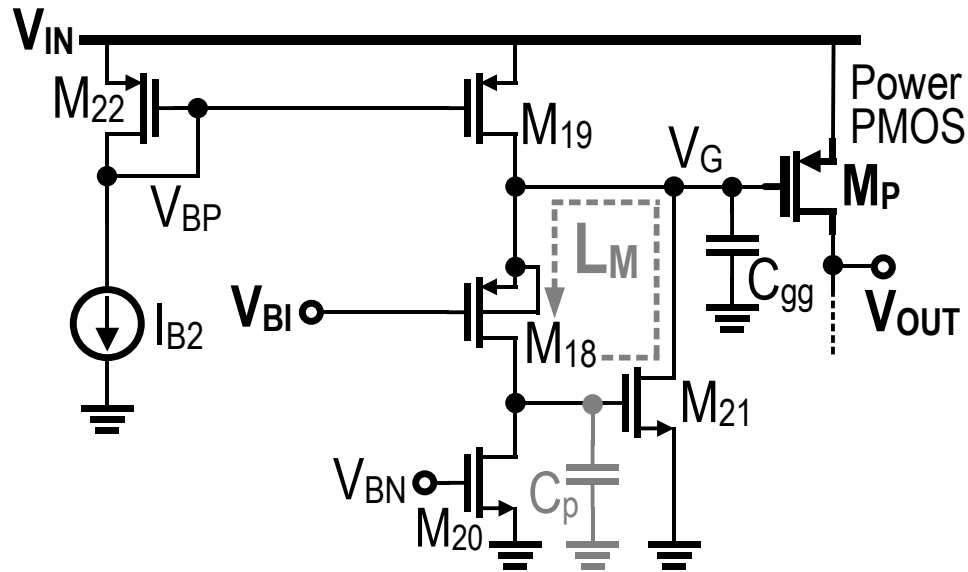
Unstable Case ☹️



Stable Case 😊

Dual-Loop Gate Buffer Design (1/4)

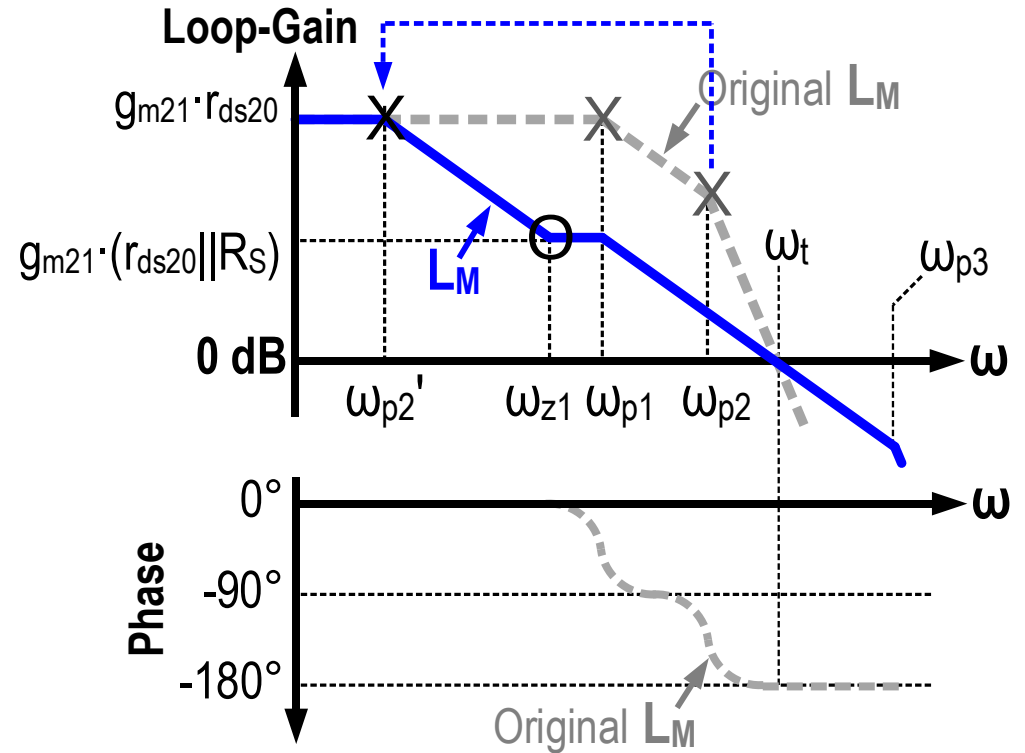
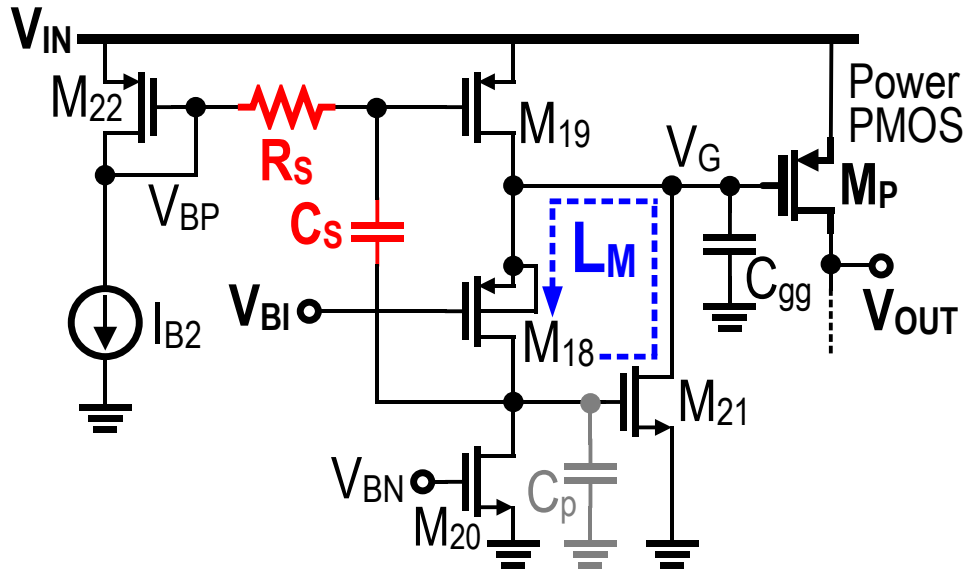
□ Kim & Kim, *IEEE TPEL* 2020



- Conventional SSF-based gate buffer
- Single-loop (L_M)
- Two poles are closely spaced $\rightarrow$ instability issue & GBW (ω_t) limit

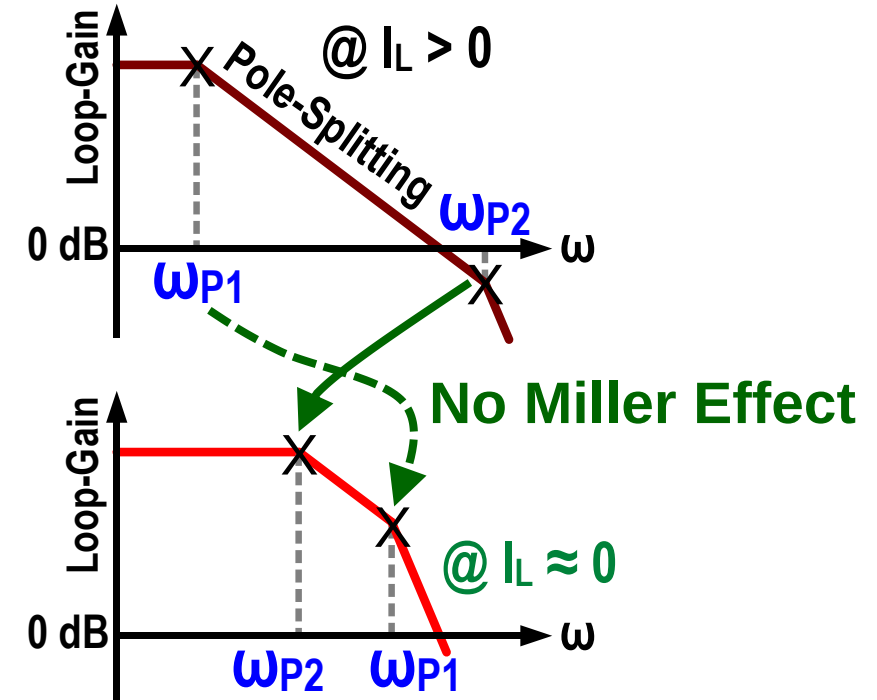
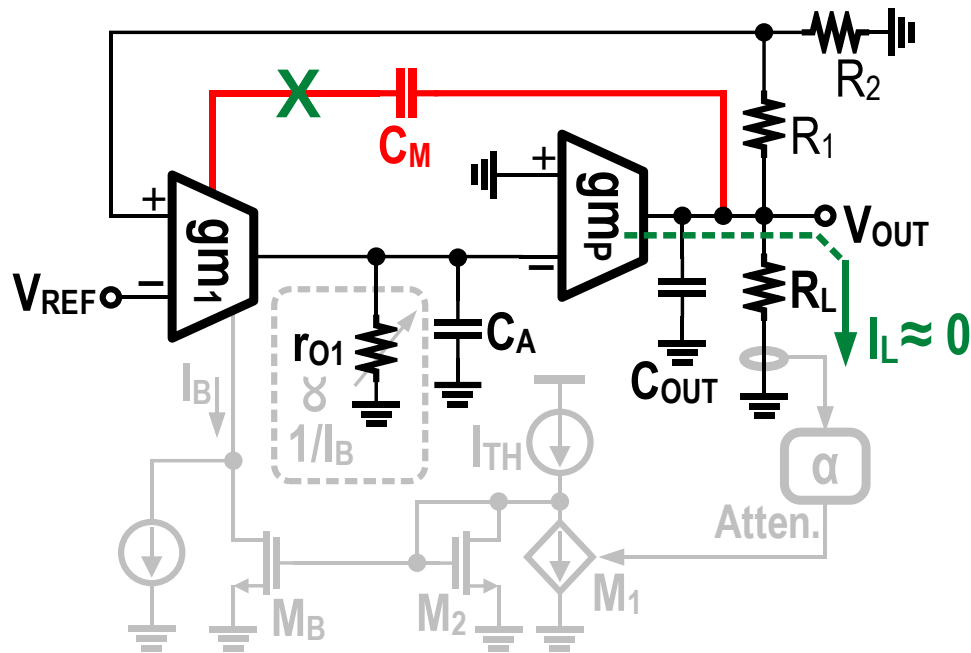
Dual-Loop Gate Buffer Design (2/4)

□ Kim & Kim, *IEEE TPEL* 2020



- Insertion of R_S and C_S
- C_S works for dominant-pole compensation ($\omega_{p2} \rightarrow \omega_{p2}'$)
- R_S introduces a zero (ω_{z1})

Stability Compensation III: Miller Effect (2/2)

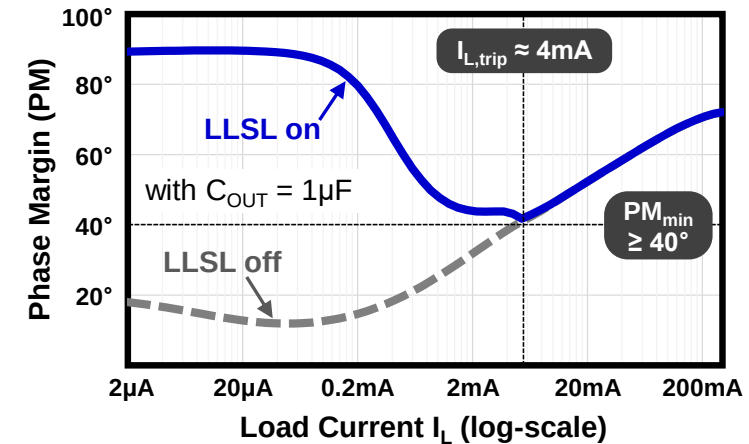
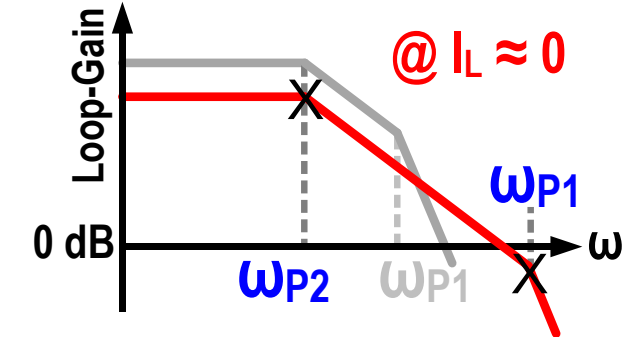
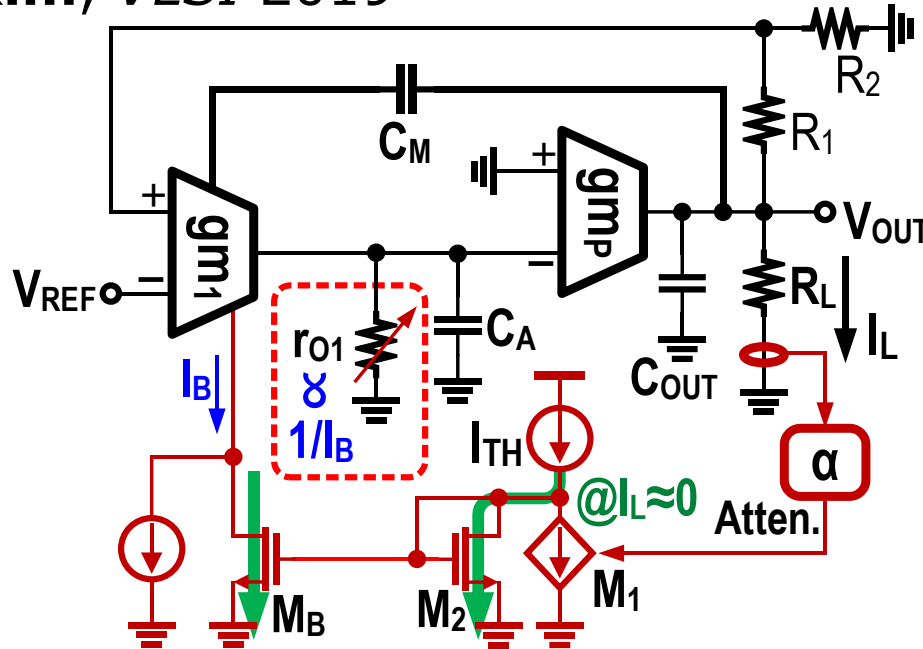


□ No Miller-Effect issue at deep light loads

- At light-load ($I_L \approx 0$) $\rightarrow g_{mp} \approx 0$ (no gain in Pass-Element) $\rightarrow$ **No Miller Effect**
- Miller compensation is difficult to cover $I_L \approx 0$ ☹️
- Otherwise, large static current via R_1 and R_2 must be consumed ☹️

Stability Compensation III: Miller w/ **LLSL** (1/3)

□ Kim & Kim, VLSI 2019

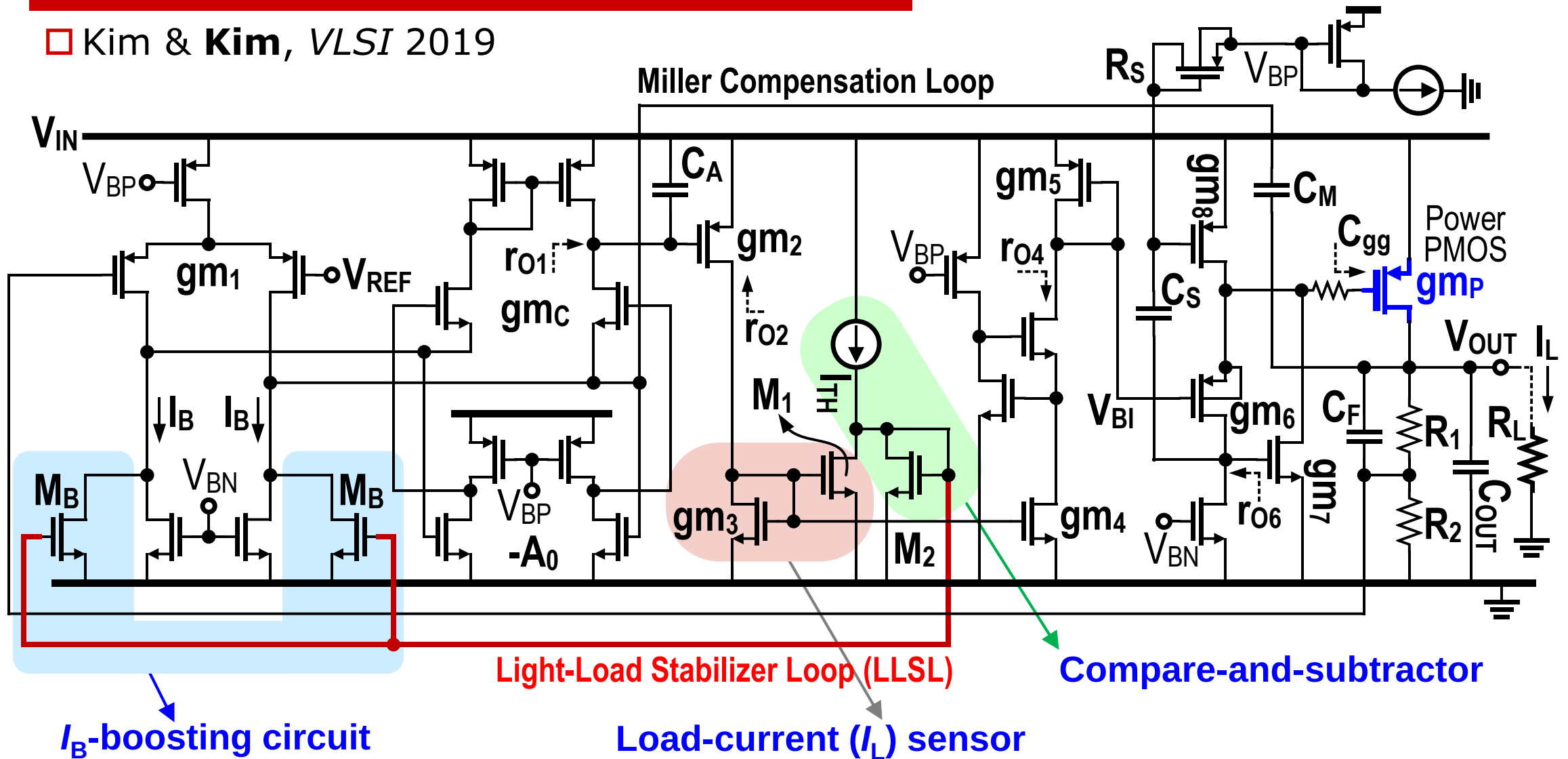


□ Light-load stabilizer loop (**LLSL**) w/ Miller

- If $I_L < \text{pre-defined } I_{TH}$, LLSL is activated and then $I_B (= I_{TH} - I_L)$ is added to g_{m1} to reduce r_{o1}
- Proposed LLSL can guarantee sufficient phase margin even at $I_L \approx 0$ 😊

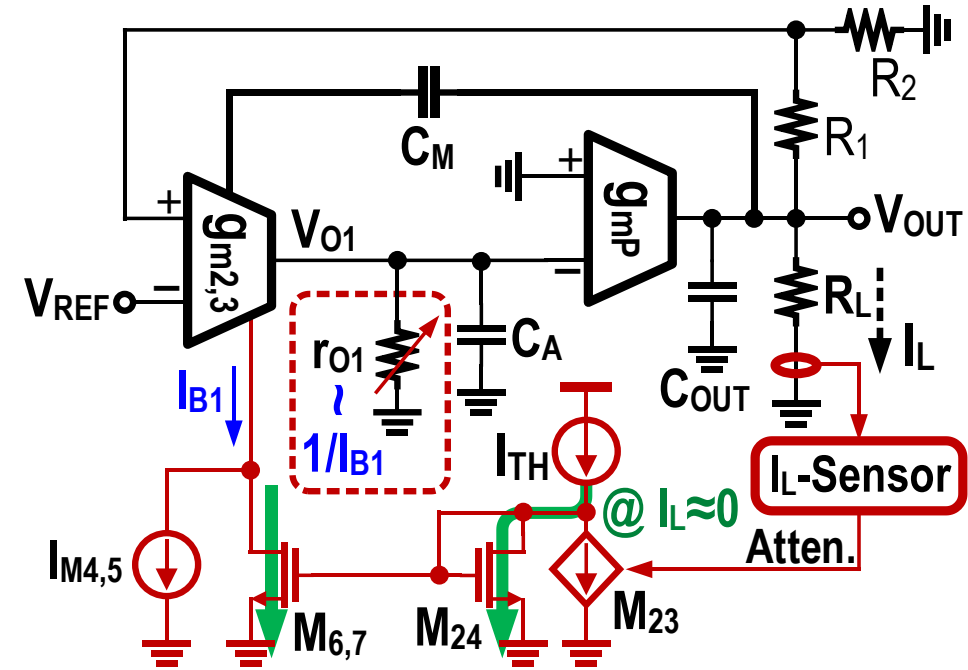
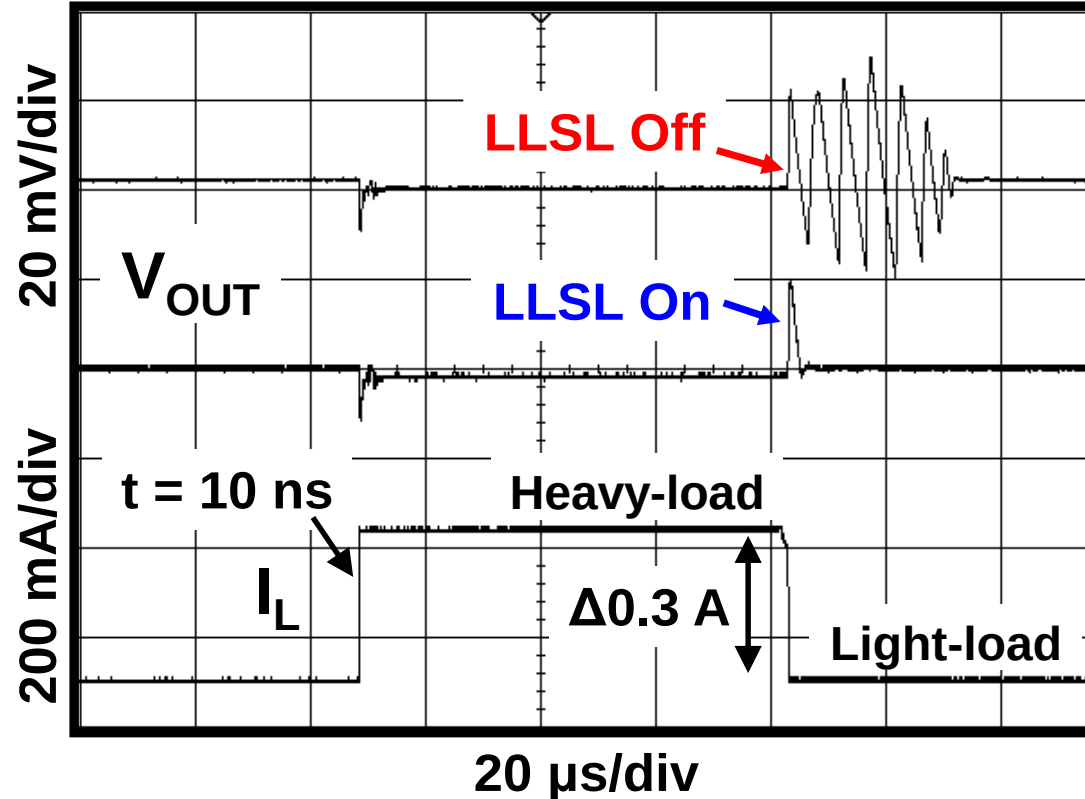
Stability Compensation III: Miller w/ **LLSL** (2/3)

□ Kim & Kim, VLSI 2019



Stability Compensation III: Miller w/ **LLSL** (3/3)

□ Kim & **Kim**, *VLSI* 2019



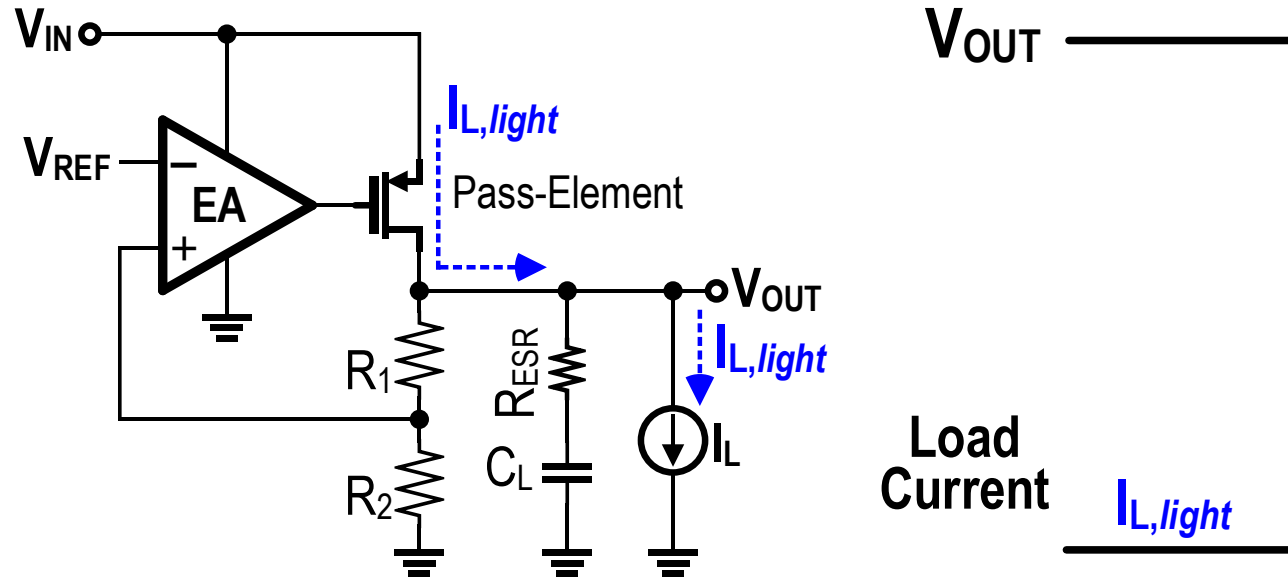
■ At load transition from 300mA to 0A, a ringing is well suppressed by **LLSL** 😊

Contents

- Basics of LDO Regulator
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- **Fast Transient-Response LDO Designs**
- Power Supply Rejection (PSR)
- Energy-Efficient Ultra-Low-Dropout (Triode) LDO
- Conclusion

Understanding Load-Transient Response (1/6)

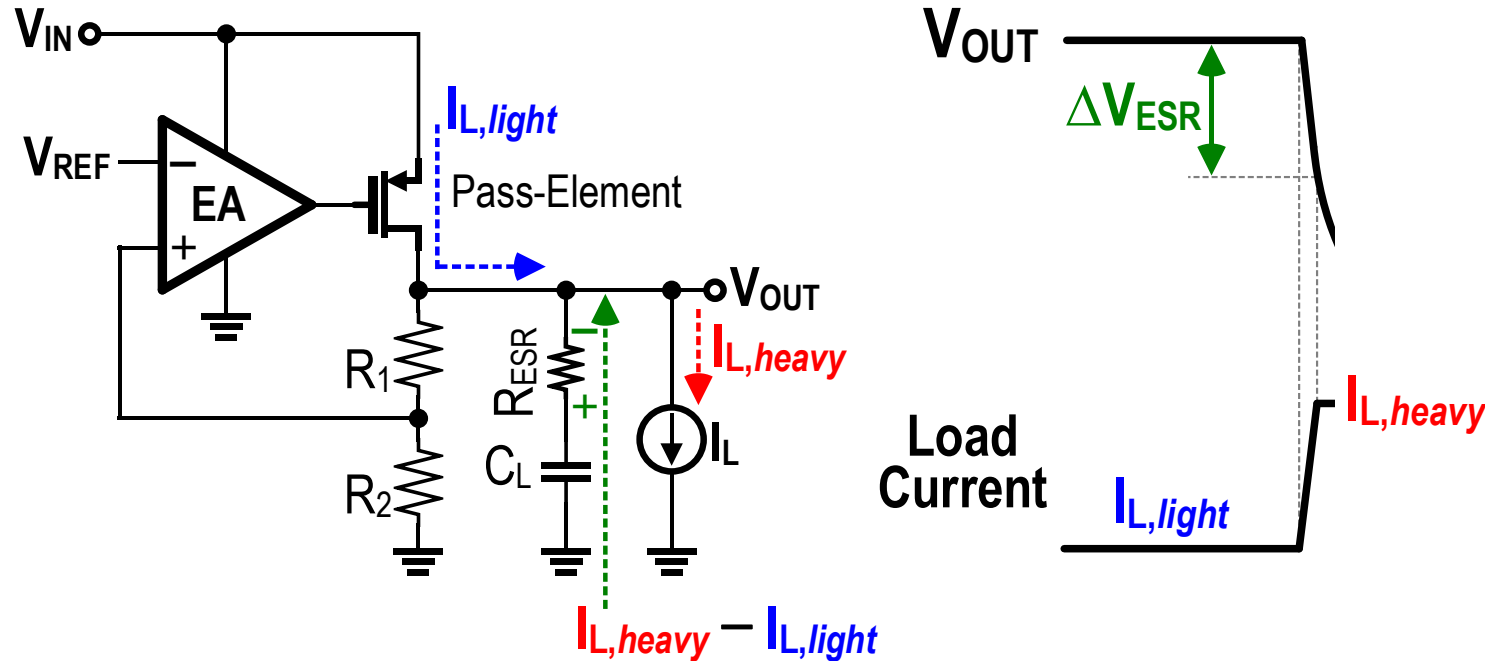
□ Light-to-heavy load transition



- At the steady-state, V_{OUT} is stabilized and no current via C_L
- PMOS's supply current = Load current = $I_{L,light}$

Understanding Load-Transient Response (2/6)

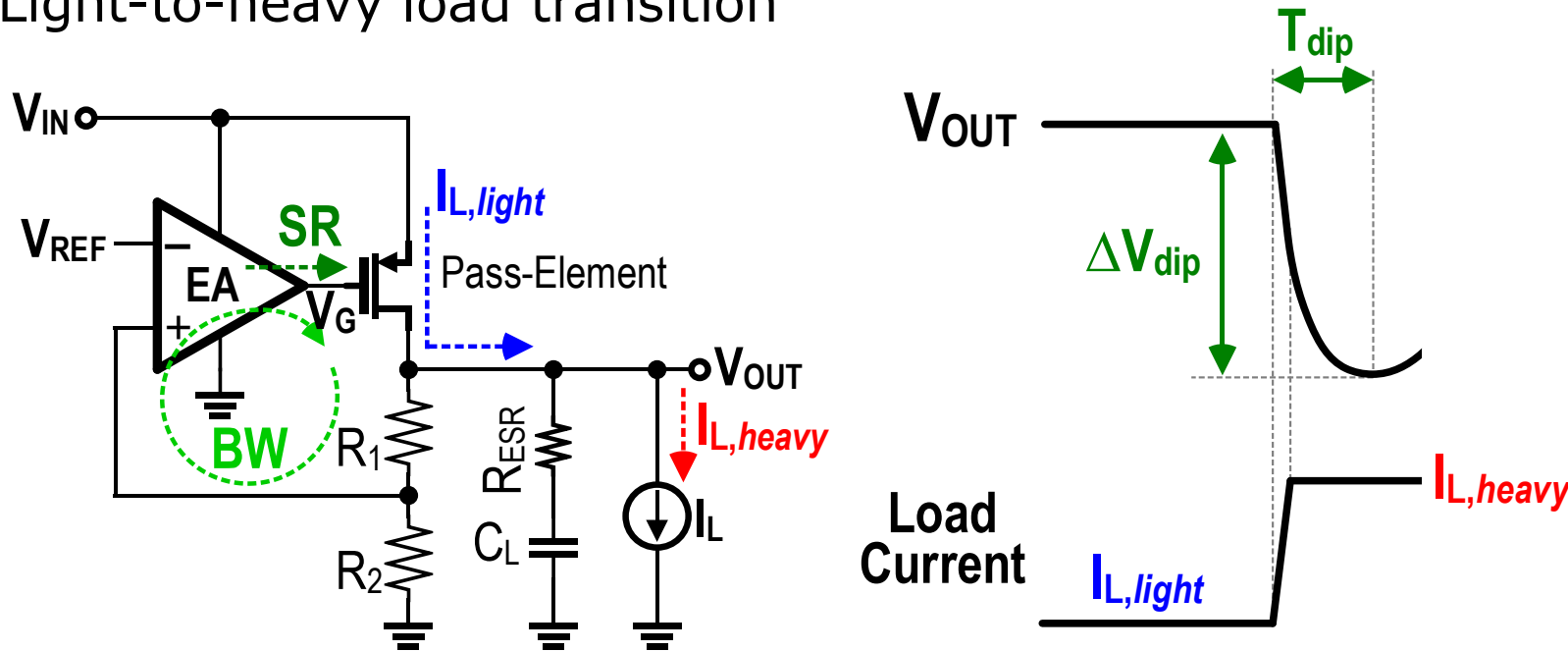
□ Light-to-heavy load transition



- A load transition from light to heavy current occurs
- Initially C_L reacts to cover the current mismatch between PMOS and load
- Large voltage droop (ΔV_{ESR}) appears at V_{OUT} : $\Delta V_{ESR} \approx R_{ESR} (i_{L,heavy} - i_{L,light})$

Understanding Load-Transient Response (3/6)

□ Light-to-heavy load transition



■ V_{OUT} falls until PMOS current reaches $I_{L,heavy}$

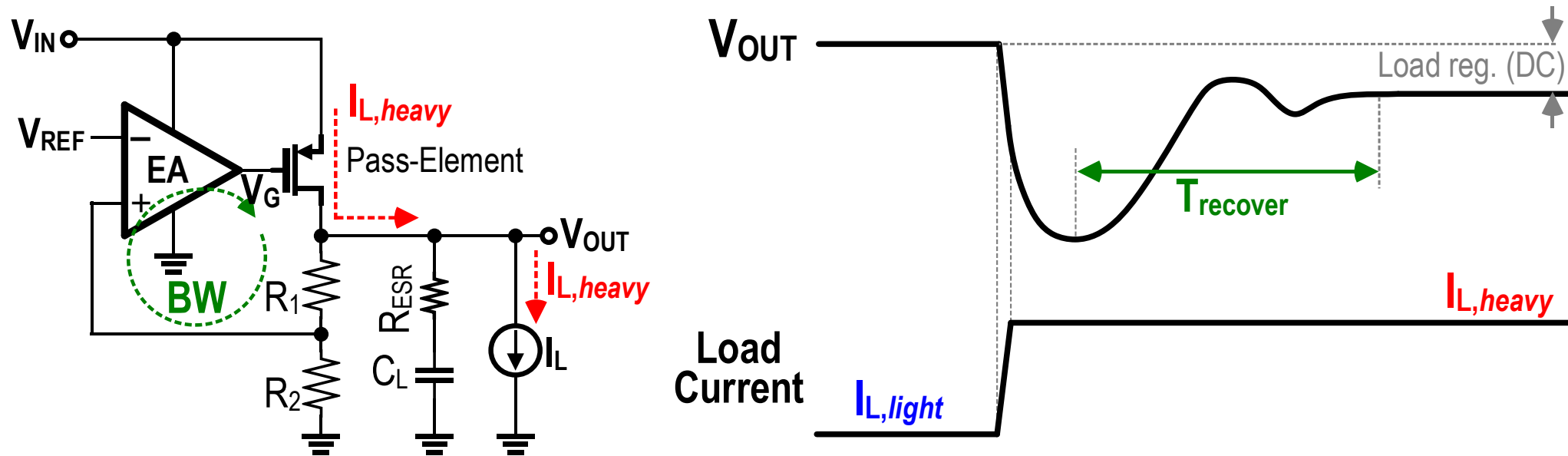
■ Voltage-droop period: $T_{dip} \approx T_{SR} + \frac{1}{BW_{@light-load}} = C_{gg} \frac{\Delta V_G}{I_{SR,EA}} + \frac{1}{BW_{@light-load}}$, where T_{SR} = slew time

■ **Droop voltage:** $\Delta V_{dip} = \frac{I_{L,heavy} - I_{L,light}}{C_L} \cdot T_{dip} + \Delta V_{ESR}$

*Note: LDO's loop-BW is dependent of I_L

Understanding Load-Transient Response (4/6)

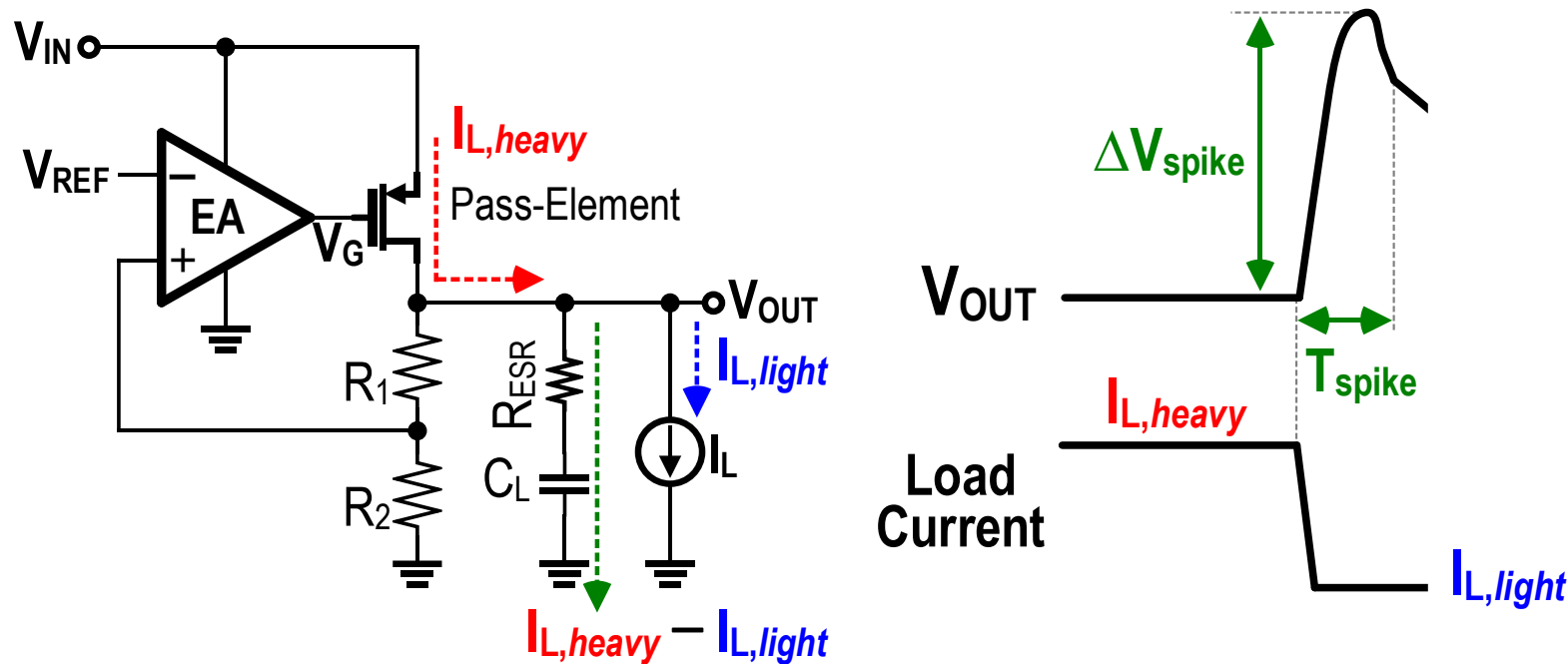
□ Light-to-heavy load transition



- Fine-settling time ($T_{recover}$) is dominated by LDO's loop-**BW** and phase margin (**PM**)
- V_{OUT} 's DC error after fine-settling is caused by limited load-regulation performance (insufficient loop-gain in DC-domain)

Understanding Load-Transient Response (5/6)

Heavy-to-light load transition

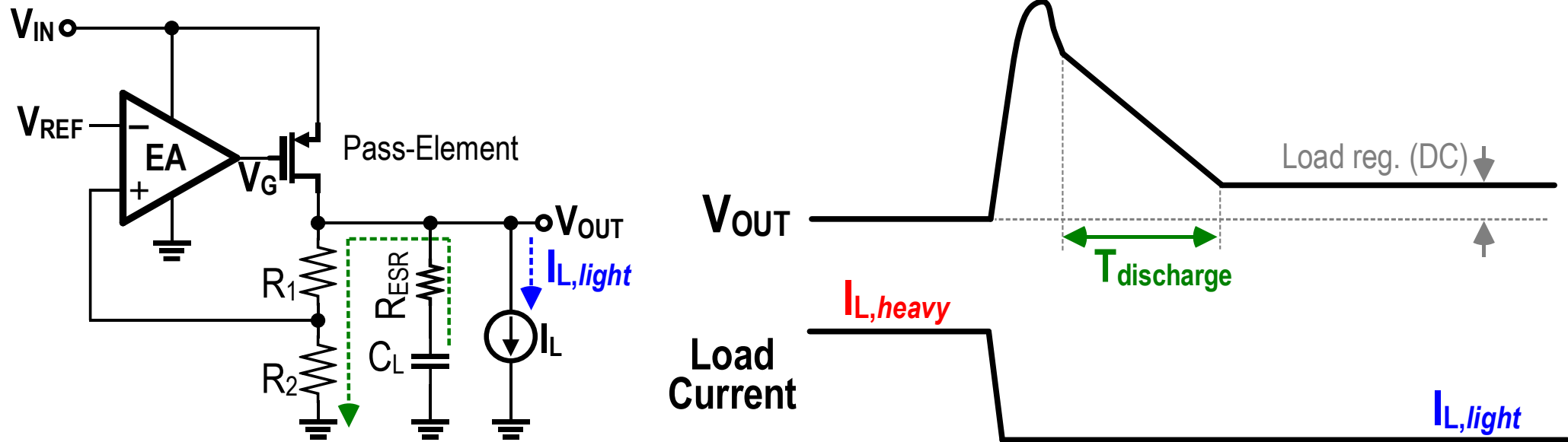


■ Overshoot period: $T_{spike} \approx T_{SR} + \frac{1}{BW_{@heavy-load}} = C_{gg} \frac{\Delta V_G}{I_{SR,EA}} + \frac{1}{BW_{@heavy-load}}$

■ **Overshoot voltage:** $\Delta V_{spike} = \frac{i_{L,heavy} - i_{L,light}}{C_L} \cdot T_{spike} + \Delta V_{ESR}$ Dominated by slew-rate (SR) and BW...

Understanding Load-Transient Response (6/6)

Heavy-to-light load transition



■ Typical LDO has no sinking capability (only pull-down path is R_1 and R_2)

■ Return-to-base period: $T_{discharge} \approx C_L \frac{\Delta V_{spike}}{i_{R2}} = C_L \frac{\Delta V_{spike}}{\frac{V_{REF}}{R_2}}$

Figure-of-Merit (FoM)

□ Load-transient response

$$\Delta V_{\text{dip}} = \frac{i_{L,\text{heavy}} - i_{L,\text{light}}}{C_L} \cdot \Delta T_{\text{dip}} + \Delta V_{\text{ESR}}$$
$$T_{\text{dip}} \approx T_{\text{SR}} + \frac{1}{\text{BW}_{@ \text{light-load}}} = C_{\text{gg}} \frac{\Delta V_{\text{G}}}{I_{\text{SR,EA}}} + \frac{1}{\text{BW}_{@ \text{light-load}}}$$

□ LDO's **FoM** in terms of load-transient response

$$\text{FoM} = \frac{C_L \cdot \Delta V_{\text{dip}} \cdot I_Q}{\Delta I_{L,\text{max}}^2}$$
$$\approx \frac{T_{\text{dip}} \cdot I_Q}{\Delta I_{L,\text{max}}}$$

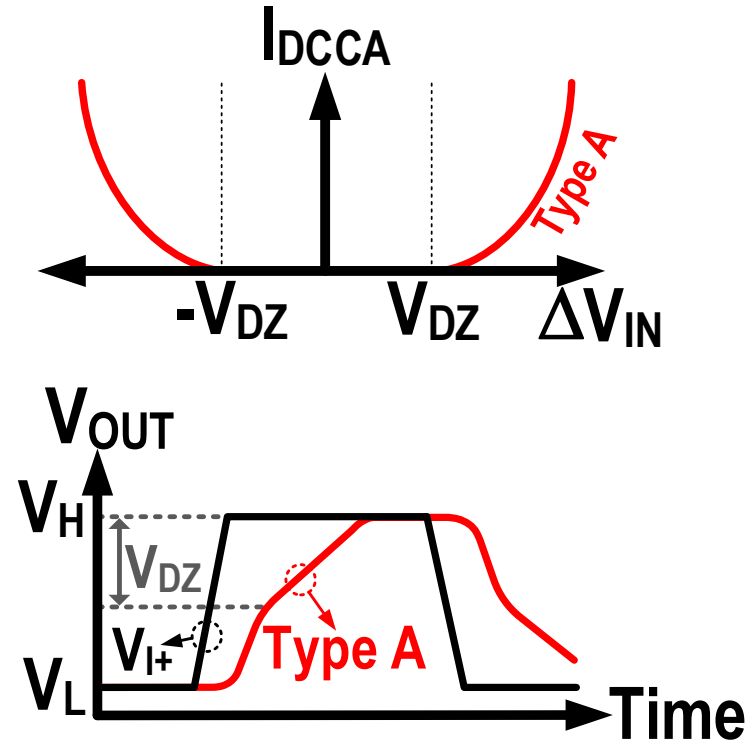
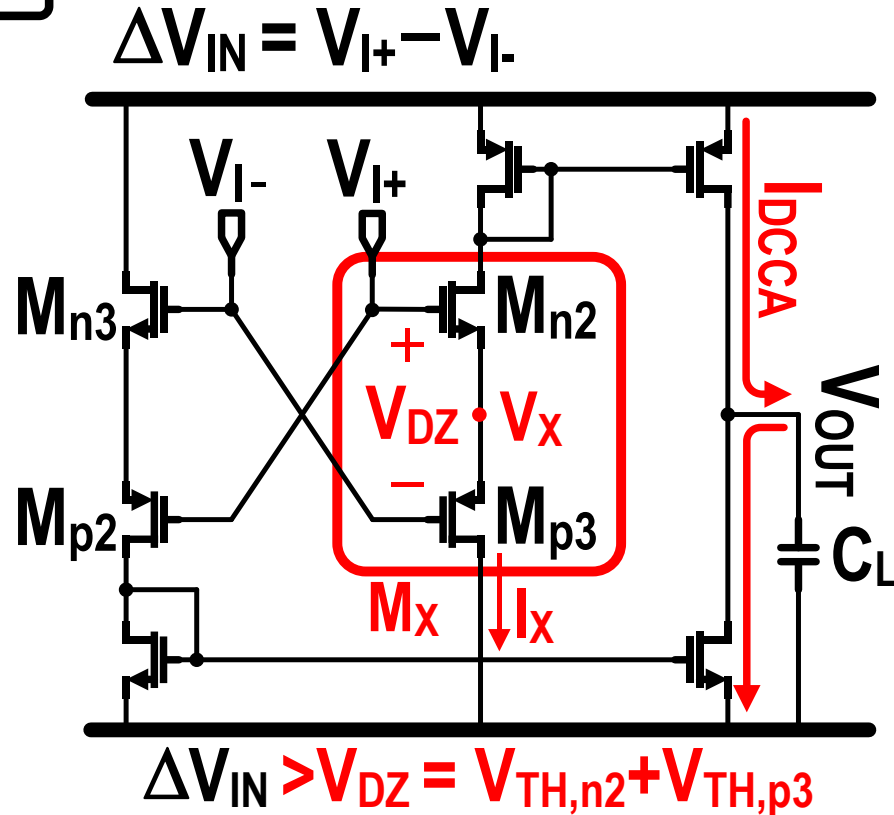
1) Pass element's C_{gg} needs to be higher to accommodate larger $I_{L,\text{max}}$

2) FoM shows the efficiencies of **SR** and **BW** against I_Q consumption...

Zero- I_Q Ultra-High Slew-Rate Buffer (1/5)

□ Koh & Kim, ISSCC'21 & JSSC'21

A

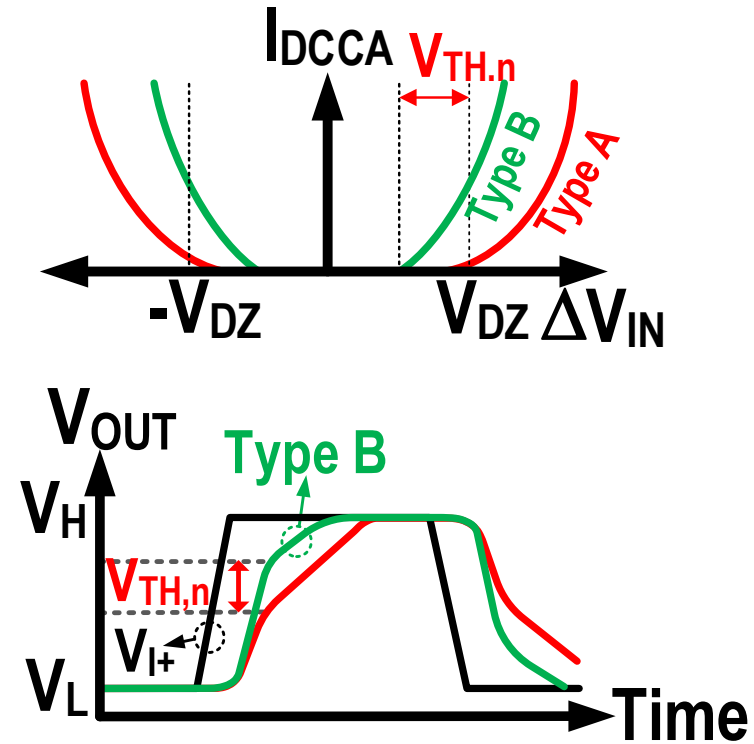
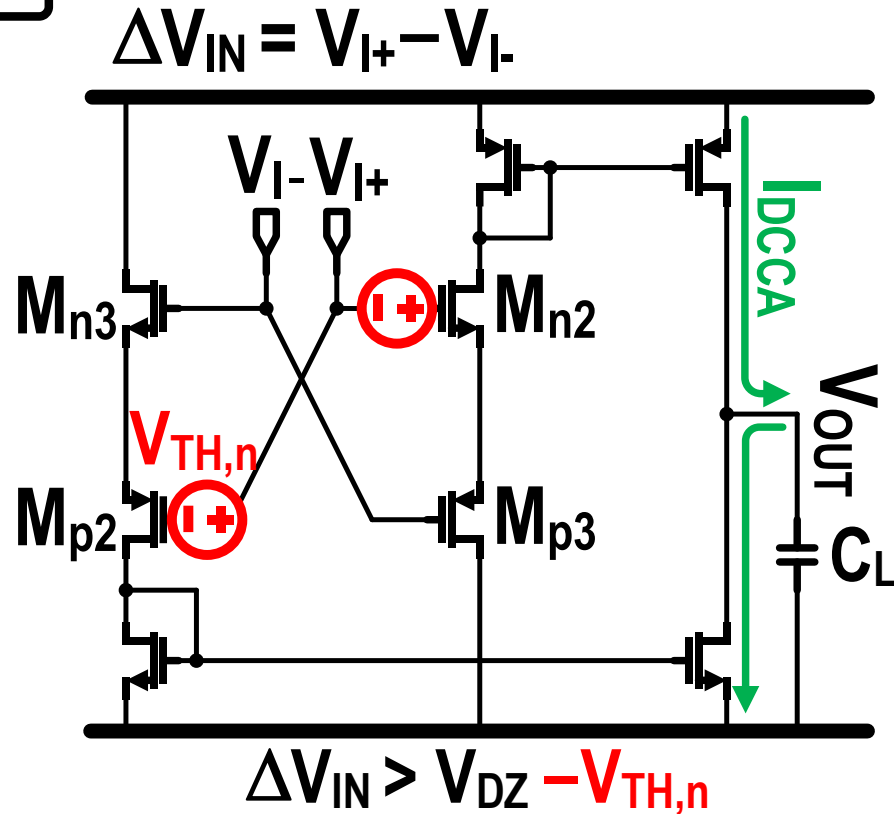


- Zero- I_Q at steady-state at $|\Delta V_{IN}| < V_{DZ}$ & Large dynamic current $I_{DCCA} \propto \Delta V_{IN}^2$ 😊
- Dead-zone (V_{DZ}) is too large ☹️

Zero- I_Q Ultra-High Slew-Rate Buffer (2/5)

□ Koh & Kim, ISSCC'21 & JSSC'21

B

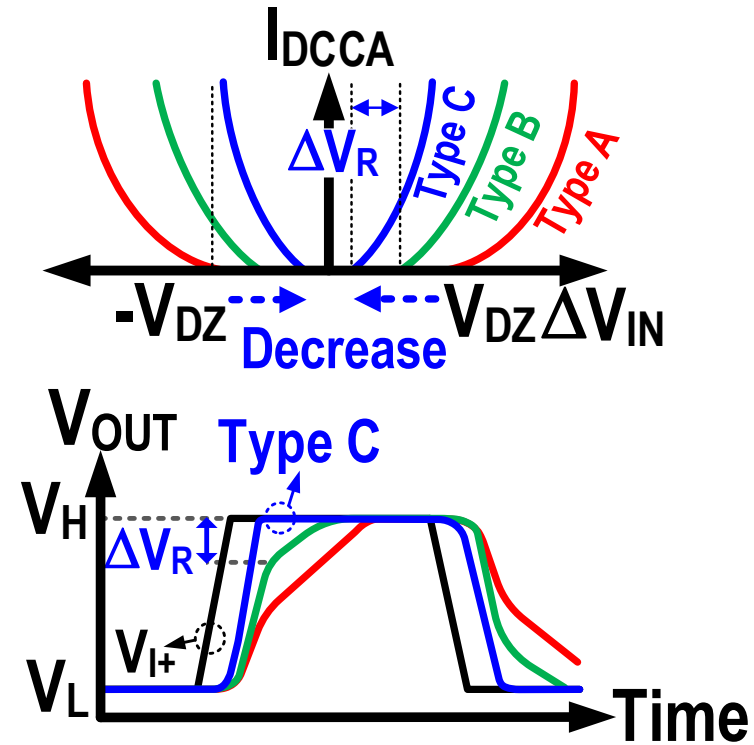
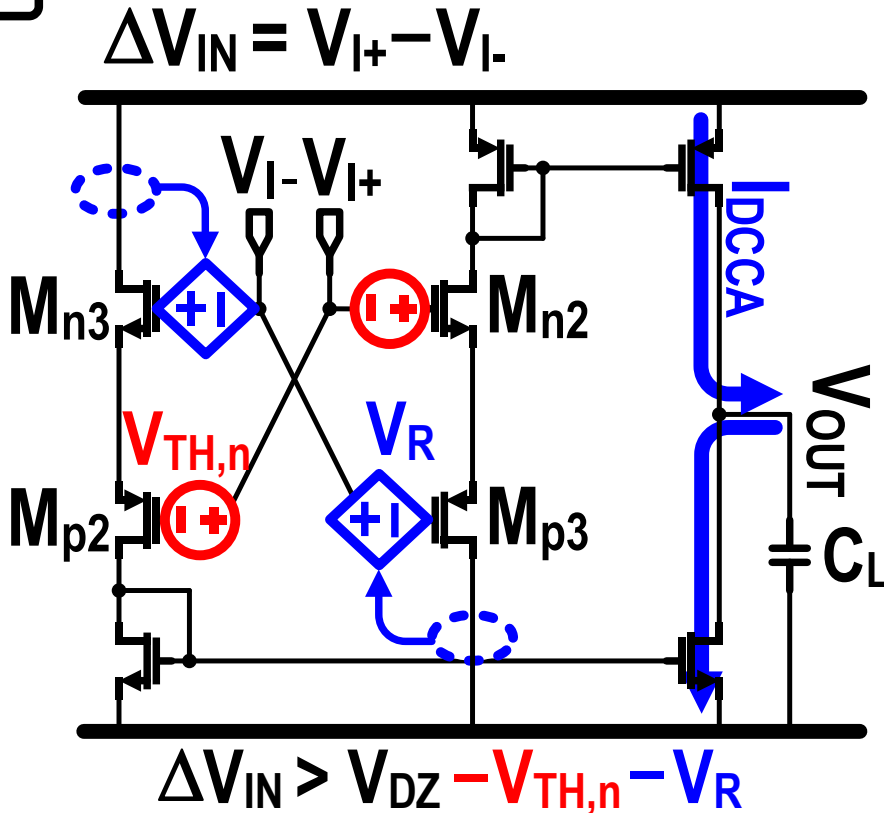


- V_{TH} of transistors can be cancelled by inserting the series voltage sources 😊
- Dead-zone (V_{DZ}) still remains ☹️

Zero- I_Q Ultra-High Slew-Rate Buffer (3/5)

□ Koh & Kim, ISSCC'21 & JSSC'21

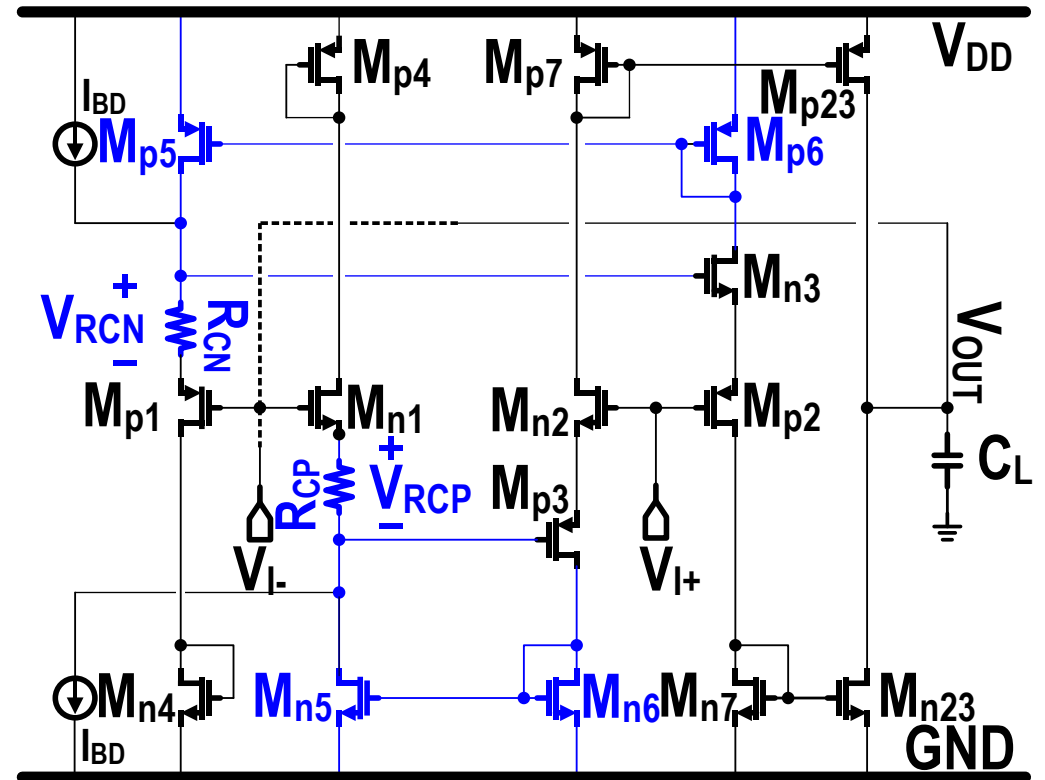
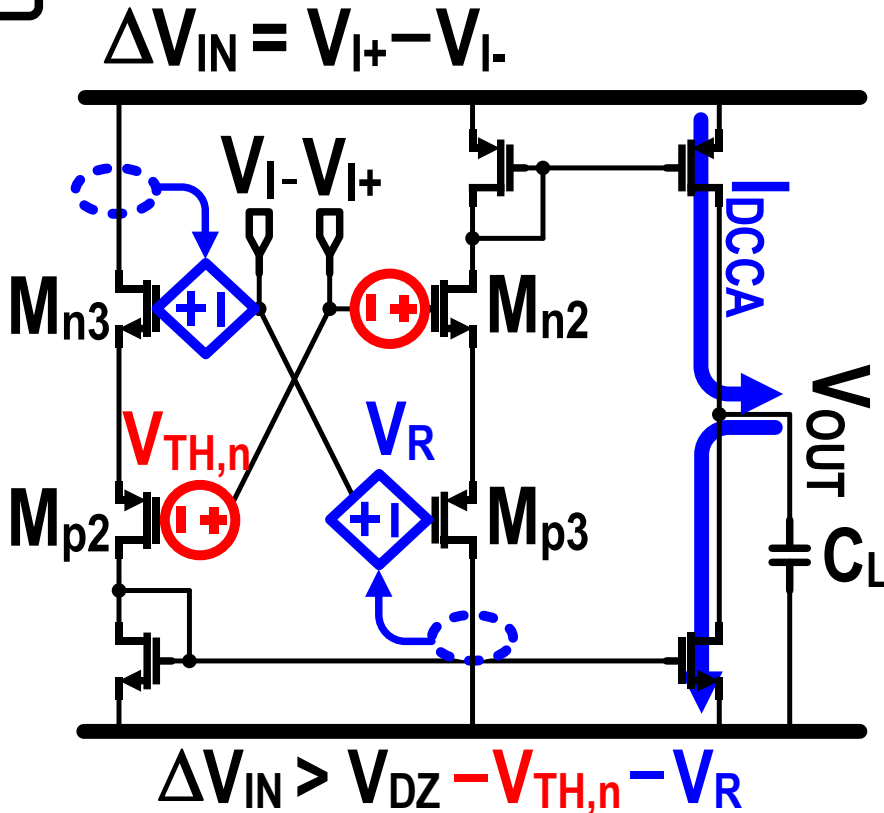
C



- Recursive-current-controlled voltage sources are introduced
- Near-zero dead-zone ($V_{DZ} \approx 0$) ☺ & Ultra-high slewing current (I_{DCCA}) ☺

Zero- I_Q Ultra-High Slew-Rate Buffer (4/5)

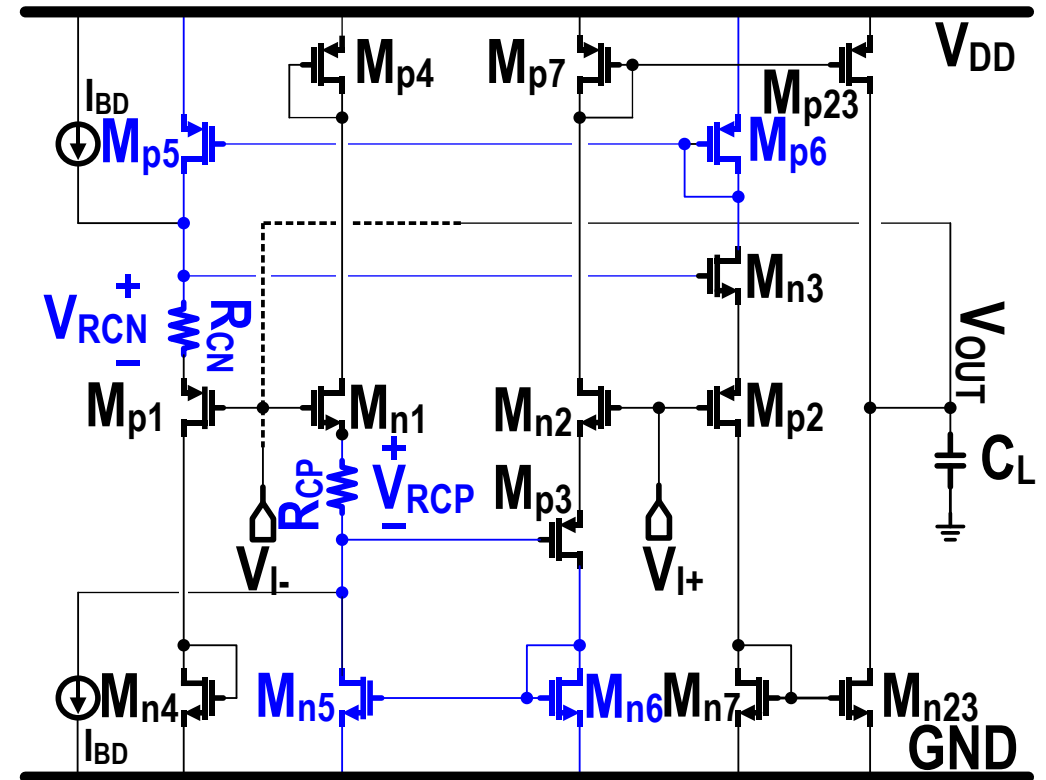
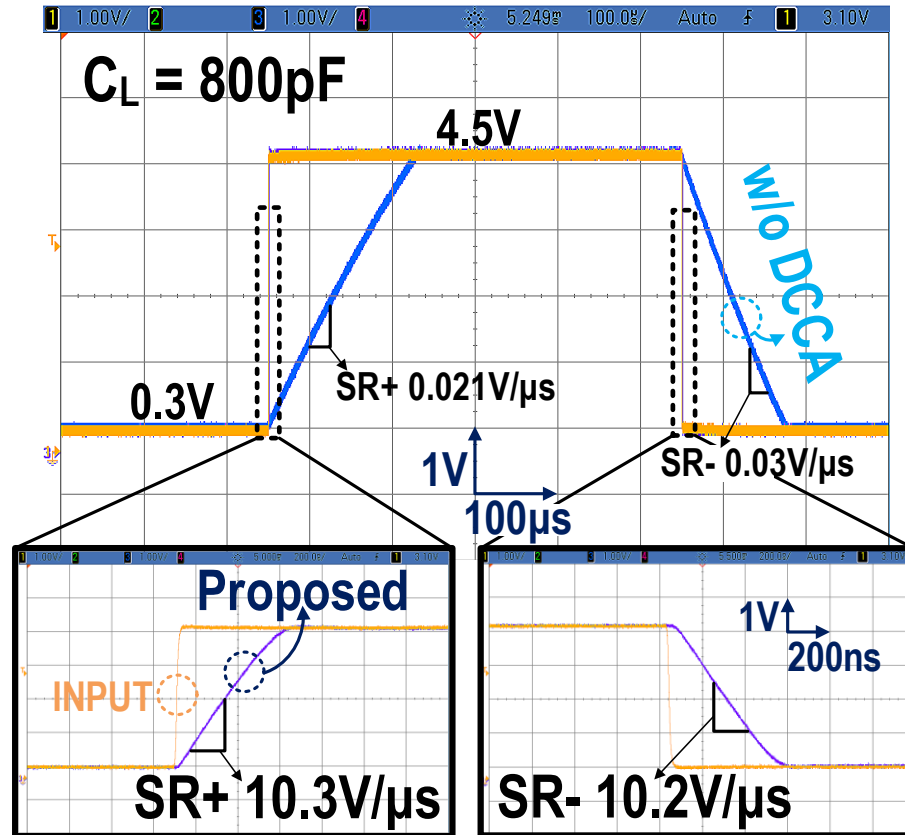
□ Koh & **Kim**, *ISSCC'21* & *JSSC'21*



- M_{n5} - M_{n6} & M_{p5} - M_{p6} forms recursive (positive) feedback loops
- Near-zero dead-zone ($V_{DZ} \approx 0$) 😊 & Ultra-high slewing current (I_{DCCA}) 😊

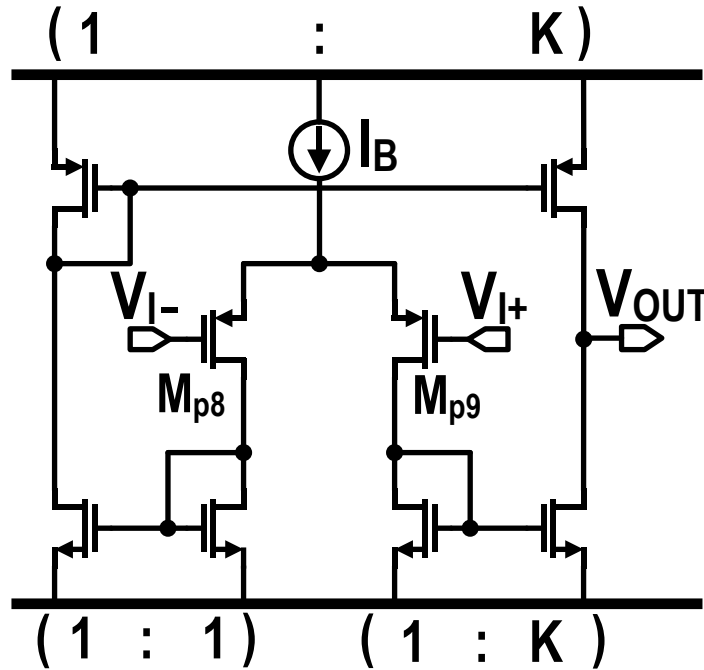
Zero- I_Q Ultra-High Slew-Rate Buffer (5/5)

□ Koh & Kim, *ISSCC'21* & *JSSC'21*



- SR+ (rise) 10.3V/ μs at 800pF, SR- (fall) 10.2V/ μs at 800pF
- I_Q consumption = 30nA & Max. slewing current = 9mA (= 300,000 $\times I_Q$) ☺

Low- I_Q G_m -Boosted OTA Design (1/4)

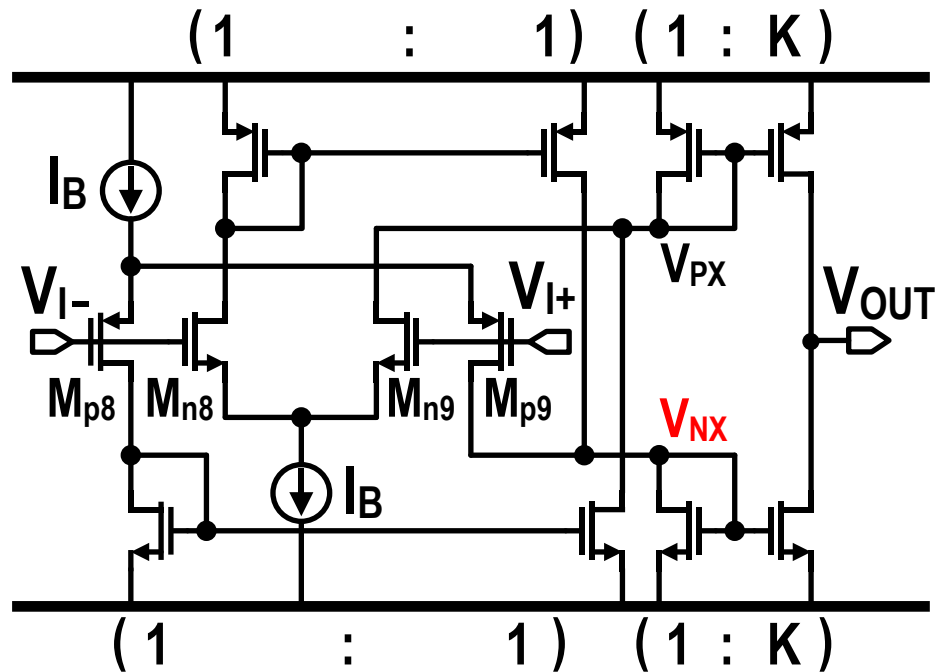


I_{DC}	$[(3+K)/2] \cdot I_B$
G_m	$K \cdot g_{mp}$

□ Typical mirror-type OTA design (A)

- Effective G_m can only be amplified by output-stage mirror-ratio (K)
- To increase G_m , larger I_{DC} is required ($G_m \propto I_{DC} \rightarrow$ low FoM) ☹️

Low- I_Q G_m -Boosted OTA Design (2/4)



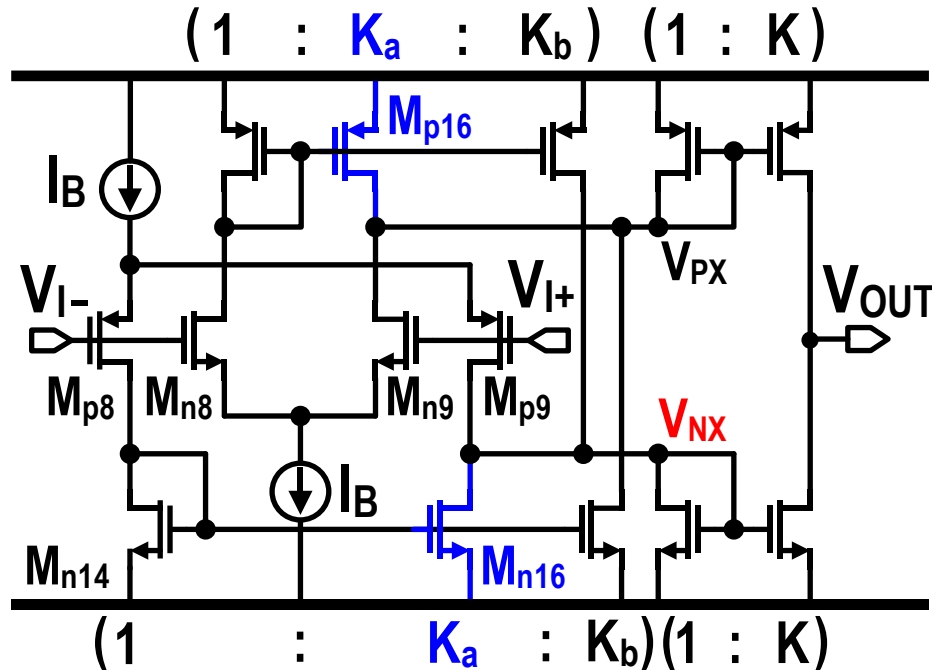
I_{DC}	$(3+K) \cdot I_B$
G_m	$K \cdot (g_{mn} + g_{mp})$

□ Rail-to-rail mirror-type OTA design (B)

- N-type & P-type differential pairs synergies to improve the effective G_m
- But, the quiescent current (I_{DC}) is also increased ☹️

Low- I_Q G_m -Boosted OTA Design (3/4)

□ Koh & Kim, ISSCC'21 & JSSC'21



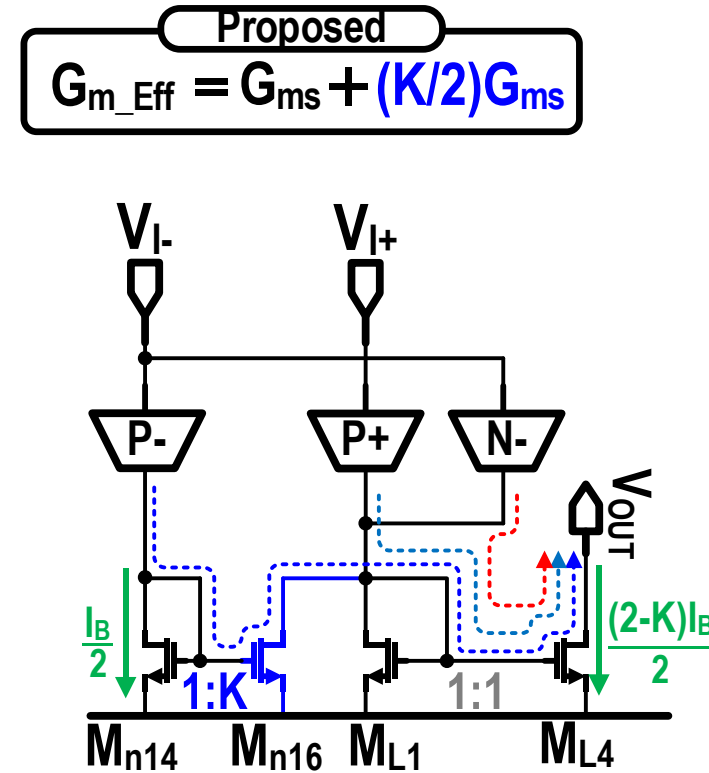
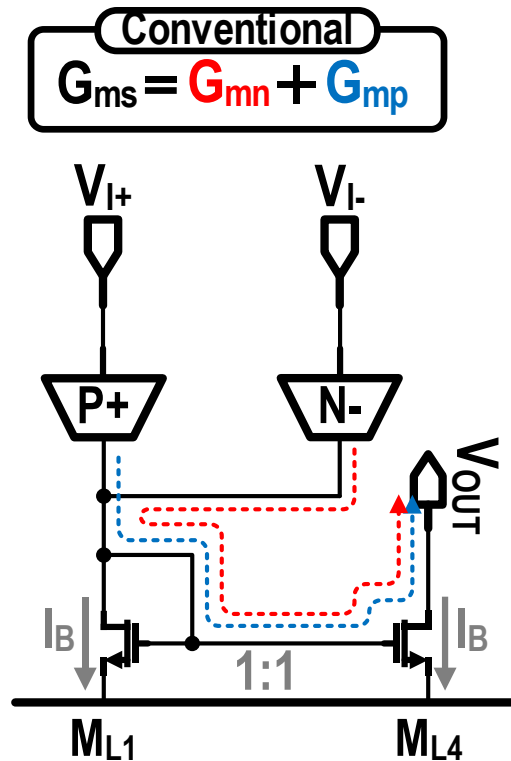
I_{DC}	$[2 + K_b + K \cdot (1 + K_b - K_a) / 2] \cdot I_B$
G_m	$K \cdot (1 + K_a + K_b) \cdot (g_{mn} + g_{mp}) / 2$

□ Proposed low- I_Q **G_m -boosted OTA** design (C)

- Signal can be amplified by K_a , but the DC current is reduced by a ratio of K_a ☺
- If $K_a = K_b = 1 \rightarrow 1.5x$ higher G_m compared to (B) for the same I_{DC}

Low- I_Q G_m -Boosted OTA Design (4/4)

□ Koh & Kim, ISSCC'21 & JSSC'21



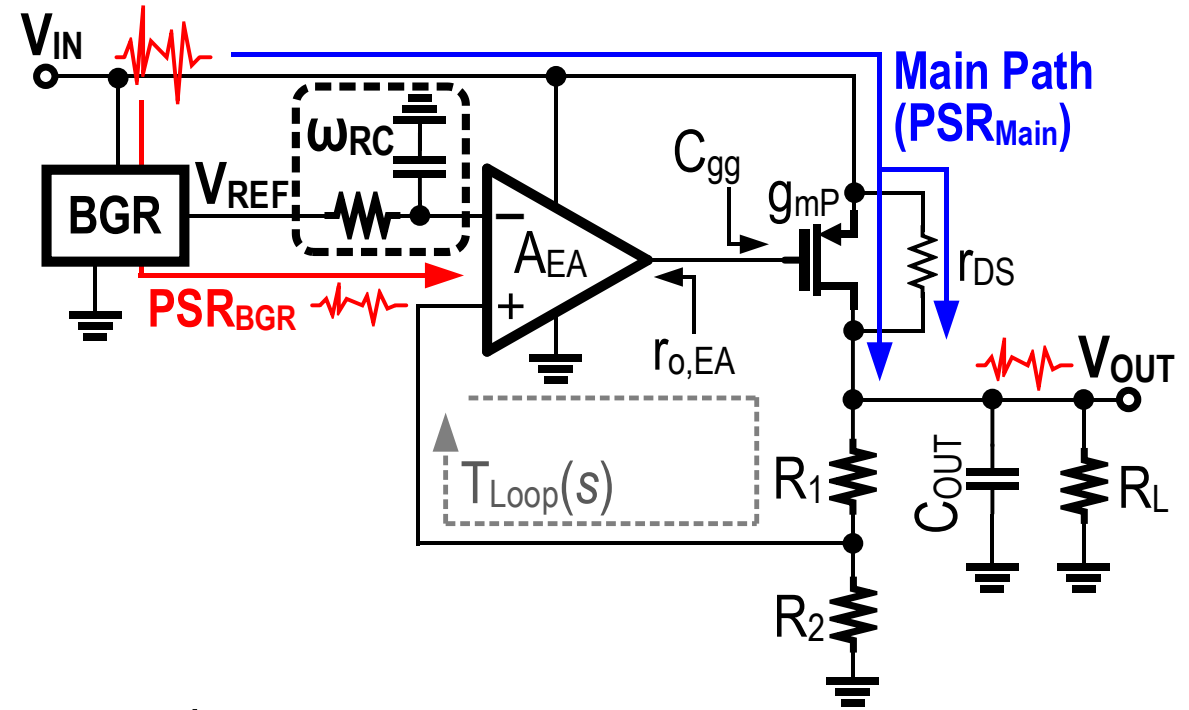
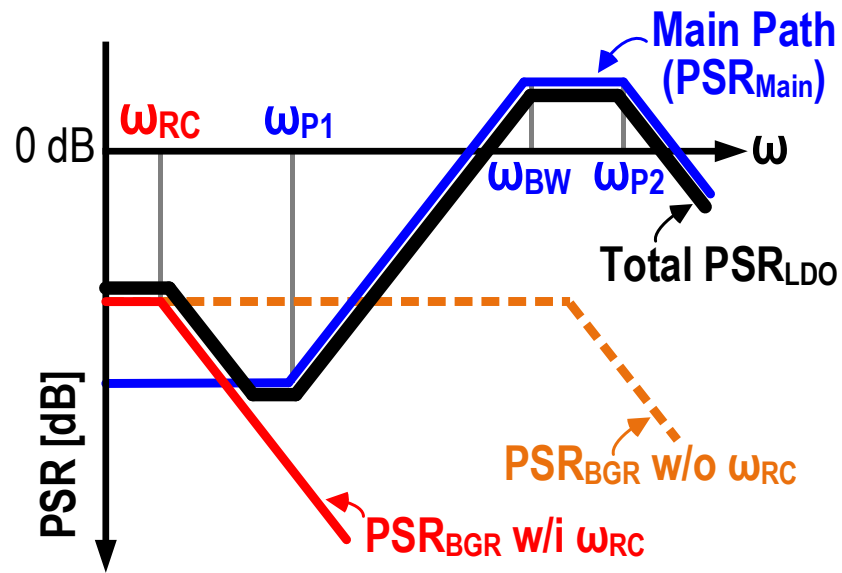
□ Proposed low- I_Q **G_m -boosted OTA** design (C)

■ Signal can be amplified by K , but the DC current is reduced to $(2-K)/2$

Contents

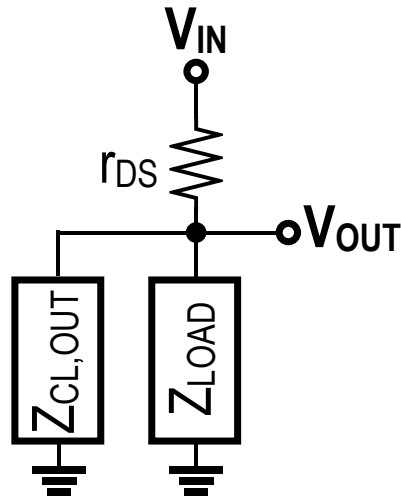
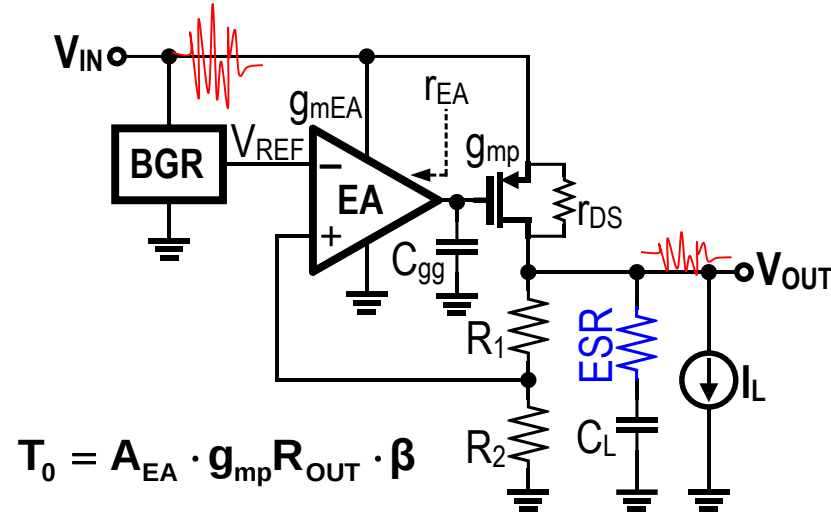
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Supply-Ripple Delivery Paths



- Power-supply rejection: $PSR(s) = \Delta V_{OUT} / \Delta V_{IN}$
- Supply-ripple delivery paths from V_{IN} to V_{OUT}
 - **Main path:** r_{DS} & g_{mP} of power PMOS
 - **Reference path** (V_{REF}): limited PSR of the reference (BGR) voltage circuit

Ripple Delivery via r_{DS} : PSR Model (1/6)



- Open-loop load impedance:

$$Z_{LOAD} = \left(\frac{1}{sC_L} + R_{ESR} \right) \parallel (R_1 + R_2) \parallel R_L$$

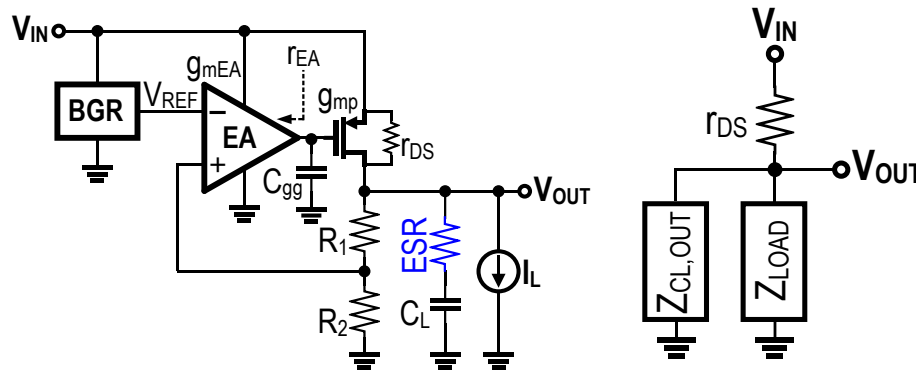
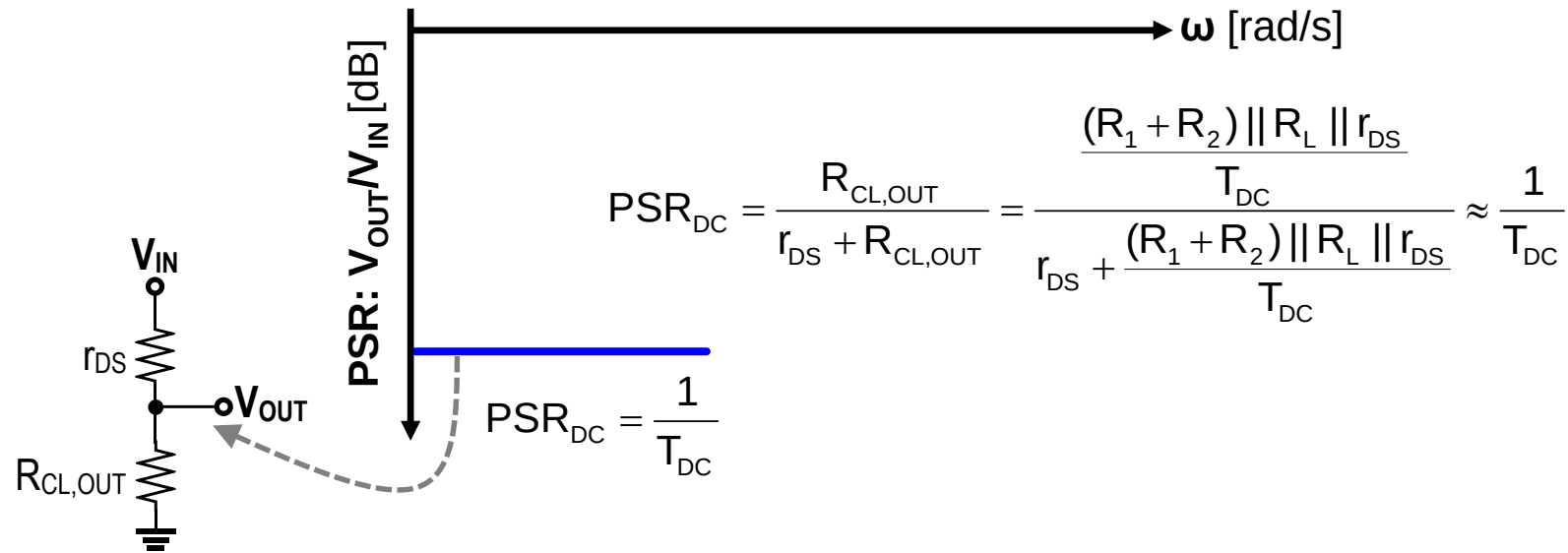
- Closed-loop output impedance:

$$Z_{CL,OUT} = \frac{Z_{LOAD} \parallel r_{DS}}{T(s)}$$

- **PSR transfer function (via r_{DS}):**

$$PSR(s) = \frac{v_{OUT}}{v_{IN}}(s) = \frac{(Z_{LOAD} \parallel Z_{CL,OUT})}{r_{DS} + (Z_{LOAD} \parallel Z_{CL,OUT})}$$

Ripple Delivery via r_{DS} : PSR Model (2/6)

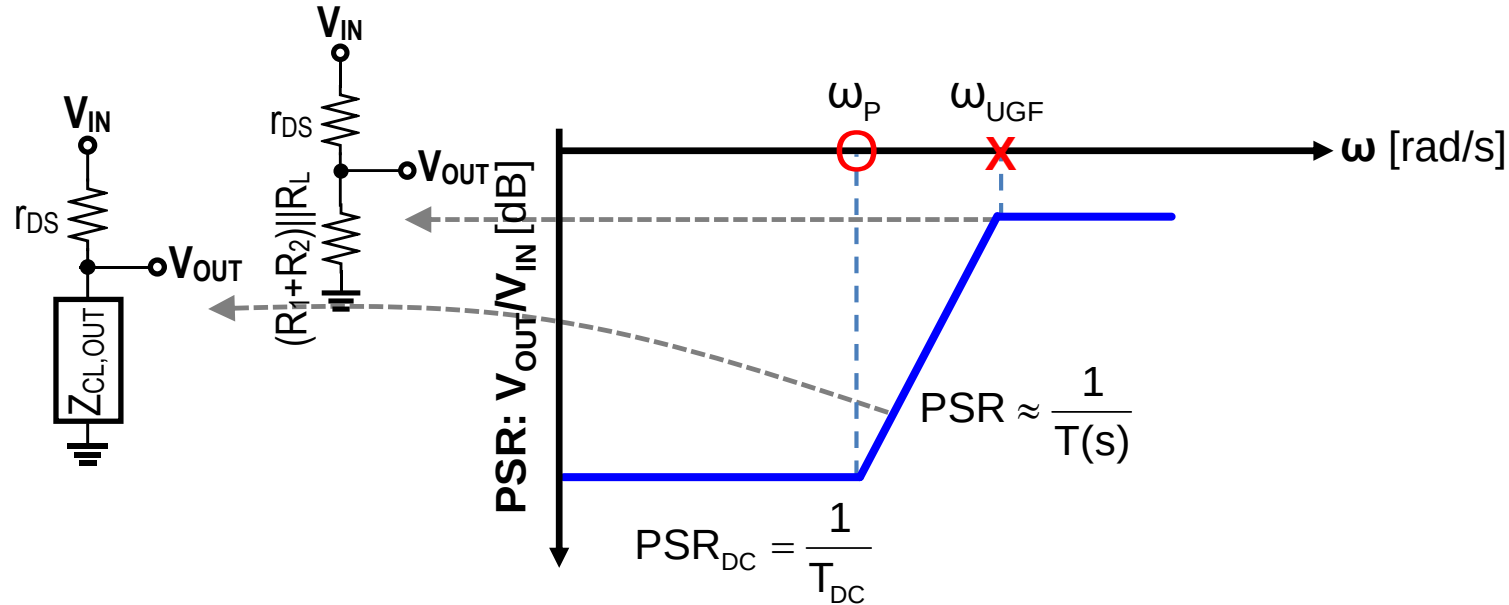


$$PSR(s) = \frac{v_{OUT}}{v_{IN}}(s) = \frac{(Z_{LOAD} \parallel Z_{CL,OUT})}{r_{DS} + (Z_{LOAD} \parallel Z_{CL,OUT})}$$

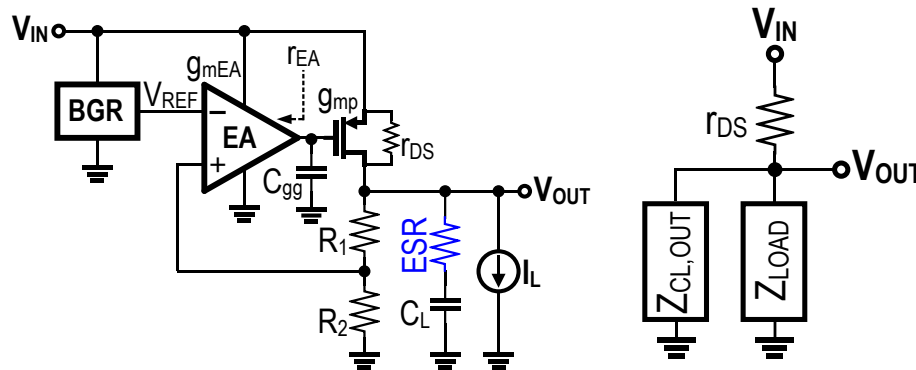
$$Z_{CL,OUT} = \frac{Z_{LOAD} \parallel r_{DS}}{T(s)}$$

$$Z_{LOAD} = \left(\frac{1}{sC_L} + R_{ESR} \right) \parallel (R_1 + R_2) \parallel R_L$$

Ripple Delivery via r_{DS} : PSR Model (3/6)



Assumed that the dominant pole ($\omega_P \approx 1/(r_{EA} \cdot C_{gg})$)

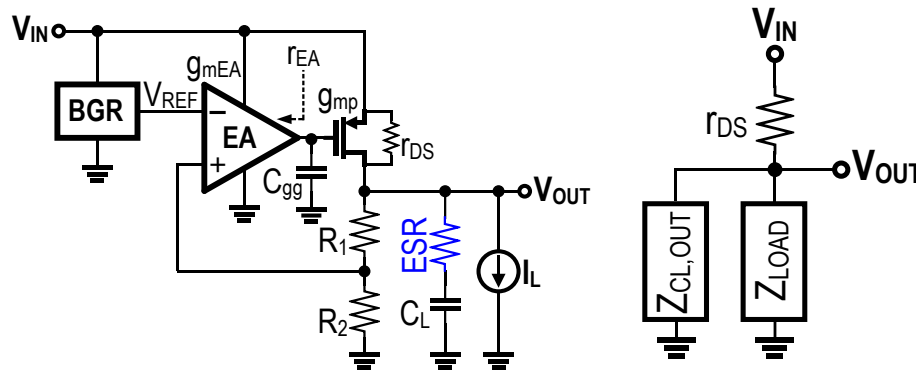
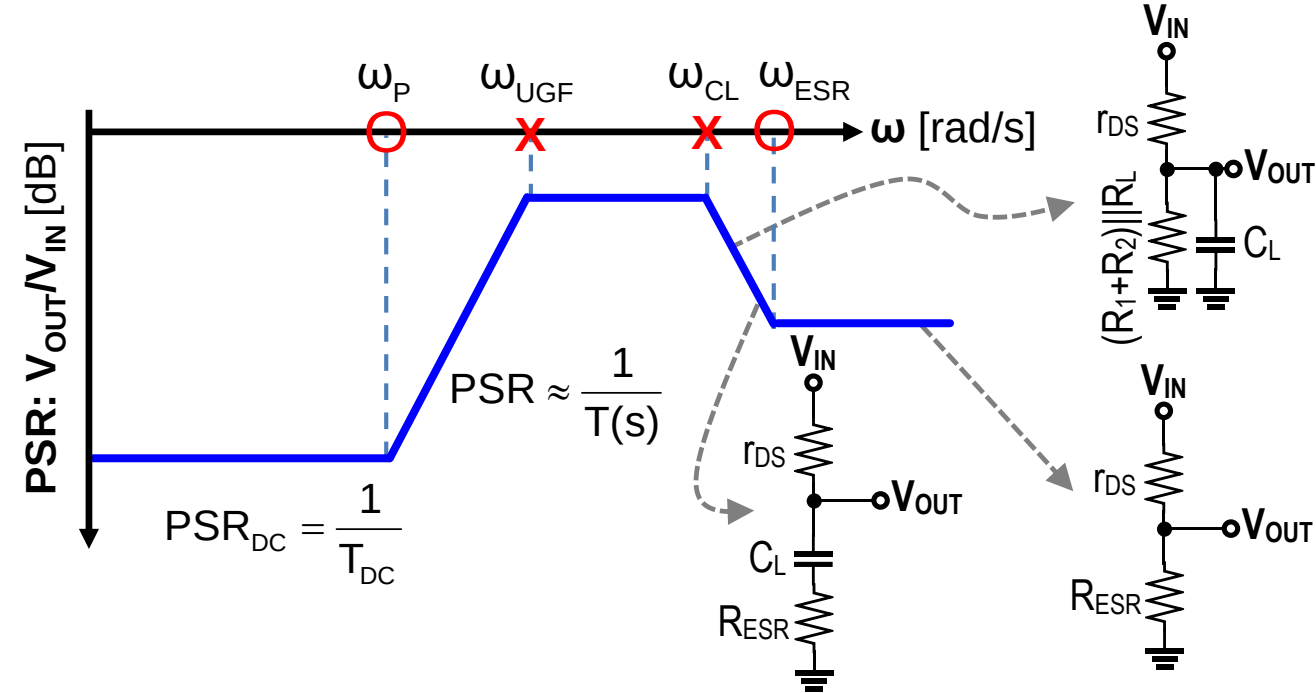


$$PSR(s) = \frac{v_{OUT}}{v_{IN}}(s) = \frac{(Z_{LOAD} \parallel Z_{CL,OUT})}{r_{DS} + (Z_{LOAD} \parallel Z_{CL,OUT})}$$

$$Z_{CL,OUT} = \frac{Z_{LOAD} \parallel r_{DS}}{T(s)}$$

$$Z_{LOAD} = \left(\frac{1}{sC_L} + R_{ESR} \right) \parallel (R_1 + R_2) \parallel R_L$$

Ripple Delivery via r_{DS} : PSR Model (4/6)

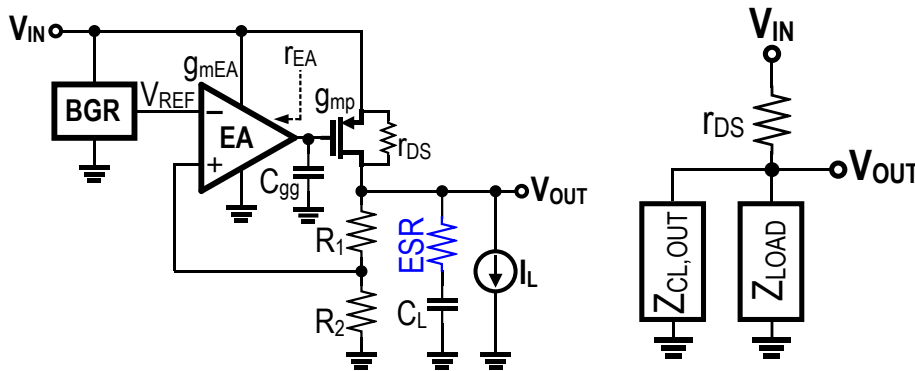
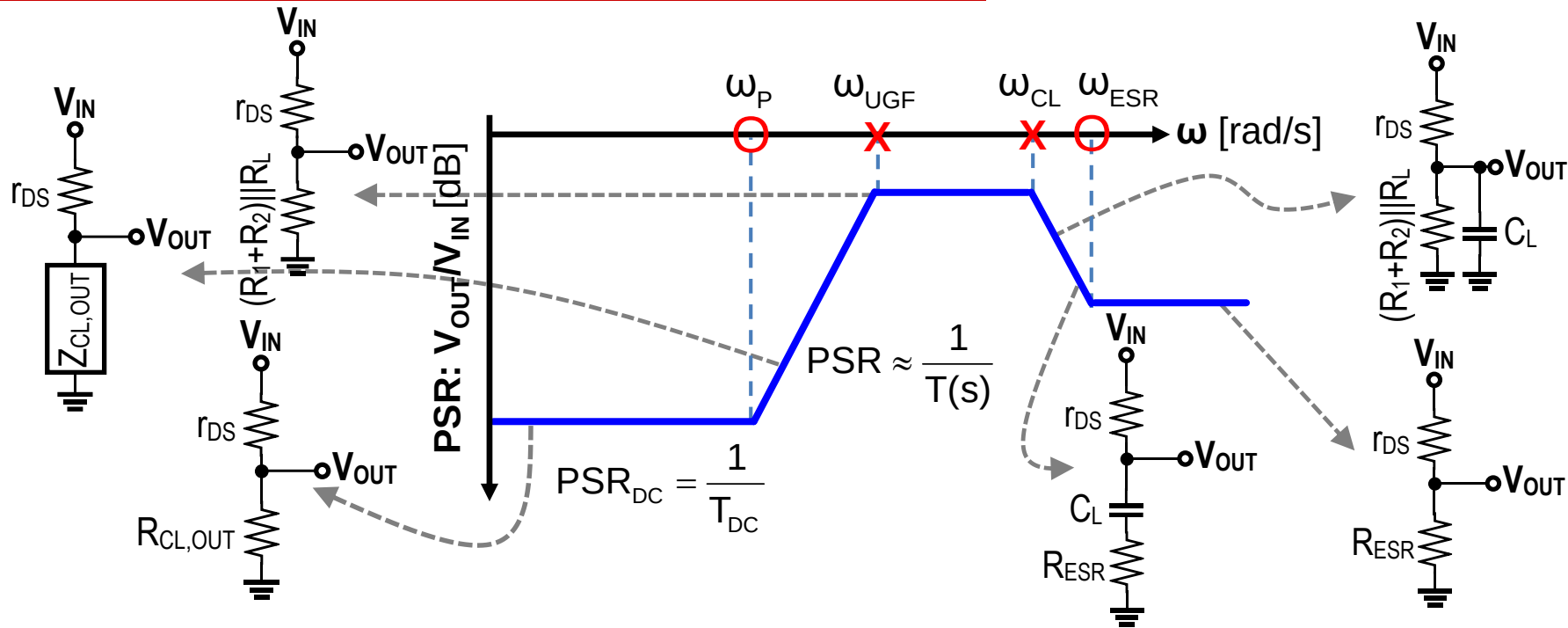


$$PSR(s) = \frac{v_{OUT}}{v_{IN}}(s) = \frac{(Z_{LOAD} \parallel Z_{CL,OUT})}{r_{DS} + (Z_{LOAD} \parallel Z_{CL,OUT})}$$

$$Z_{CL,OUT} = \frac{Z_{LOAD} \parallel r_{DS}}{T(s)}$$

$$Z_{LOAD} = \left(\frac{1}{sC_L} + R_{ESR} \right) \parallel (R_1 + R_2) \parallel R_L$$

Ripple Delivery via r_{DS} : PSR Model (5/6)

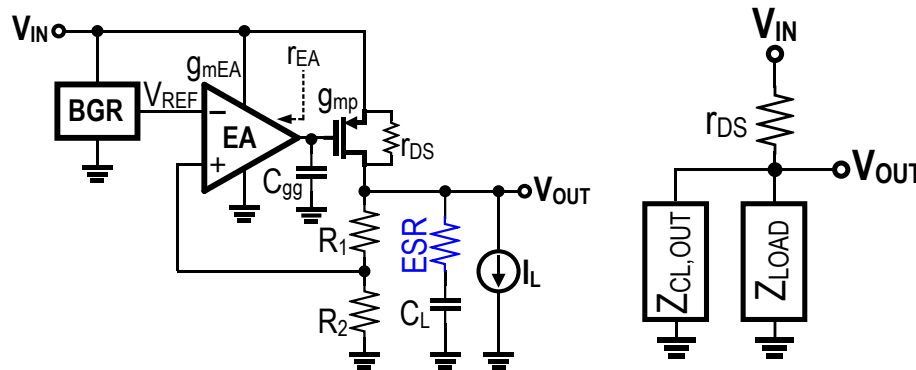
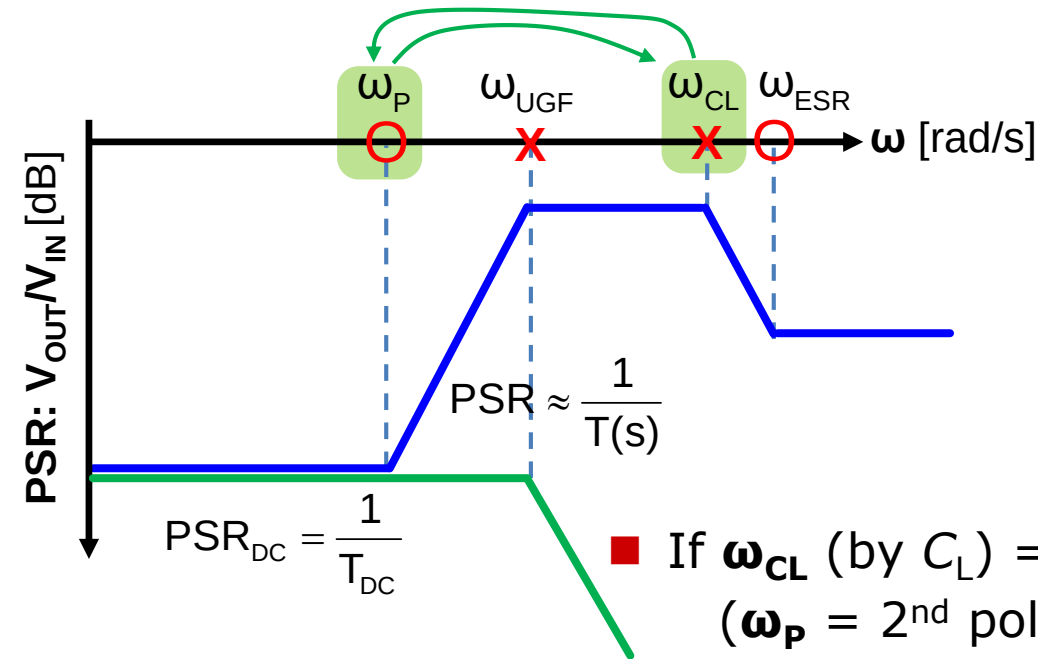


$$\text{PSR}(s) = \frac{v_{\text{OUT}}}{v_{\text{IN}}}(s) = \frac{(Z_{\text{LOAD}} \parallel Z_{\text{CL,OUT}})}{r_{\text{DS}} + (Z_{\text{LOAD}} \parallel Z_{\text{CL,OUT}})}$$

$$Z_{\text{CL,OUT}} = \frac{Z_{\text{LOAD}} \parallel r_{\text{DS}}}{T(s)}$$

$$Z_{\text{LOAD}} = \left(\frac{1}{sC_L} + R_{\text{ESR}} \right) \parallel (R_1 + R_2) \parallel R_L$$

Ripple Delivery via r_{DS} : PSR Model (6/6)



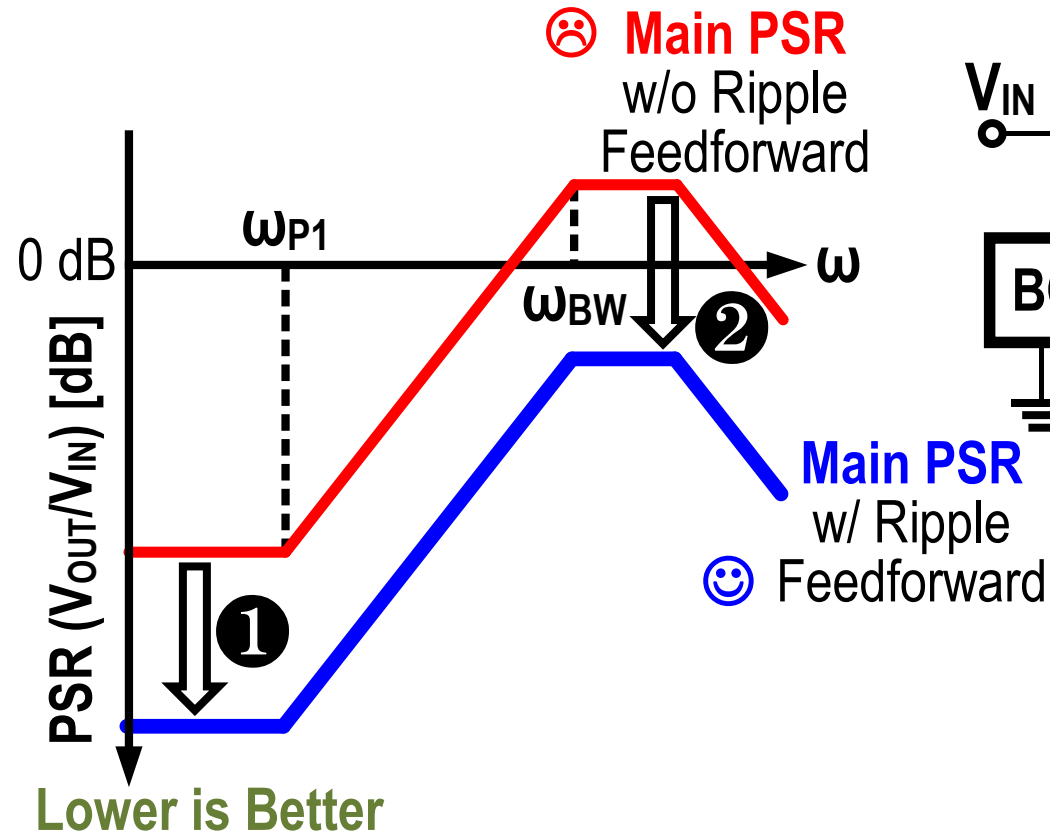
$$PSR(s) = \frac{v_{OUT}}{v_{IN}}(s) = \frac{(Z_{LOAD} \parallel Z_{CL,OUT})}{r_{DS} + (Z_{LOAD} \parallel Z_{CL,OUT})}$$

$$Z_{CL,OUT} = \frac{Z_{LOAD} \parallel r_{DS}}{T(s)}$$

$$Z_{LOAD} = \left(\frac{1}{sC_L} + R_{ESR} \right) \parallel (R_1 + R_2) \parallel R_L$$

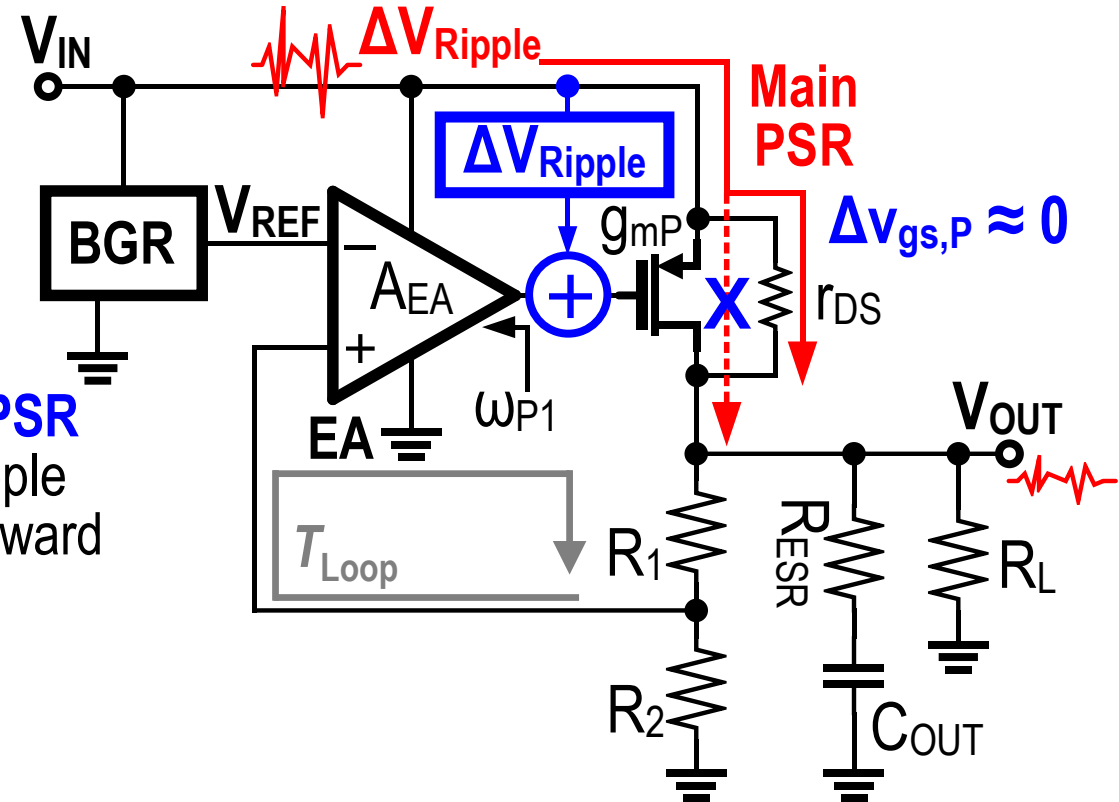
Ripple Delivery via g_{mP} : Ripple Feedforward (1/2)

□ Kim & Kim, VLSI'19 & TPEL'20



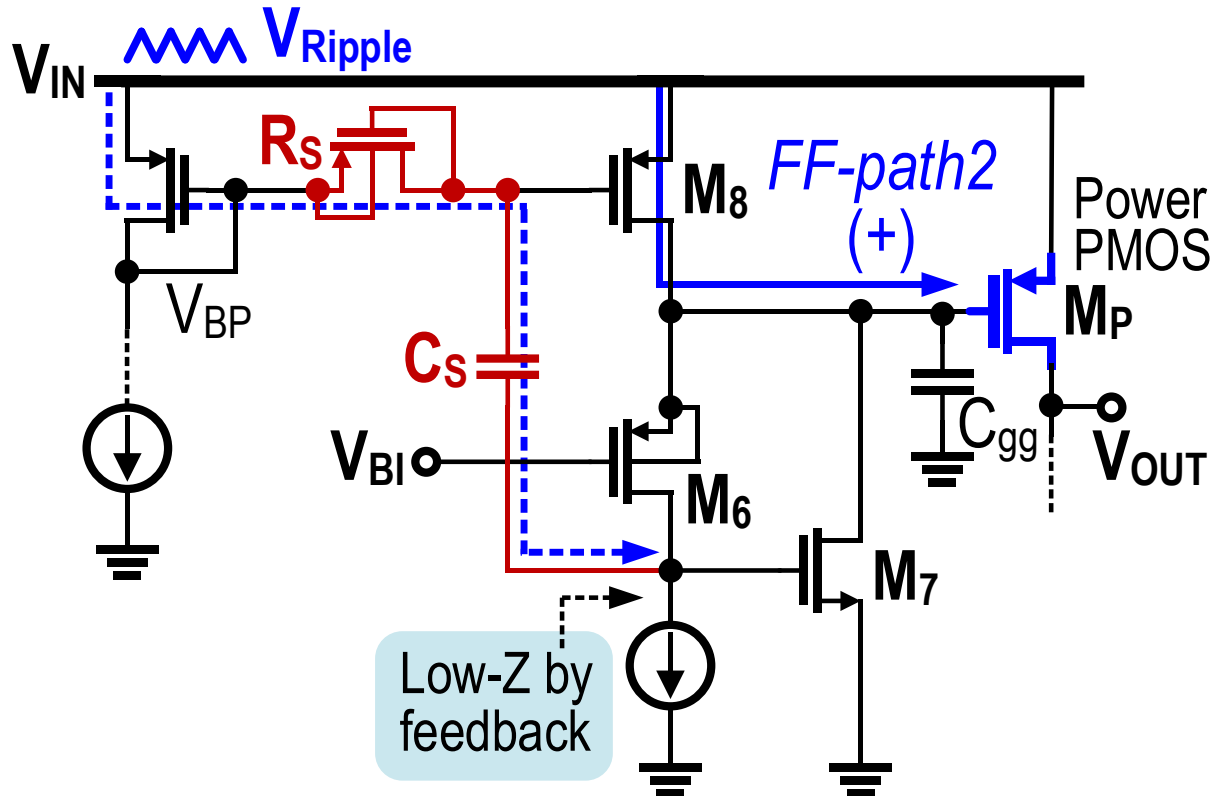
$$PSR_{Low} = \frac{1}{A_{EA} \cdot \frac{R_2}{R_1 + R_2}} \Rightarrow \boxed{\frac{1}{T_{Loop,DC}}} \quad \text{①}$$

$$PSR_{High} = g_{mP}(r_{DS} || R_L) \Rightarrow \boxed{\frac{R_L}{r_{DS} + R_L}} \quad \text{②}$$



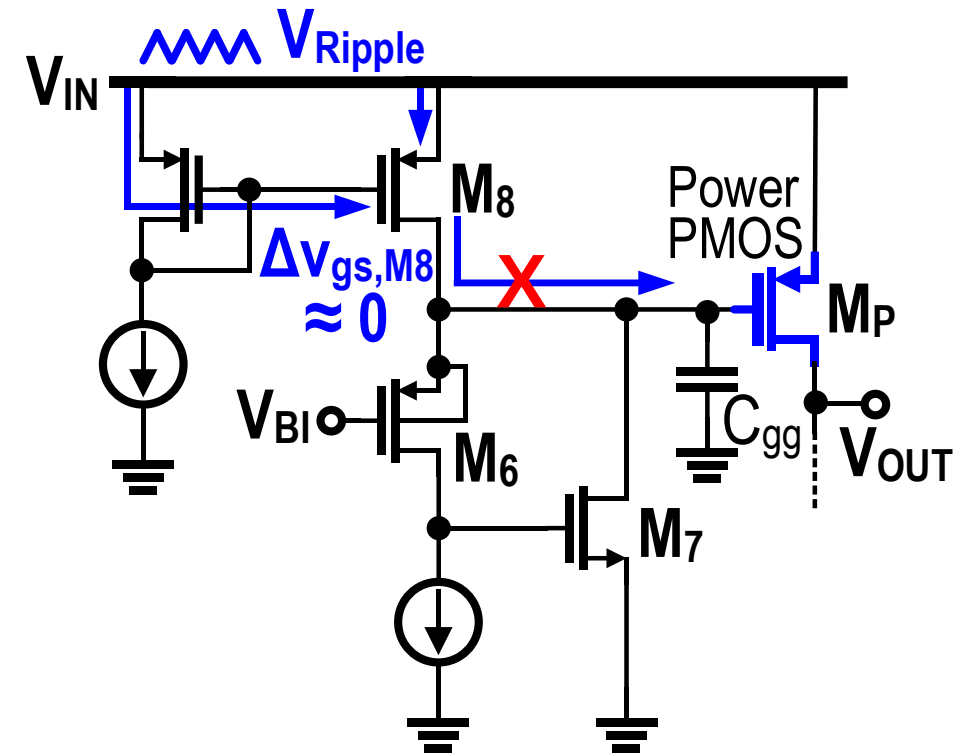
Ripple Delivery via g_{mP} : Ripple Feedforward (2/2)

□ Kim & Kim, VLSI'19 & TPEL'20



□ *Proposed* ripple-feedforward buffer

■ V_{Ripple} is absorbed into low- Z via C_s

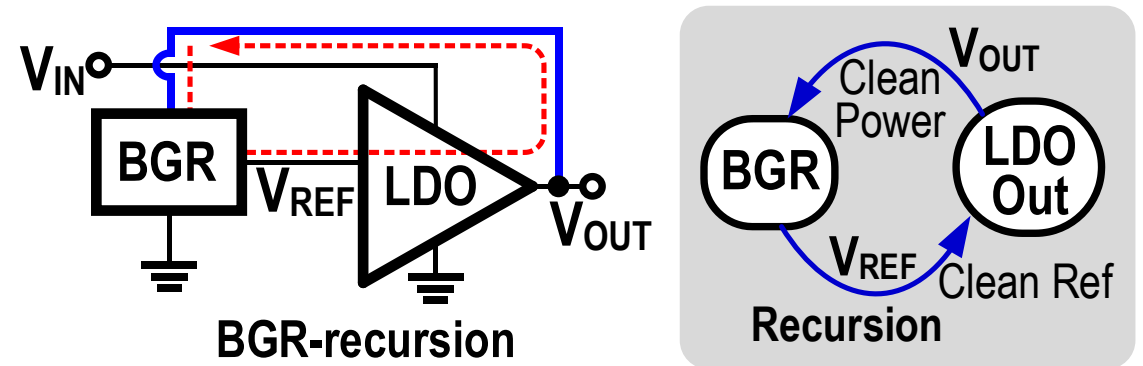
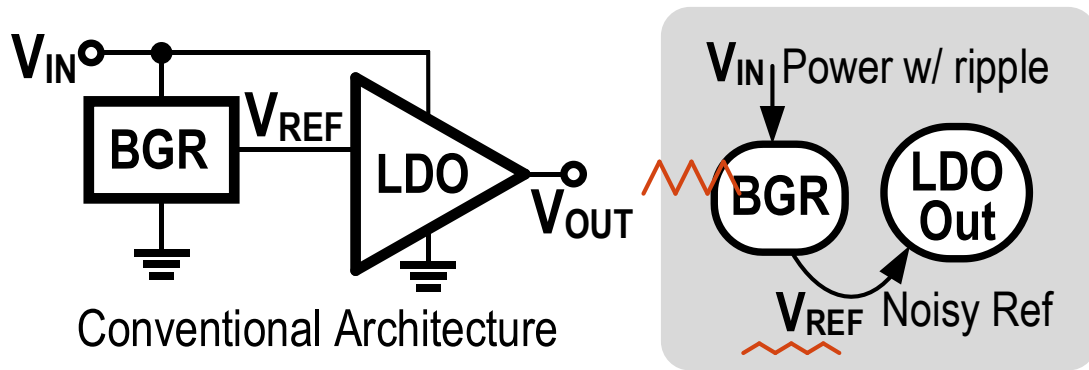


□ Conventional buffer

■ No feedforward by $\Delta v_{gs,M8} \approx 0$

Ripple Delivery via V_{REF} : Recursive LDO (1/3)

□ Kim & Kim, VLSI'19 & TPEL'20

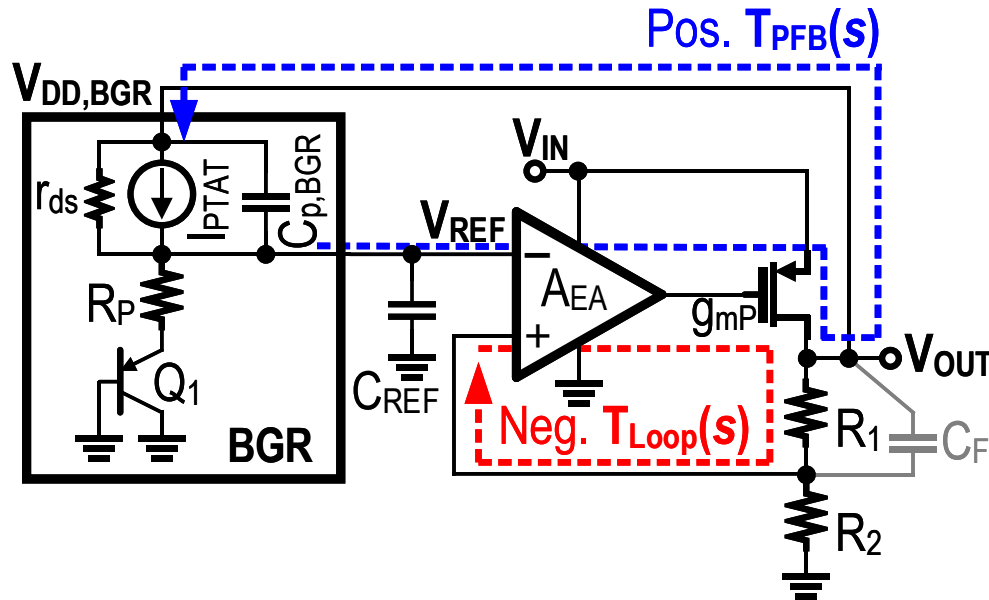


Total PSR of LDO $\approx \text{Worst}(\text{Main-PSR}, \text{BGR-PSR})$

Total PSR of LDO $\approx \text{Main-PSR}$

Ripple Delivery via V_{REF} : Recursive LDO (2/3)

□ Kim & Kim, VLSI'19 & TPEL'20



$$PSR_{BGR} = \frac{R_P + r_{e,Q1}}{r_{ds} + R_P + r_{e,Q1}} \frac{1 + sC_{p,BGR}r_{ds}}{1 + s(C_{p,BGR} + C_{REF})[r_{ds} \parallel (R_P + r_{e,Q1})]}$$

$$\beta(s) = \frac{R_2}{R_1 + R_2} \frac{1 + sC_F R_1}{1 + sC_F (R_1 \parallel R_2)}$$

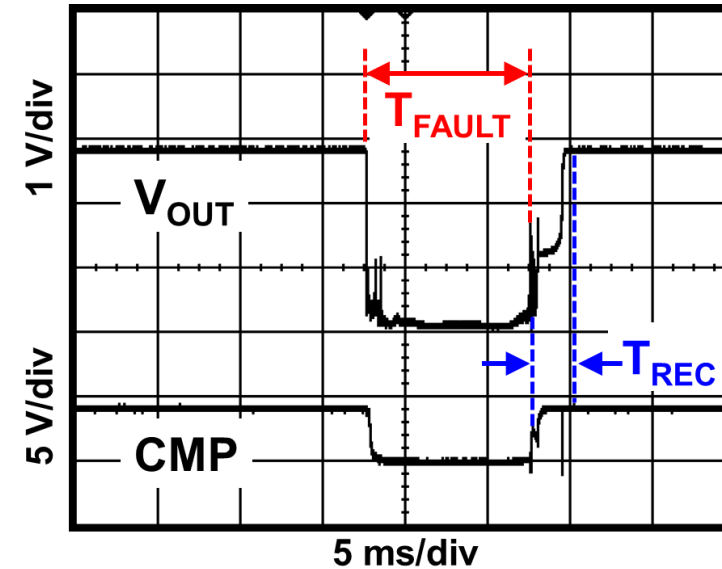
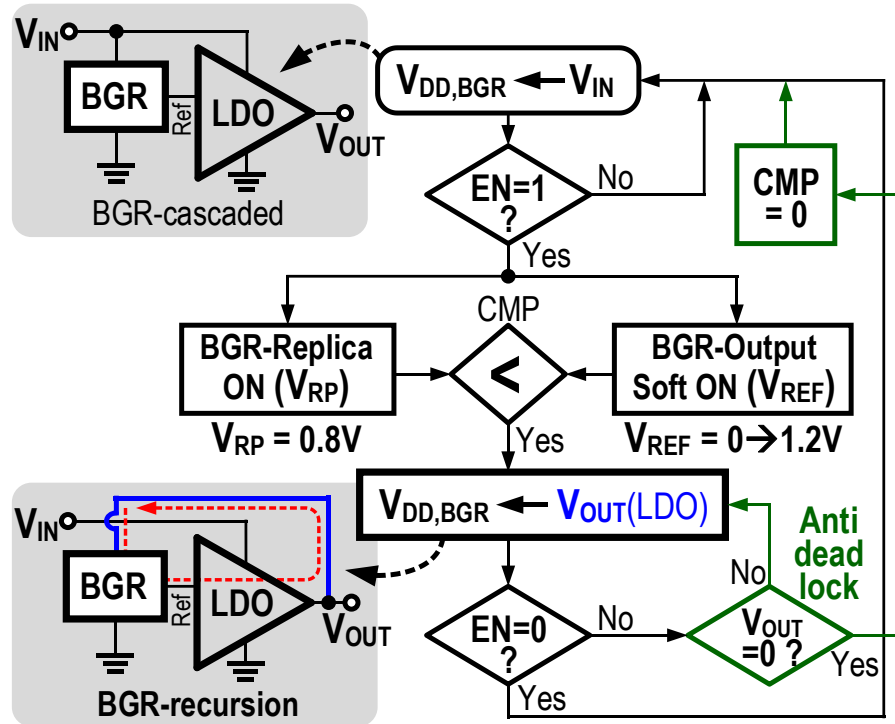
$$T_{tot}(s) \approx T_{Loop}(s) + T_{PFB}(s) = T_{Loop}(s) \left[1 - \frac{PSR_{BGR}(s)}{\beta(s)} \right]$$

□ Design Consideration (1) in recursive LDO: **stability**

- $T_{Loop}(s)$ = main LDO regulation loop, $T_{PFB}(s)$ = BGR-recursive loop
- For stability, $T_{PFB}(s) < T_{Loop}(s)$

Ripple Delivery via V_{REF} : Recursive LDO (3/3)

□ Kim & Kim, VLSI'19 & TPEL'20



□ Design Consideration (2) in recursive LDO: **startup & deadlock**

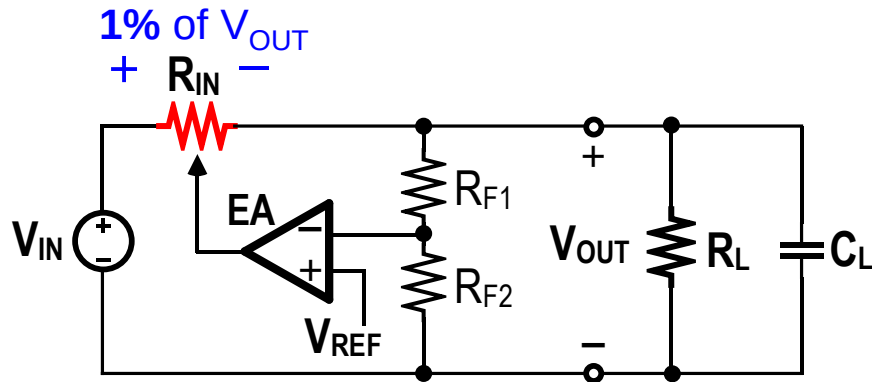
- As a result of V_{OUT} fault, the recursive LDO may fall in a deadlock
- When fault is detected, the recursive-loop is unlocked & temporarily V_{IN} -supplied

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- Conclusion

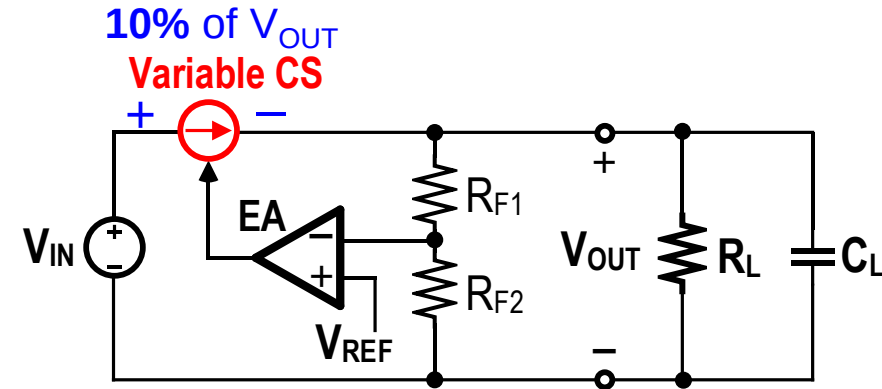
Triode LDO vs. Saturation (CS) LDO

Triode LDO



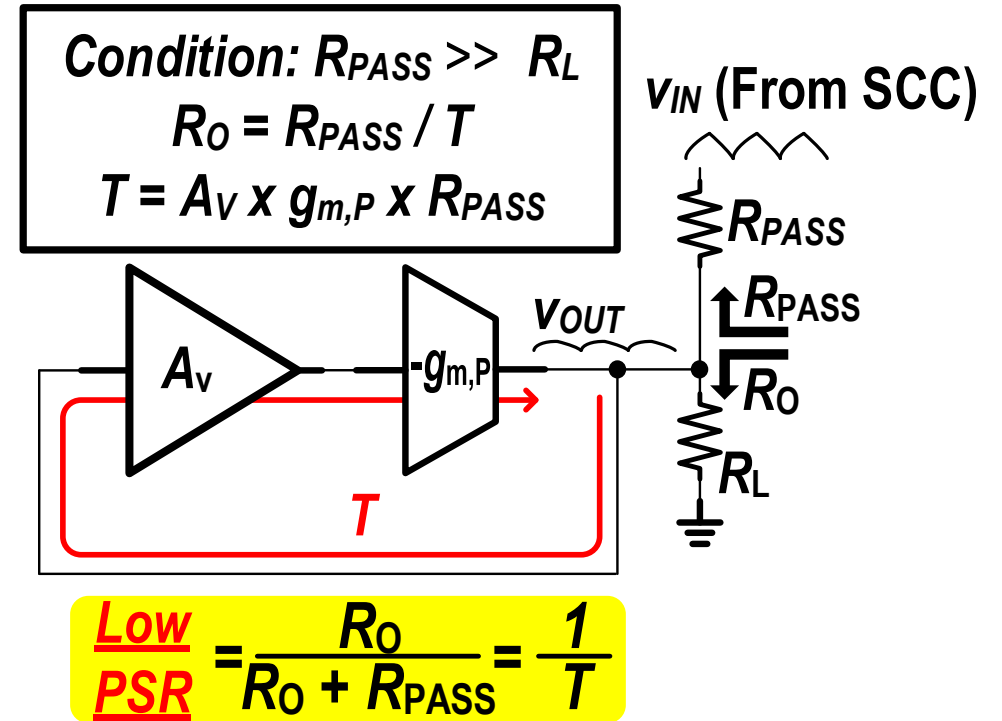
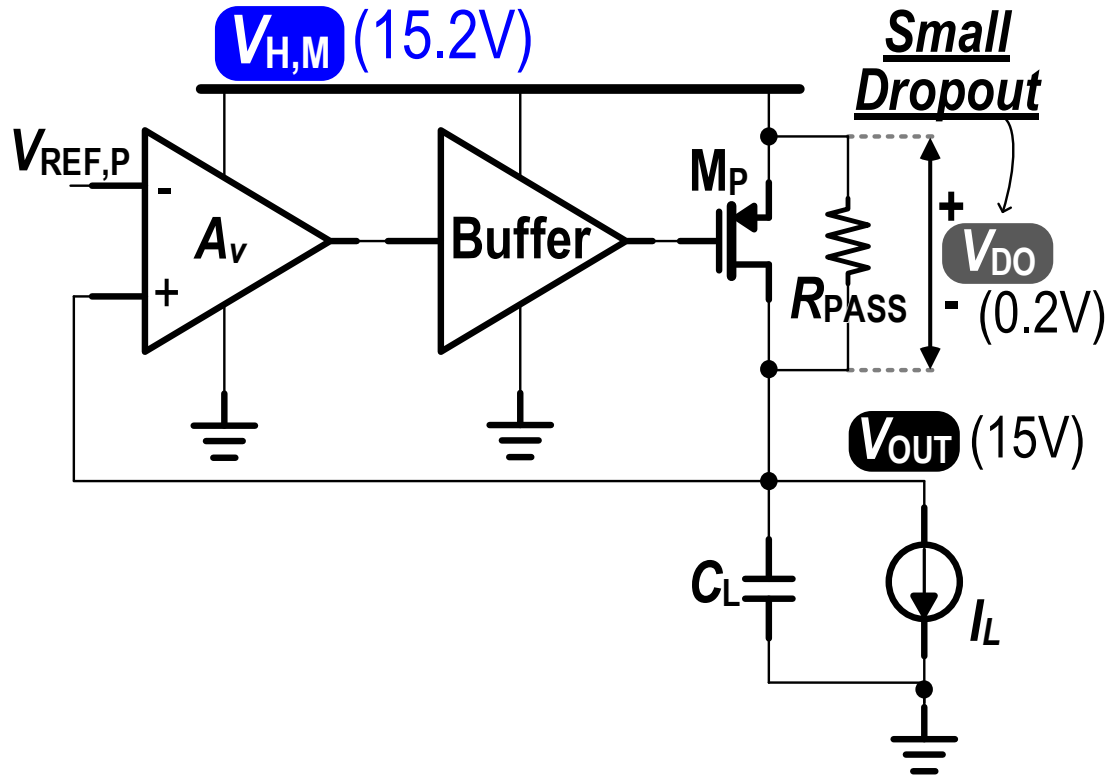
- Pass element works in triode region
- Small V_{DO} (high η) 😊
- Low regulation due to low loop-gain (T) 😞
- Bad PSR due to small R_{DS} 😞
- Normally not preferred

Saturation LDO



- Pass element works in saturation region
- High V_{DO} (lower η) 😞
- Good regulation performance due to high loop-gain (T) 😊
- Good PSR owing to large R_{DS} 😊
- Adopted in most LDOs

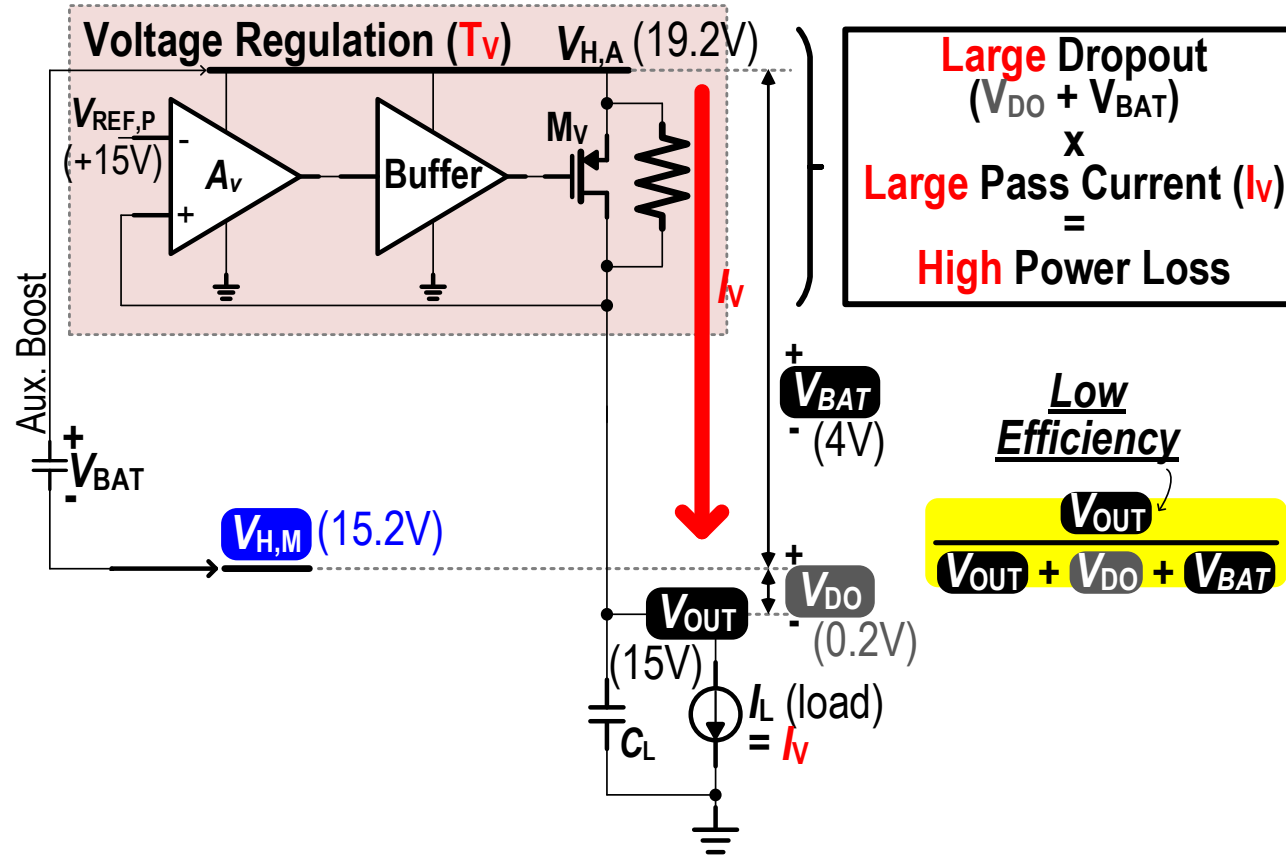
Low PSR of Triode LDO



- ❑ Extremely excellent efficiency 😊: $V_{DO} = 0.2V$ [$\approx 1.3\%$ of V_{OUT} (15V)]
- ❑ Low PSR due to small R_{PASS} ($= R_{DS}$ of M_P) ☹️
 - M_P (power PMOS) working in triode region has a lower R_{DS}

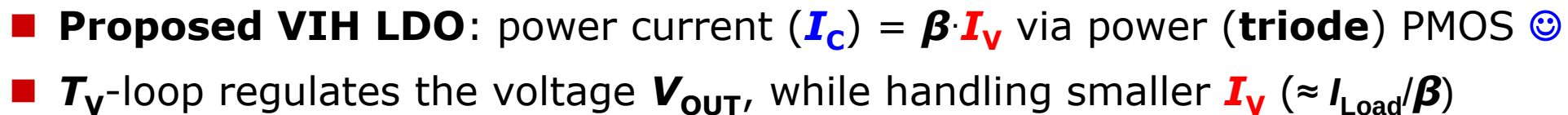
Voltage-Current-Hybrid (VIH) LDO (1/6)

□ Ko & Kim, ISSCC'21 & JSSC'23



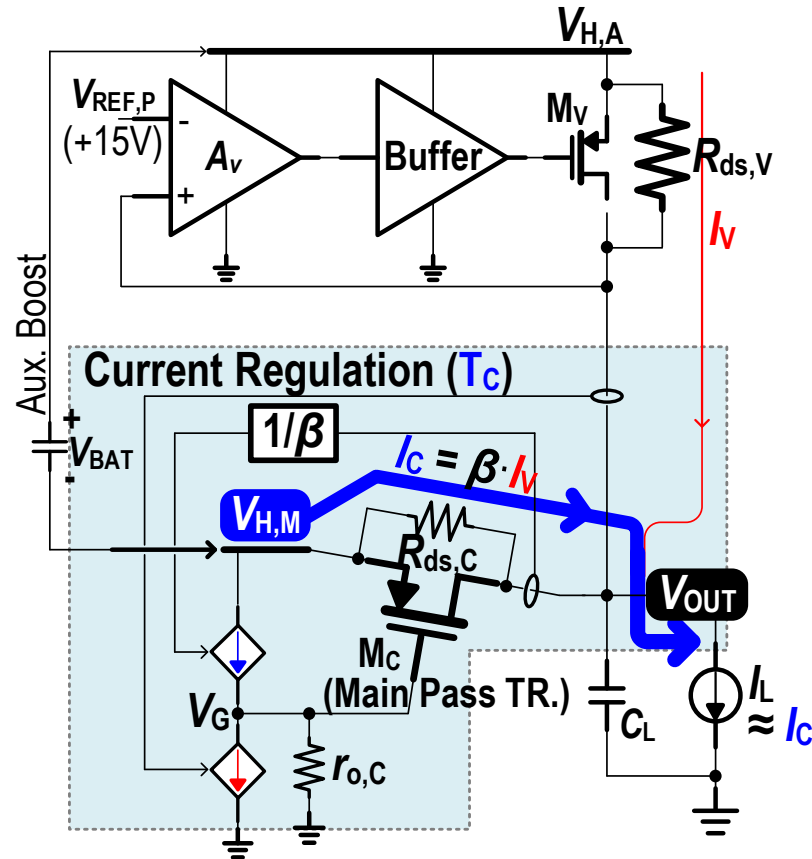
- V_{IN} ($V_{H,A} = 4V + V_{OUT}$) is boosted by charge-pump to ensure a sufficient V_{DO}
- Significant power loss ☹️ (= large V_{DO} x pass current I_V)

□ Ko & **Kim**, *ISSCC'21* & *JSSC'23*

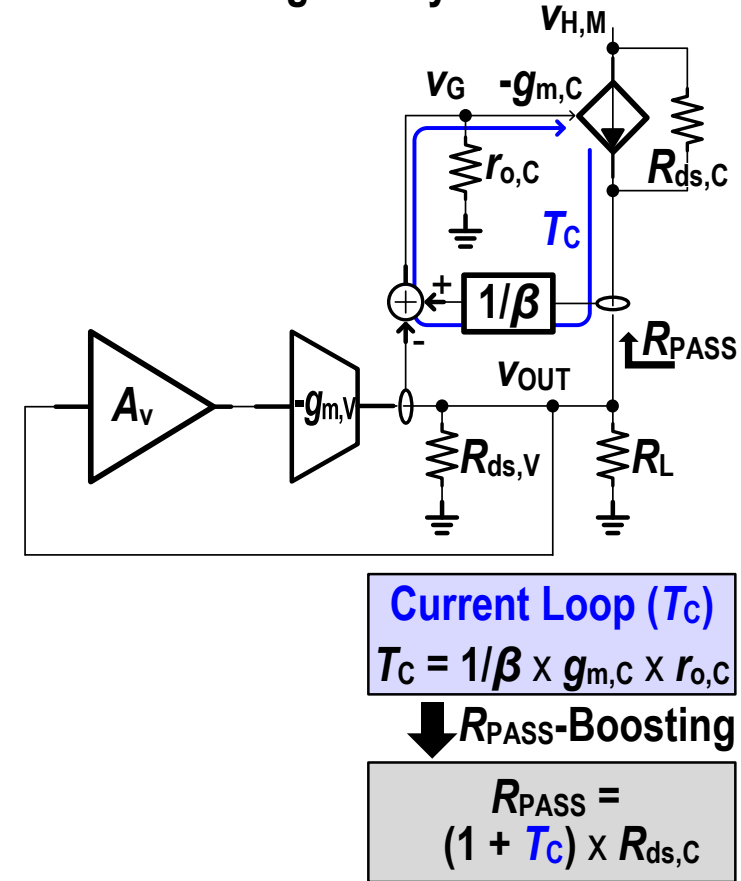


Voltage-Current-Hybrid (VIH) LDO (3/6)

□ Ko & Kim, *ISSCC'21* & *JSSC'23*



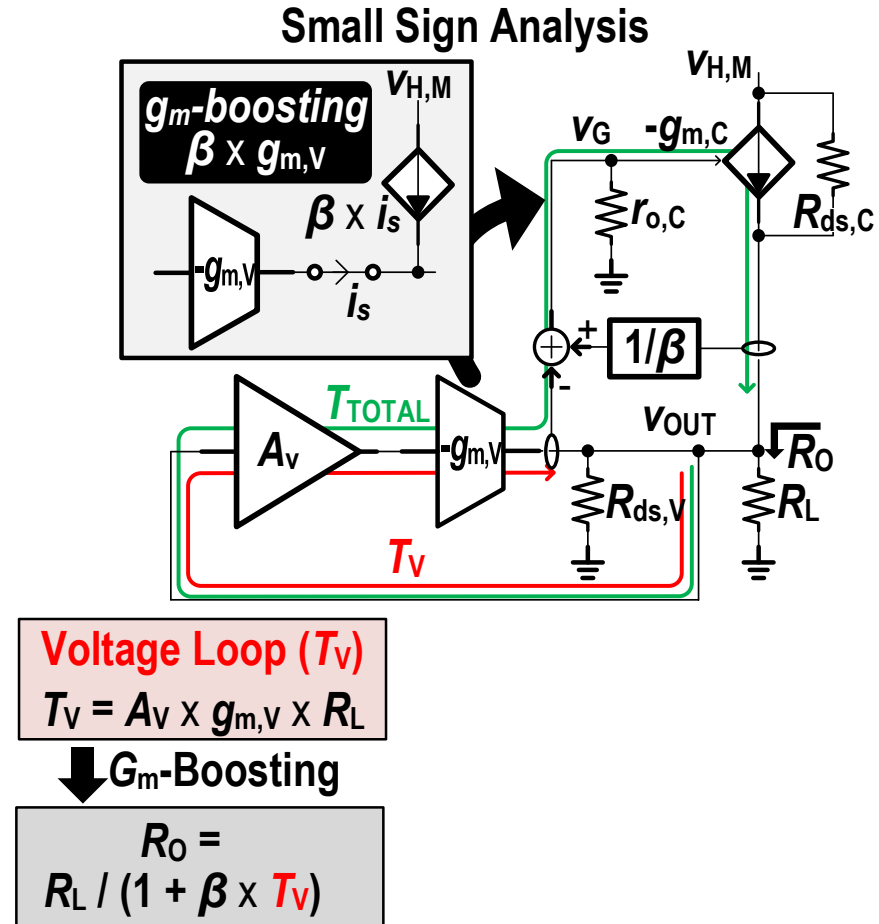
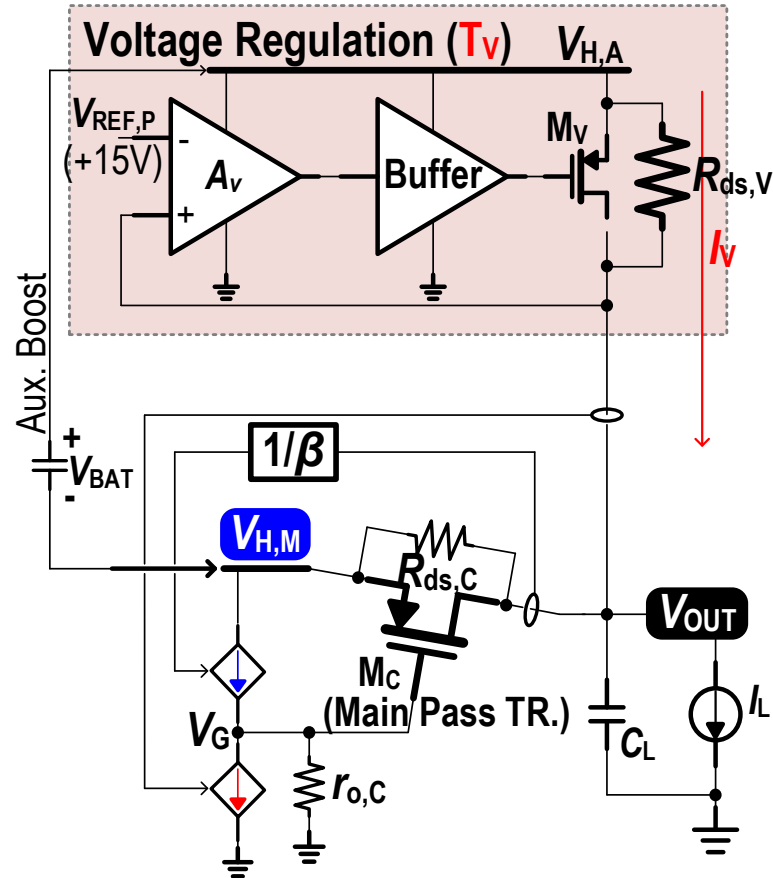
Small Sign Analysis



- PMOS's R_{DS} is small because of working in triode region, BUT
- R_{PASS} is highly boosted by current-loop gain (T_C): $R_{DS} \rightarrow (1 + T_C)R_{DS}$

Voltage-Current-Hybrid (VIH) LDO (4/6)

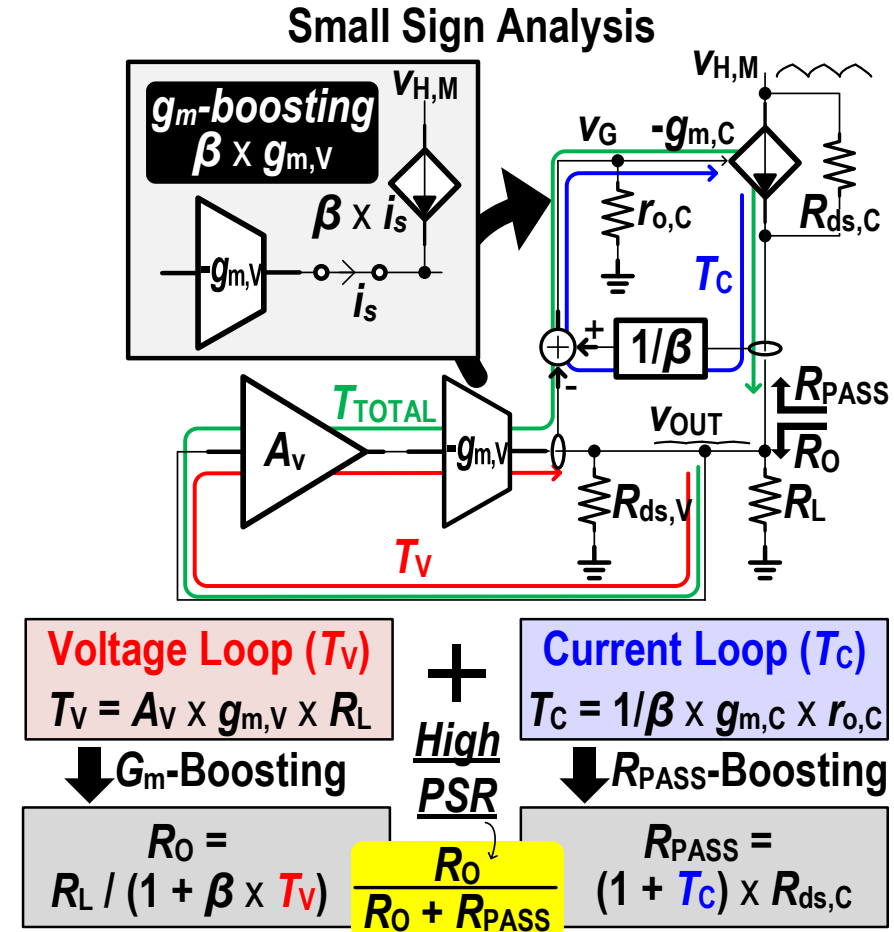
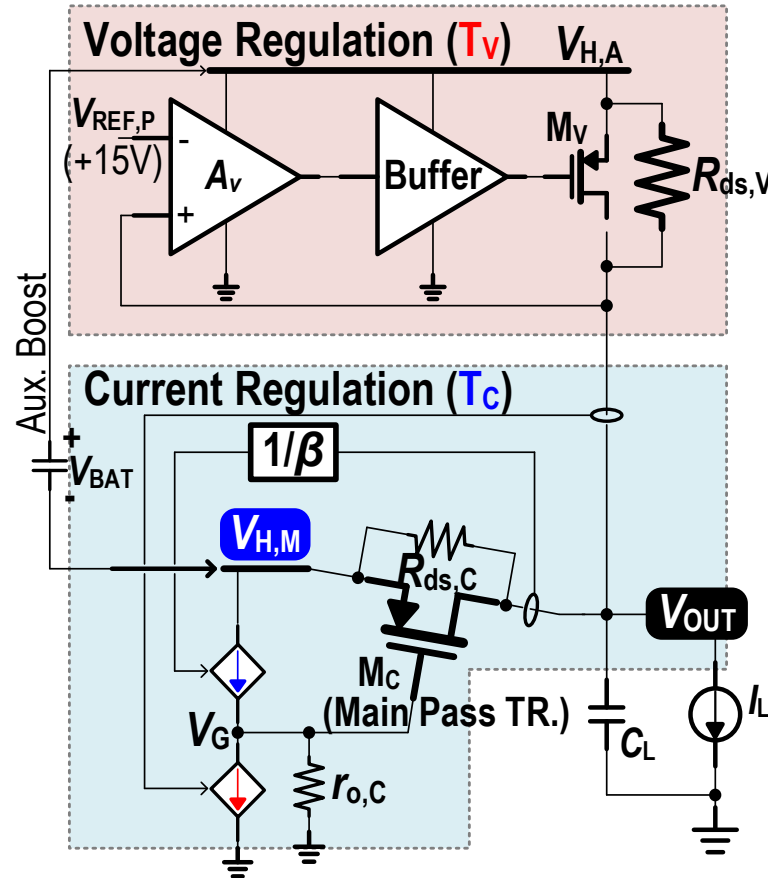
□ Ko & Kim, ISSCC'21 & JSSC'23



■ R_O is further reduced owing to G_M -boosting effect (β): $T_V \rightarrow \beta \cdot T_V$

Voltage-Current-Hybrid (VIH) LDO (5/6)

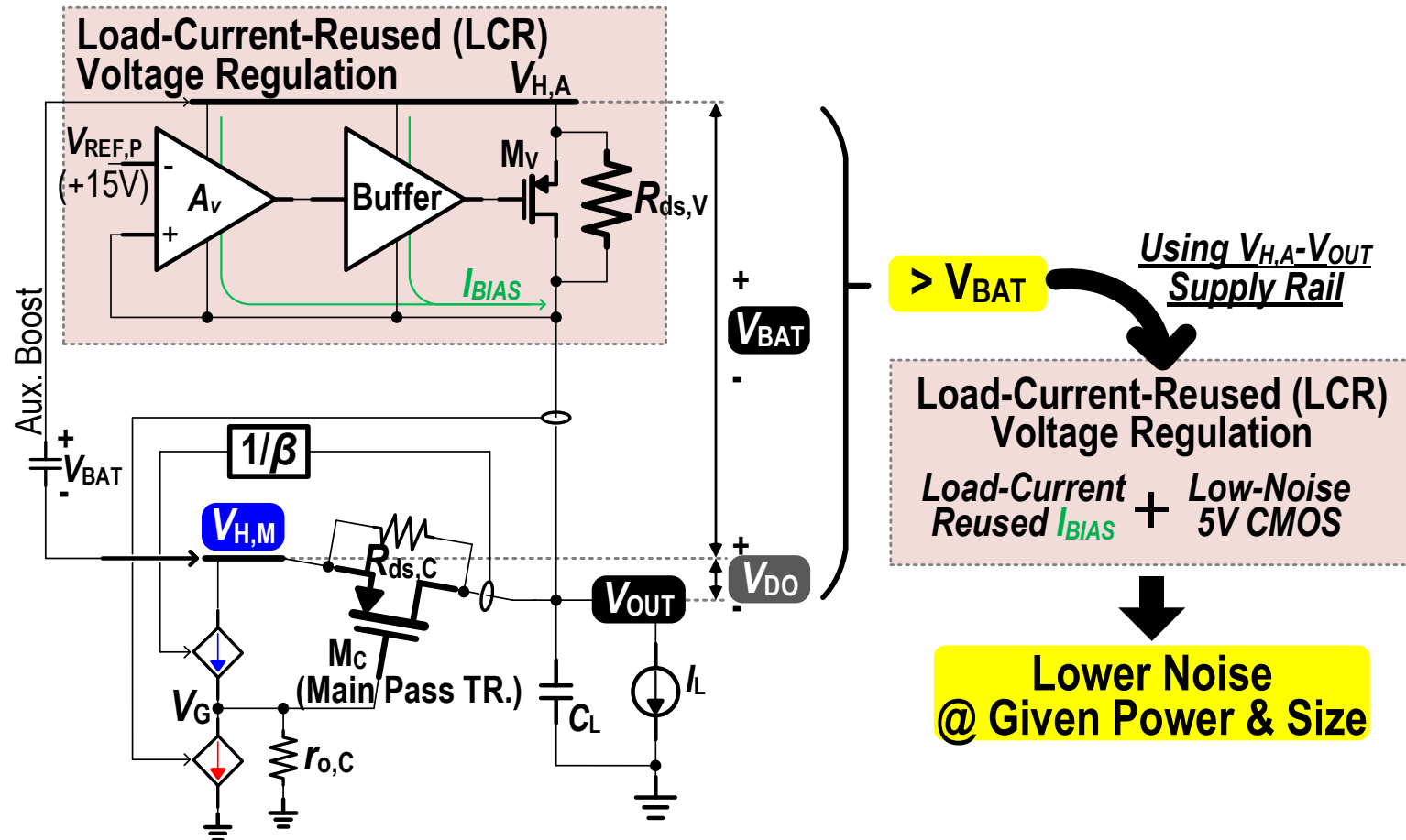
□ Ko & Kim, ISSCC'21 & JSSC'23



■ Small R_O (by T_V -loop) & Boosted R_{PASS} (by T_C -loop) → high PSR even low V_{DO} 😊

Voltage-Current-Hybrid (VIH) LDO (6/6)

□ Ko & Kim, ISSCC'21 & JSSC'23

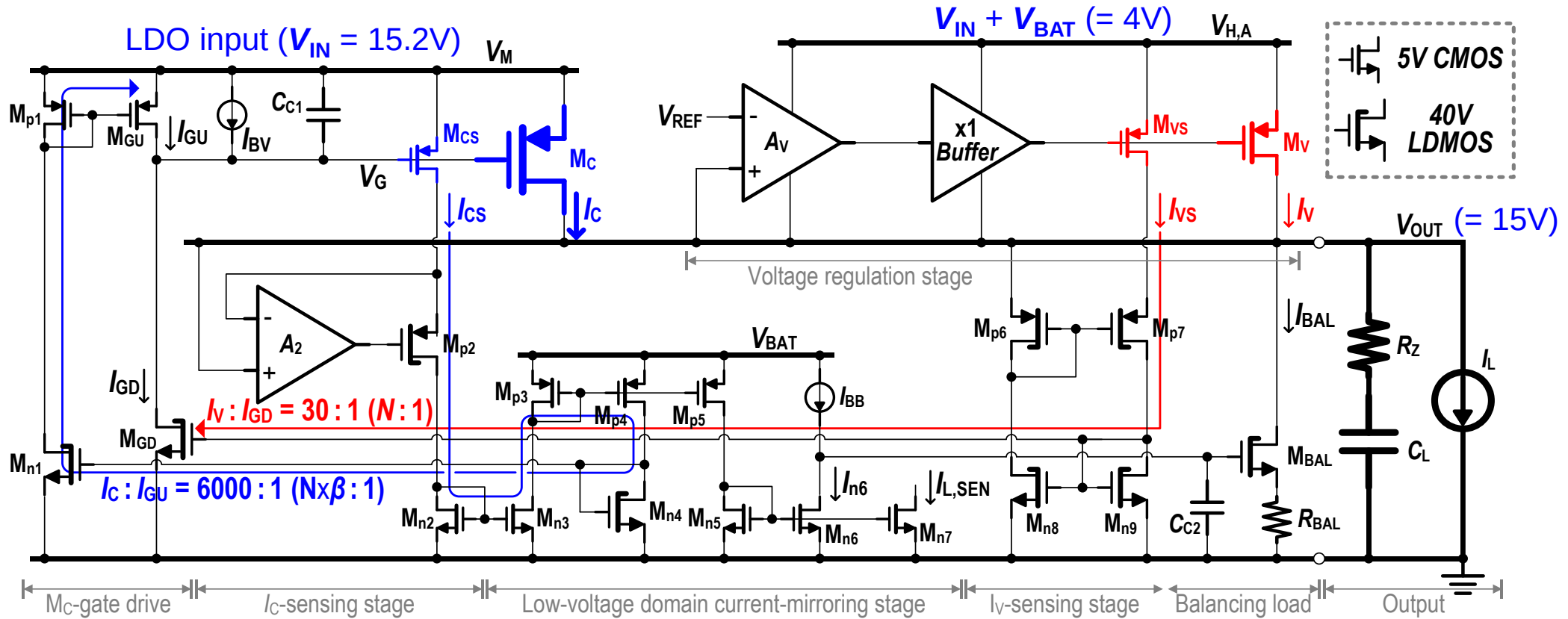


- Bias current (I_{BIAS}) consumed in T_v -loop is re-used as a part of I_{Load}
- Thus, high bias current (I_{BIAS}) can contribute on low noise of V_{OUT} 😊

Implementation of VIH LDO

β (current gain) = 200 A/A

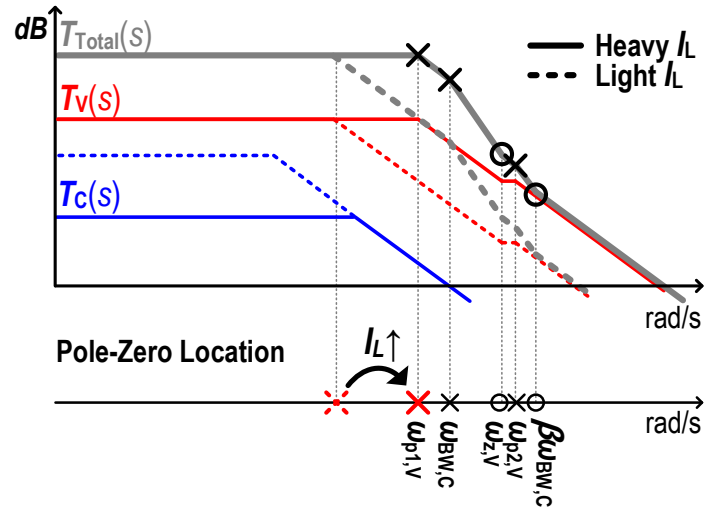
□ Ko & Kim, ISSCC'21 & JSSC'23



$M_C : M_{CS}$	300 : 1	$M_{N2} : M_{N3}$	2 : 1	$M_{P3} : M_{P4} : M_{P5}$	10 : 1 : 1	$M_V : M_{VS}$	30 : 1
$M_{N5} : M_{N6}$	1 : 1	$M_{N5} : M_{N6}$	1 : 10	$M_{N5} : M_{N7}$	1 : 5	$M_{N9} : M_{N8} : M_{GD}$	1 : 1 : 1
$I_C : I_{GU}$	6000 : 1	$I_C : I_{N6}$	600 : 1	$I_C : I_{L,SEN}$	1200 : 1	$I_V : I_{GD} (= N : 1)$	30 : 1

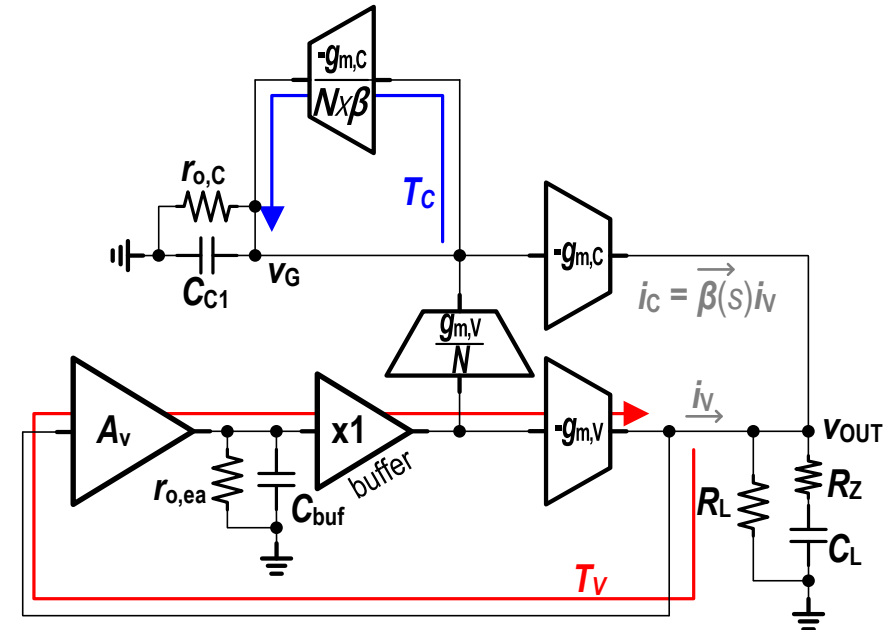
Frequency & Stability of VIH LDO

□ Ko & Kim, ISSCC'21 & JSSC'23



Notation	Pole/Zero	Frequency [rad/s]
$\omega_{p1,V}$	Pole	$1/(R_L \cdot C_L)$
$\omega_{BW,C}$	Pole	$g_{m,C}/(N \cdot \beta \cdot C_{C1})$
$\omega_{p2,V}$	Pole	$1/(r_{o,ea} \cdot C_{buf})$
$\omega_{z,V}$	Zero	$1/(R_Z \cdot C_L)$
$\beta \cdot \omega_{BW,C}$	Zero	$g_{m,C}/(N \cdot C_{C1})$

- Overall $T_{Total}(s)$ has 3 poles and 2 zeros
- VIH LDO can cover a wide load range



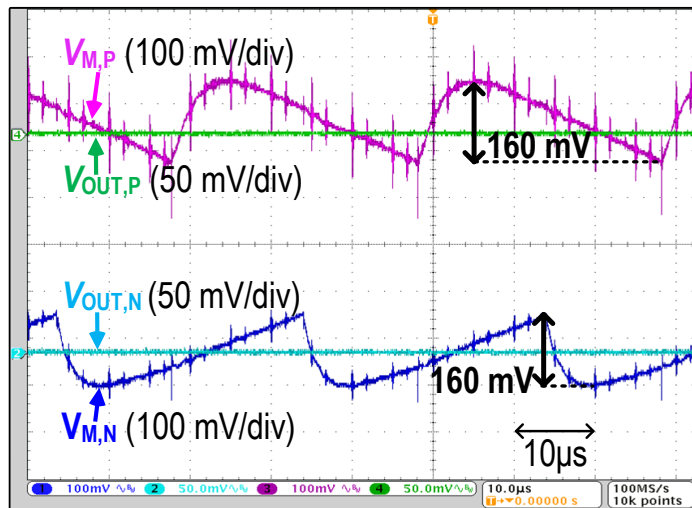
$$\beta(s) = \frac{i_C}{i_V} = \beta \cdot \frac{T_C(s)}{1 + T_C(s)} \approx \beta \cdot \frac{1}{1 + s/\omega_{BW,C}}$$

$$T_{Total}(s) = [1 + \beta(s)] \cdot T_V(s)$$

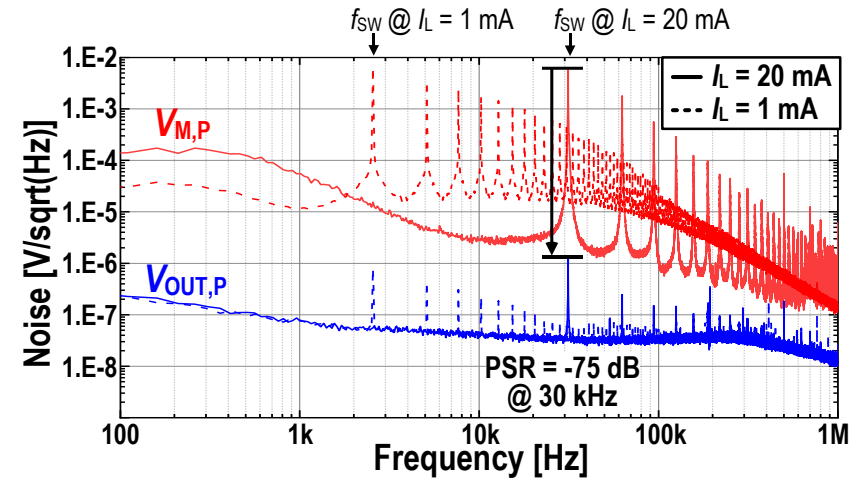
$$\approx \beta \cdot A_v \cdot g_{m,V} \cdot R_L \cdot \frac{(1 + s/\omega_{z,V}) [1 + s/(\beta \cdot \omega_{BW,C})]}{(1 + s/\omega_{p1,V}) (1 + s/\omega_{BW,C}) (1 + s/\omega_{p2,V})}$$

Effects of VIH LDO

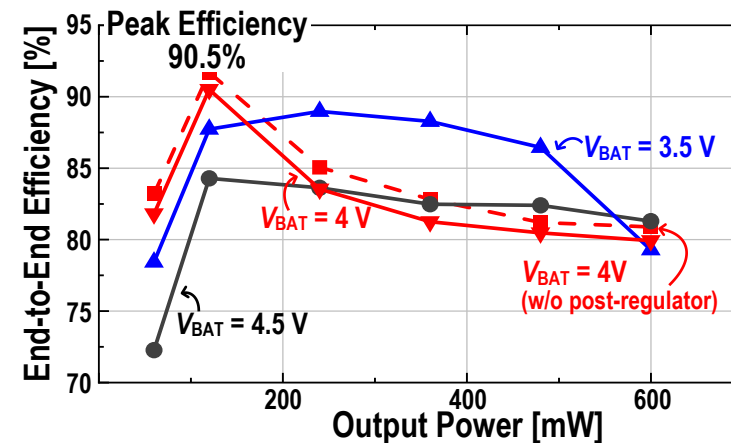
□ Ko & Kim, *ISSCC'21* & *JSSC'23*



- VIH LDO effectively rejects the switching ripple

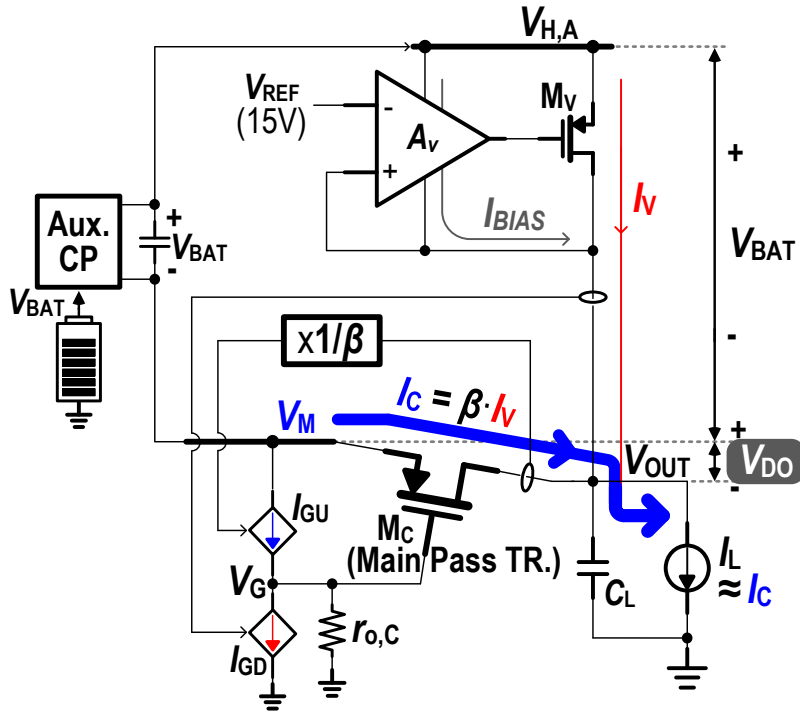


- **-75dB PSR @ 30kHz & 28.7 μ V_{RMS} noise**



- Efficiency loss is only **1.3%**

Summary of VIH (triode) LDO □ Ko & Kim, ISSCC'21 & JSSC'23



0.34mm² in 0.18-μm BCD
V_{IN} = 15.2V, V_{OUT} = 15V

- Dedicated separate two loops
 - T_v (voltage) regulates V_{OUT} by handling low I_v
 - T_c (current) supplies $I_c (= \beta \cdot I_v)$ via ultra-low V_{DO}
- High PSR even at ultra-low V_{DO}
 - R_{PASS} is boosted: $R_{DS} \rightarrow (1 + T_c)R_{DS}$
 - G_m -boosting (β) effect further reduces R_o
- Fabricated results
 - Power efficiency = 98.7%
 - PSR = -75dB @ 30kHz
 - V_{OUT} noise = 28.7μV_{RMS} integrated in 0.1 – 500kHz

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Conclusion

- LDO performs DC-DC step-down conversion via a *resistive* power transfer media
 - Error amplifier compares V_{OUT} with V_{REF} and controls the gate volt. of pass transistor
 - Features: no ripple (clean V_{OUT}), high PSR, small footprint (no bulky L), low efficiency
- Transient response of LDO is determined by loop bandwidth and slew-rate
 - Dynamic range of load current that LDO can accommodate affects the loop stability
 - Good FoM in LDO = bandwidth & slew rate **vs.** I_Q consumption
- Supply-ripple (ΔV_{IN}) may leak to V_{OUT} via three paths: 1) r_{DS} , 2) g_{mP} , 3) V_{REF}
 - Wide loop-BW helps to suppress ΔV_{IN} that is transferred through r_{DS}
 - Ripple-feedforward to gate can restrain ΔV_{GS} of power PMOS, improving PSR
 - Recursive (self-biased) LDO design is introduced to be free from PSR of V_{REF}
- Ultra-low-dropout (triode) LDO resolves a major drawback of low efficiency
 - Voltage-current-hybrid (VIH) design is a good option for both high eff. & high PSR

Thank you!

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